
Offline Two-Player Zero-Sum Markov Games with KL Regularization

Claire Chen¹ Yuheng Zhang² Xinyu Liu³ Zixuan Xie³ Shuze Daniel Liu^{4,5} Nan Jiang²

Abstract

We study the problem of learning Nash equilibria in offline two-player zero-sum Markov games. While existing approaches often rely on explicit pessimism to address distribution shift, we show that KL regularization alone suffices to stabilize learning and guarantee convergence. We first introduce Regularized Offline Sequential Equilibrium (ROSE), a theoretical framework that achieves a fast $\tilde{O}(1/n)$ convergence rate under *unilateral concentrability* and the stated coupled regularization/reference-policy condition, improving over the standard $\tilde{O}(1/\sqrt{n})$ rates in unregularized settings. We then propose Sequential Offline Self-Play Mirror Descent (SOS-MD), a practical model-free algorithm based on least-squares value estimation and iterative self-play updates. We prove that the last iterate of SOS-MD attains the same $\tilde{O}(1/n)$ statistical rate up to a vanishing optimization error of order $\tilde{O}(1/\sqrt{T})$ in the number of self-play iterations T .

1. Introduction

Offline Reinforcement Learning (RL) has emerged as a critical paradigm for developing decision-making policies in safety-critical or high-cost domains where online exploration is prohibitive, such as healthcare, autonomous driving, and industrial control (Fujimoto et al., 2019; Liu et al., 2019; Levine et al., 2020). While standard offline RL focuses on single-agent optimization, many real-world challenges inherently involve strategic interactions or adversarial dynamics that require a game-theoretic perspective (Liu et al., 2021b; 2026b;a). Two-player zero-sum Markov games provide the foundational framework for these settings, modeling scenarios where a learner must compete against an opponent

to ensure robust performance (Littman, 1994; Zhang et al., 2021). This formulation is particularly essential for risk-sensitive decision-making, where the adversary represents environmental disturbances or model uncertainties that the agent must withstand (Pinto et al., 2017; Cui & Du, 2022a). In this offline game-theoretic setting, the objective is to identify a Nash equilibrium (NE) solely from previously obtained offline data. Defined as a stable saddle point where no player can improve their outcome by changing their strategy alone, an NE ensures the learned policy remains robust even against the adversary (Osborne & Rubinstein, 1994; Başar & Olsder, 1998).

However, the offline nature of these games introduces the severe challenge of distribution shift, where the learned policy deviates from the behavior distribution represented in the dataset (Fujimoto et al., 2019; Xie et al., 2021). In such cases, learners often suffer from extrapolation error, assigning overoptimistic value estimates to state-action pairs poorly supported by the data (Kumar et al., 2020). To address the challenges of distribution shift, standard approaches typically rely on the principle of explicit pessimism (Liu et al., 2020; Jin et al., 2021; Rashidinejad et al., 2021). Methods such as Conservative Q-Learning (CQL) (Kumar et al., 2020) and Lower Confidence Bound (LCB) approaches (Jin et al., 2021; Cui & Du, 2022a) incorporate manually constructed *pessimistic bonuses* to suppress the values of out-of-distribution actions. While effective, these methods often require extensive manual tuning of penalties and complex implementation details (Cui & Du, 2022a; Zhong et al., 2022).

Recently, KL regularization with respect to a reference policy has emerged as a fundamental objective for incorporating prior knowledge and enforcing behavioral constraints (Xiong et al., 2024; Munos et al., 2024). It anchors updates to a reference distribution, such as a pre-trained model or human demonstrations. A prominent example is Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022), where a KL penalty is explicitly added to the reward function to prevent the aligned model from deviating excessively from the pre-trained language model. This regularization ensures that learned policies remain within the support of the reference distribution (Ye et al., 2024), preventing overoptimistic value estimates in regions without prior guidance (Nayak et al., 2025). However, despite

¹California Institute of Technology. ²University of Illinois Urbana-Champaign. ³University of Virginia. ⁴Massachusetts Institute of Technology. ⁵Purdue University. Correspondence to: Claire Chen <clairechen@caltech.edu>, Yuheng Zhang <yuhengz2@illinois.edu>.

Table 1. Comparison with previous offline Markov game results. Ye et al. (2024) and our paper study KL-regularized games, while other works consider the unregularized setting. We provide the first $\tilde{O}(1/n)$ statistical rate analysis in the multi-step setting, improving over the standard $\tilde{O}(1/\sqrt{n})$ rate in prior literature.

Method	KL-reg.	Multi-step	Statistical Rate
PNVI (Cui & Du, 2022a)	✗	✓	$\tilde{O}(1/\sqrt{n})$
PMVI (Zhong et al., 2022)	✗	✓	$\tilde{O}(1/\sqrt{n})$
BCEL (Zhang et al., 2023)	✗	✓	$\tilde{O}(1/\sqrt{n})$
Ye et al. (2024)	✓	✗	$\tilde{O}(1/\sqrt{n})$
Ours (SOS-MD)	✓	✓	$\tilde{O}(1/n)$

the widespread empirical success of KL regularization, its theoretical impact on sample efficiency remains poorly understood. Existing analyses typically derive convergence rates identical to those without KL regularization (Ye et al., 2024), thereby missing the opportunity to theoretically justify the accelerated convergence observed in practice.

In this work, we demonstrate that a faster statistical rate of $\tilde{O}(1/n)$ is achievable in KL-regularized offline two-player zero-sum Markov games. Our analysis reveals that anchoring to a fixed reference policy is sufficient to stabilize offline learning in this setting, eliminating the need for explicit pessimistic bonuses. We first propose Regularized Offline Sequential Equilibrium (ROSE, Algorithm 1), a theoretical framework that directly computes the regularized Nash equilibrium via backward induction. ROSE achieves this fast rate relying only on *unilateral concentrability* as its coverage condition, the minimal coverage condition to learn the NE in the worst case (Cui & Du, 2022a), matching the learnability requirements of offline Markov games. To address the computational intractability of solving exact equilibria in large-scale settings, we further propose Sequential Offline Self-Play Mirror Descent (SOS-MD, Algorithm 2), a practical model-free method that approximates the equilibrium via iterative mirror descent updates. Our primary contributions are as follows:

1. **Pessimism-Free Theoretical Framework:** We introduce ROSE, which establishes that KL regularization eliminates the need for explicit pessimism in offline Markov games. This approach bypasses the technical overhead of constructing and tuning confidence bonuses, a requirement central to prior pessimistic approaches (Jin et al., 2021; Zhong et al., 2022).
2. **Fast Statistical Rate:** We establish that ROSE achieves a fast $\tilde{O}(1/n)$ statistical rate for the duality gap. As summarized in Table 1, this significantly improves upon the standard $\tilde{O}(1/\sqrt{n})$ rates found in prior offline game-theoretic literature (Cui & Du, 2022a; Zhong et al., 2022; Zhang et al., 2023). While KL regularization has been studied in single-step or online

settings (Ye et al., 2024), our work is the first to unlock the fast-rate phenomenon in the *multi-step* offline game setting. Our analysis reveals that the strong convexity of the KL-regularized objective ensures the regret scales quadratically with the estimation error, extending the fast rates from single-agent systems (Hu et al., 2021) and regularized contextual games (Zhang et al., 2026) to Markov games.

3. **Computationally Efficient Approximation:** We prove that the last iterate of our practical instantiation, SOS-MD, preserves the same $\tilde{O}(1/n)$ statistical rate up to a vanishing optimization error of order $\tilde{O}(1/\sqrt{T})$. Unlike standard game-theoretic approaches that rely on iterate averaging to guarantee convergence (Freund & Schapire, 1999), we show that the regularity induced by KL regularization enables the *last iterate* of mirror descent to converge directly. By establishing a total error bound that incorporates both optimization and statistical errors, we demonstrate that SOS-MD bridges the gap between theoretical solvability and computational efficiency.

2. Related Work

Offline Reinforcement Learning. The primary challenge in offline reinforcement learning (RL) arises from the distribution shift between the data collection policy (behavior policy) and the target policy (Liu et al., 2025a; Liu & Zhang, 2024; Liu, 2025; Chen et al., 2025; Liu et al., 2025b; Mahadevan et al., 2026). To address the insufficient coverage of the dataset, the principle of *pessimism* has been established as a standard technique for provably efficient learning (Liu et al., 2020; Rashidinejad et al., 2021; Jin et al., 2021; Xie et al., 2021; Uehara & Sun, 2022; Zhan et al., 2022). The key idea is to penalize state-action pairs that are rarely visited in the dataset to avoid overestimation bias. This pessimistic framework has been shown to be minimax optimal under single-policy concentrability (Li et al., 2024). In contrast to this prevailing paradigm, we propose a new algorithm for KL-regularized Markov games that operates without explicit

pessimism, and our analysis establishes that it also achieves provably efficient learning.

Markov Games. Markov games, also known as stochastic games, generalize Markov decision processes (MDPs) to multi-agent settings, where multiple players interact strategically over time under shared environment dynamics (Littman, 1994). These games have been extensively studied in the online setting, covering both two-player zero-sum games (Wei et al., 2017; Bai et al., 2020; Liu et al., 2021a; Bai et al., 2022) and multi-player general-sum games (Zhang et al., 2021; Jin et al., 2023; Song et al., 2022; Mao & Başar, 2023). More recently, there has been growing interest in *offline* Markov games, where agents must learn equilibrium strategies from a fixed dataset without further interaction. Existing works typically address this challenge by extending the principle of pessimism to game-theoretic objectives (Cui & Du, 2022a; Zhong et al., 2022; Cui & Du, 2022b; Zhang et al., 2023). A key distinction of offline Markov games from single-agent RL is the coverage requirement: while single-agent tasks often require only single-policy coverage, learning Nash equilibria in offline Markov games necessitates covering all unilateral deviations, formalized as the *unilateral concentrability* condition (Cui & Du, 2022a; Chen & Zhang, 2026a;b). Our analysis adopts this assumption, which is established as the weakest sufficient coverage condition in the worst case.

KL-regularized Objectives. Motivated by the widespread success of Reinforcement Learning from Human Feedback (RLHF) in aligning large language models (Ouyang et al., 2022), there has been a surge of research interest in studying KL-regularized objectives in RL, which penalize deviations from a reference policy. The KL-regularized objectives have been theoretically analyzed in both the single-agent setting (Xiong et al., 2024; Xie et al., 2025; Zhao et al., 2025) and the multi-agent setting (Munos et al., 2024; Ye et al., 2024; Zhang et al., 2025b; Nayak et al., 2025). However, much of this existing theory focuses on bandit settings (single-step) or online interactions. In contrast, our work investigates the multi-step dynamics inherent in sequential decision-making, formulating the problem as an offline KL-regularized Markov game.

3. Preliminaries

We consider the offline learning problem in episodic two-player zero-sum Markov games with KL regularization. In this section, we define the problem setup, the regularized objective, and the solution concepts used throughout the paper. For convenience, a summary of the notation used throughout this paper is provided in Table 2 of Appendix A.

3.1. Two-Player Zero-Sum Markov Games

We define a finite-horizon Markov game by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r, H)$, where $\mathcal{S}$ is the possibly infinite state space, and $\mathcal{A}_1, \mathcal{A}_2$ are the finite action spaces for Player 1 and Player 2, respectively. We take $(\mathcal{S}, \Sigma)$ to be standard Borel, with ρ a probability measure, each P_h a measurable kernel, and the rewards, reference policies, and every $q \in \mathcal{Q}_h$ measurable; the finite action sets make the Gibbs, saddle, and $\arg \max$ policies measurable in s . The induced trajectory laws are Ionescu–Tulcea measures, and the occupancy laws below are their marginals. Accordingly, the change-of-measure steps in Appendices C.3 and C use Radon–Nikodym derivatives; point probabilities such as $\mathbb{P}_\pi(S_h = s)$ are shorthand for the countable case. The game proceeds over H steps. At each step $h \in [H]$, given the current state s_h , both players simultaneously choose actions $a_{h,1} \in \mathcal{A}_1$ and $a_{h,2} \in \mathcal{A}_2$. The system transitions to the next state $s_{h+1} \sim P_h(\cdot | s_h, a_{h,1}, a_{h,2})$, and Player 1 receives a deterministic reward $r_h^*(s_h, a_{h,1}, a_{h,2}) \in [0, 1]$. As the game is zero-sum, Player 2 receives reward $-r_h^*(s_h, a_{h,1}, a_{h,2})$.

We denote a policy for Player $k \in \{1, 2\}$ as $\pi_k = \{\pi_{h,k}\}_{h=1}^H$, where $\pi_{h,k} : \mathcal{S} \rightarrow \Delta(\mathcal{A}_k)$ maps states to distributions over actions. A joint policy is denoted by $\pi = (\pi_1, \pi_2)$. For brevity, we denote the joint action by $a = (a_1, a_2)$ and the product distribution of the policies by $\pi(a|s) = \pi_1(a_1|s)\pi_2(a_2|s)$. We use $\mathbb{E}_{a \sim \pi}[\cdot]$ to represent the expectation with respect to this joint distribution. For a joint policy π , let d_h^π denote the *state-action* law of $(s_h, a_{h,1}, a_{h,2})$ under the trajectory induced by π and P , starting from $s_1 \sim \rho$, and let $d_{h,S}^\pi$ denote its state marginal. In kernel notation,

$$d_h^\pi(ds, a_1, a_2) = d_{h,S}^\pi(ds) \pi_{h,1}(a_1|s) \pi_{h,2}(a_2|s).$$

For countable $\mathcal{S}$, this reduces to $d_h^\pi(s, a_1, a_2) = \mathbb{P}_\pi(s_h = s, a_{h,1} = a_1, a_{h,2} = a_2)$ and $d_{h,S}^\pi(s) = \sum_{a_1, a_2} d_h^\pi(s, a_1, a_2)$. We reserve d_h^π for the state-action law, and for a function f of the state alone use the shorthand $\mathbb{E}_{s \sim d_h^\pi}[f(s)] := \mathbb{E}_{(s, a_1, a_2) \sim d_h^\pi}[f(s)] = \mathbb{E}_{s \sim d_{h,S}^\pi}[f(s)]$.

3.2. KL-Regularized Value Functions

To incorporate prior knowledge and restrict the learned policy to valid regions, we employ KL regularization with respect to fixed reference policies $\pi^{\text{ref}} = (\pi_1^{\text{ref}}, \pi_2^{\text{ref}})$. For a regularization coefficient $\eta > 0$, the goal of the agent is to maximize the expected cumulative reward subject to a KL penalty at each step.

Accordingly, the regularized value functions V_h^π and Q_h^π satisfy the following Bellman recursions. The Q-function is

the expected regularized return starting from an (s, a) pair:

$$Q_h^\pi(s, a_1, a_2) := r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot|s, a_1, a_2)}[V_{h+1}^\pi(s')],$$

with $V_{H+1}^\pi(s) = 0$. The value function $V_h^\pi(s)$ incorporates the immediate regularization penalty:

$$V_h^\pi(s) = \mathbb{E}_{a \sim \pi_h(\cdot|s)}[Q_h^\pi(s, a)] - \eta^{-1} \text{KL}(\pi_{h,1}(\cdot|s) \parallel \pi_{h,1}^{\text{ref}}(\cdot|s)) \\ + \eta^{-1} \text{KL}(\pi_{h,2}(\cdot|s) \parallel \pi_{h,2}^{\text{ref}}(\cdot|s)). \quad (1)$$

Under this formulation, minimizing V_h^π with respect to π_2 is equivalent to Player 2 maximizing their own cumulative reward penalized by the KL divergence from π_2^{ref} .

Reference support and admissible policies. The KL terms in (1) are finite only for policies that are absolutely continuous with respect to the reference policy, so we build this restriction into the policy classes. For $i \in \{1, 2\}$ we define the *reference support* and the induced class of *admissible* policies

$$\mathcal{A}_{h,i}^{\text{ref}}(s) := \{a \in \mathcal{A}_i : \pi_{h,i}^{\text{ref}}(a|s) > 0\}, \\ \Pi_i^{\text{ref}} := \{\pi_i : \text{supp } \pi_{h,i}(\cdot|s) \subseteq \mathcal{A}_{h,i}^{\text{ref}}(s)\}, \quad (2)$$

imposed at every (h, s) . Throughout, we set $\Pi_i := \Pi_i^{\text{ref}}$ and define all solution concepts over $\Pi_1 \times \Pi_2$; this also keeps V_h^π well defined, with the two KL terms in (1) having opposite signs.

3.3. Regularized Nash Equilibrium

In the regularized game, a regularized *Nash equilibrium* (NE) is a saddle point of the value function. We define the best-response values for Player 1 and Player 2 respectively:

$$V_h^{\dagger, \pi_2}(s) := \max_{\pi_1' \in \Pi_1} V_h^{\pi_1', \pi_2}(s), \quad \text{and} \\ V_h^{\pi_1, \dagger}(s) := \min_{\pi_2' \in \Pi_2} V_h^{\pi_1, \pi_2'}(s).$$

Both extrema are over admissible policies (2), and both are attained: by backward induction, the stage- h problem faced by a single player against a fixed opponent is solved in closed form by a Gibbs distribution supported on $\mathcal{A}_{h,i}^{\text{ref}}(s)$ (see the proof of Lemma C.2). A joint policy $\pi^* = (\pi_1^*, \pi_2^*)$ is a regularized Nash equilibrium if, for all $s \in \mathcal{S}$ and $h \in [H]$:

$$V_h^{\pi_1^*, \dagger}(s) = V_h^{\pi^*}(s) = V_h^{\dagger, \pi_2^*}(s). \quad (3)$$

Such a π^* exists and is unique in $\Pi_1 \times \Pi_2$: given the next-step value, the stage objective in (1) is strictly concave in Player 1's action distribution and strictly convex in Player 2's on the reference support, so backward induction over $h = H, \dots, 1$ produces a unique stage saddle point at every (h, s) . We write $V_h^* := V_h^{\pi^*}$ and

$Q_h^* := Q_h^{\pi^*}$. When the reward needs to be explicit, let $J(r, \pi_1, \pi_2) := \mathbb{E}_{s_1 \sim \rho}[V_1^{\pi_1, \pi_2}(s_1; r)]$, where $V^{\pi_1, \pi_2}(\cdot; r)$ is the regularized value function under reward r ; for the true reward r^* , we abbreviate $J(\pi_1, \pi_2) := J(r^*, \pi_1, \pi_2)$. To evaluate the quality of a learned policy $\hat{\pi}$, we use the *Duality Gap*, which measures how exploitable the policy is by optimal adversaries:

$$\text{Gap}(\hat{\pi}) := \mathbb{E}_{s_1 \sim \rho} \left[V_1^{\dagger, \hat{\pi}_2}(s_1) - V_1^{\hat{\pi}_1, \dagger}(s_1) \right],$$

where ρ is the initial state distribution. Since $V_1^{\dagger, \hat{\pi}_2} \geq V_1^{\hat{\pi}_1, \dagger}$ pointwise, $\text{Gap}(\hat{\pi}) \geq 0$, and $\text{Gap}(\hat{\pi}) = 0$ if and only if $\hat{\pi}$ is a regularized Nash equilibrium from ρ . Note that this is weaker than the two-sided condition (3) at every (h, s) : each of the two equalities is forced only along its own best-response process, and those two reachable sets differ.

3.4. Offline Learning with Function Approximation

We operate in the offline setting, where the learner has access to a static dataset $\mathcal{D}$ collected by a behavior policy. We assume the dataset consists of n trajectories drawn independently from the behavior process, so that independence holds across the trajectories i , while the H stages within a single trajectory are dependent. The dataset is denoted as:

$$\mathcal{D} = \{(s_{i,h}, a_{i,h,1}, a_{i,h,2}, r_{i,h}, s'_{i,h})\}_{i=1, h=1}^{n, H},$$

where $s'_{i,h} = s_{i,h+1}$ is the state at the next time step. The rewards in the dataset are noisy realizations of the true deterministic reward (Liu et al., 2025c):

$$r_{i,h} = r_h^*(s_{i,h}, a_{i,h,1}, a_{i,h,2}) + \xi_{i,h},$$

where, conditionally on $(s_{i,h}, a_{i,h,1}, a_{i,h,2})$, the noise $\xi_{i,h}$ is zero-mean and 1-sub-Gaussian, independently across the trajectories i .

We estimate the Q -values with finite stagewise classes $\mathcal{Q}_h \subset (\mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathbb{R})$, writing $\mathcal{Q}$ when the step is clear and $|\mathcal{Q}| := \max_h |\mathcal{Q}_h|$. Let $(\mathcal{T}_h f)(s, a) := r_h^*(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[f(s')]$ be the stage- h Bellman backup. We adopt the standard completeness assumption (Xie et al., 2021; Zhang et al., 2023) stagewise: with $\mathcal{V}_{h+1}^{\text{alg}}$ the step- $(h+1)$ value class induced by the algorithm under analysis from arbitrary $\tilde{Q}_{h'} \in \mathcal{Q}_{h'}$, we require $\mathcal{T}_h V_{h+1} \in \mathcal{Q}_h$ for all $V_{h+1} \in \mathcal{V}_{h+1}^{\text{alg}}$.

4. Method: Pessimism-Free Learning via KL Regularization

In this section, we propose Regularized Offline Sequential Equilibrium (ROSE, Algorithm 1), a model-free framework that utilizes KL regularization to achieve sample-efficient

Algorithm 1 Regularized Offline Sequential Equilibrium (ROSE)

Require: Regularization $\eta > 0$, reference policies $\pi_1^{\text{ref}}, \pi_2^{\text{ref}}$, offline dataset $\mathcal{D}$, stagewise function classes $\{\mathcal{Q}_h\}_{h=1}^H$. Here $\mathcal{D}_h = \{(s_{i,h}, a_{i,h,1}, a_{i,h,2}, r_{i,h}, s'_{i,h})\}_{i=1}^n$ denotes the stage- h slice of $\mathcal{D}$, which contains one tuple per trajectory.

Initialization: Set $\hat{V}_{H+1}(s) = 0$ for all s .

Backward Induction: For $h = H, \dots, 1$ do:

 1. **Regression Targets:**

$$y_{i,h} = r_{i,h} + \hat{V}_{h+1}(s'_{i,h}), \quad \forall i \in [n].$$

 2. **Q-Function Estimation:** Compute $\hat{Q}_h$ by solving the least-squares regression:

$$\hat{Q}_h \in \arg \min_{f \in \mathcal{Q}_h} \sum_{i \in [n]} (f(s_{i,h}, a_{i,h,1}, a_{i,h,2}) - y_{i,h})^2.$$

 3. **Equilibrium Computation:** For every $s \in \mathcal{S}$, define the regularized stage objective

$$F_{h,s}(\pi_1, \pi_2) := \mathbb{E}_{a \sim \pi_1 \times \pi_2} [\hat{Q}_h(s, a)] - \eta^{-1} \text{KL}(\pi_1 \| \pi_{h,1}^{\text{ref}}) + \eta^{-1} \text{KL}(\pi_2 \| \pi_{h,2}^{\text{ref}}),$$

and compute its unique saddle point $(\hat{\pi}_{h,1}(\cdot|s), \hat{\pi}_{h,2}(\cdot|s))$, characterized by

$$\begin{aligned} F_{h,s}(\pi_1, \hat{\pi}_{h,2}(\cdot|s)) &\leq F_{h,s}(\hat{\pi}_{h,1}(\cdot|s), \hat{\pi}_{h,2}(\cdot|s)) \\ &\leq F_{h,s}(\hat{\pi}_{h,1}(\cdot|s), \pi_2) \end{aligned}$$

for all $\pi_i \in \Delta(\mathcal{A}_{h,i}^{\text{ref}}(s))$. Existence and uniqueness follow from strict concavity–convexity on the reference supports.

 4. **Value Update:** Update the value function $\hat{V}_h(s)$ as the value of the unique regularized equilibrium:

$$\begin{aligned} \hat{V}_h(s) &= \mathbb{E}_{a \sim \hat{\pi}_{h,1} \times \hat{\pi}_{h,2}} [\hat{Q}_h(s, a)] \\ &\quad - \eta^{-1} \text{KL}(\hat{\pi}_{h,1} \| \pi_{h,1}^{\text{ref}}) + \eta^{-1} \text{KL}(\hat{\pi}_{h,2} \| \pi_{h,2}^{\text{ref}}). \end{aligned}$$

Output: Learned policy $\hat{\pi} = \{\hat{\pi}_h\}_{h=1}^H$.

learning without explicit pessimistic bonuses. It integrates Least-Squares Value Iteration with an exact Regularized Nash Equilibrium solver to establish the theoretical foundations of our approach. We then discuss the intuition behind why KL regularization effectively handles distribution shift in offline games.

4.1. Algorithm Description

Our approach, outlined in Algorithm 1, adopts a fitted Q-iteration approach. The core idea is to iteratively estimate the regularized Q-function using the offline dataset $\mathcal{D}$ and then solve for the regularized Nash equilibrium (NE) of the estimated game at each step.

The algorithm proceeds via backward induction from $h = H$ to 1. At each step h , we first construct regression targets $y_{i,h}$ using the reward and the estimated value from the next step, $\hat{V}_{h+1}$. We then estimate the Q-function $\hat{Q}_h$ by solving a least-squares regression problem over the stage- h function class $\mathcal{Q}_h$: $\hat{Q}_h \in \arg \min_{f \in \mathcal{Q}_h} \sum_{i=1}^n (f(s_{i,h}, a_{i,h,1}, a_{i,h,2}) - y_{i,h})^2$. Given the estimated $\hat{Q}_h$, we compute the joint policy $(\hat{\pi}_{h,1}, \hat{\pi}_{h,2})$ that forms a Regularized Nash Equilibrium for the current stage game. Algorithm 1 uses the *raw* estimated Q-values directly, without requiring explicit pessimistic penalties or modifications to the regression targets. The regularization is handled entirely by the equilibrium solver step, which maximizes the KL-regularized objective defined in Equation (1).

4.2. Implicit Pessimism via KL Regularization

A key distinction of our work is the absence of explicit uncertainty quantification. While standard offline RL methods (Kumar et al., 2020; Jin et al., 2021) and their game-theoretic extensions (Cui & Du, 2022a; Zhong et al., 2022) rely on manually constructed penalty terms (e.g., LCBs) to suppress out-of-distribution actions, Algorithm 1 achieves robustness intrinsically through the geometry of the KL-regularized objective.

To see this, consider the stage- h best-response problem for Player 1 in the regularized game defined by $\hat{Q}_h$, against a fixed opponent policy $\pi_{h,2}$. Writing $\hat{q}_{h,1}(s, a_1) := \mathbb{E}_{a_2 \sim \pi_{h,2}(\cdot|s)} [\hat{Q}_h(s, a_1, a_2)]$ for the induced marginal payoff, the closed-form solution is the Gibbs distribution

$$\hat{\pi}_{h,1}(a_1|s) \propto \pi_{h,1}^{\text{ref}}(a_1|s) \cdot \exp(\eta \hat{q}_{h,1}(s, a_1)), \quad (4)$$

which for $\pi_{h,2} = \hat{\pi}_{h,2}$ is exactly the Player-1 component computed by Algorithm 1. The reference policy thus acts as a probabilistic anchor: the update re-weights π^{ref} multiplicatively rather than adding a penalty. Two consequences follow. (i) The re-weighting is support-preserving: an action with $\pi_{h,1}^{\text{ref}}(a_1|s) = 0$ gets probability exactly 0, whatever $\hat{Q}_h$ assigns it. (ii) The normalizer of (4) is a $\pi_{h,1}^{\text{ref}}(\cdot|s)$ -average of $\exp(\eta \hat{q}_{h,1}(s, \cdot))$ and so lies between its extrema over $\mathcal{A}_{h,1}^{\text{ref}}(s)$; hence the log-deviation from the anchor is controlled by the payoff oscillation: for every $a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)$,

$$\left| \log \frac{\hat{\pi}_{h,1}(a_1|s)}{\pi_{h,1}^{\text{ref}}(a_1|s)} \right| \leq \eta \text{osc}(\hat{q}_{h,1}(s, \cdot)),$$

with osc taken over $\mathcal{A}_{h,1}^{\text{ref}}(s)$. The learned policy can still

inflate the reference mass of a supported action, but only by this bounded factor: the barrier is a bounded-logit one, not an absolute one. It is this quantified property — not any claim that $\hat{\pi}_{h,1}$ is insensitive to large $\hat{Q}_h$ — that the analysis uses, via Lemmas D.4 and C.5. Under the coupled condition in Theorem 6.3, the resulting trajectory-level density-ratio transfers have an absolute constant cost.

Equation (4) characterizes the optimality condition for a fixed opponent. In the full game setting, both players optimize their strategies simultaneously, leading to a coupled equilibrium problem. However, computing this exact equilibrium is computationally prohibitive, particularly in large-scale settings. This necessitates the iterative self-play algorithm (SOS-MD) detailed in Section 5.

5. Sequential Offline Self-Play Mirror Descent

While ROSE (Algorithm 1) establishes the theoretical foundation of the KL-regularized framework, computing the exact regularized Nash equilibrium at each step is often computationally intractable in practice. To bridge this gap, we introduce Sequential Offline Self-Play Mirror Descent (SOS-MD), a computationally efficient instantiation that approximates the equilibrium via iterative updates.

Instead of relying on an exact solver, SOS-MD employs a dual Mirror Descent approach to iteratively refine the policy. By exploiting the geometry of the KL-divergence, our method produces simple, closed-form updates based on exponentiated gradients. Notably, for sufficiently small policy updates, our KL-regularized mirror descent rule reduces to the Natural Policy Gradient (NPG) (Kakade, 2001) update. This connection aligns our offline game-theoretic framework with successful online policy optimization algorithms such as TRPO (Schulman et al., 2015).

5.1. Algorithm Details

As outlined in Algorithm 2, SOS-MD operates via backward induction. At each stage h , the algorithm alternates between estimating the game structure and solving the induced game.

Stagewise Q-Function Regression. We employ the same least-squares regression procedure as Algorithm 1 to estimate $\hat{Q}_h$, minimizing the squared error over $\mathcal{Q}_h$ with targets derived from the next-step value estimate $\hat{V}_{h+1}$.

Self-Play Optimization via Mirror Descent. SOS-MD resolves the coupled optimization by iterative self-play. Both players start at their reference policies π^{ref} and, for $T \geq 1$ iterations, update using the *marginal payoffs* induced by the current $\hat{Q}_h$. The update (5)–(6) is a KL-regularized mirror descent step: in log-space it takes a convex combination of $\pi^{(t)}$ and π^{ref} and shifts it by γ_t times the marginal

payoff, so the policy improves its payoff while staying close to the reference.

Value Function Update. We compute $\hat{V}_h(s)$ from the last iterates $\pi_h^{(T)}$; regularization-induced strong convexity makes them converge fast, so no time-averaging is needed. Since $\pi_h^{(T)}$ is only an approximate equilibrium of $\hat{Q}_h$, SOS-MD backs up the objective value at the last iterate rather than the saddle value, so its fitted sequence differs from that of Algorithm 1; we write $(\tilde{Q}_h, \tilde{V}_h)$ when the two must be distinguished.

6. Theoretical Analysis

In this section, we establish the theoretical guarantees for our approach. We first analyze the statistical sample complexity of the Algorithm 1, demonstrating the “implicit pessimism” property. We then analyze the convergence of SOS-MD (Algorithm 2) and derive the final total error bound, which accounts for both statistical and optimization errors.

6.1. Statistical Efficiency

We begin by bounding the suboptimality of the policy learned by Algorithm 1. This analysis decouples the statistical error from the optimization error, focusing exclusively on the suboptimality arising from finite sampling and function approximation.

To characterize the coverage of the offline dataset, we first introduce the D^2 -divergence as a function-class-aware pointwise extrapolation factor, following Ye et al. (2024); Zhao et al. (2025).

Definition 6.1 (D^2 -divergence). Let $\mu = \{\mu_h\}_{h=1}^H$ denote the underlying distribution of the offline dataset $\mathcal{D}$, where $\mu_h \in \Delta(\mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2)$ is the marginal state-action distribution at step h induced by the behavior policy. The D^2 -divergence at (s, a_1, a_2) relative to μ_h is defined as

$$D_{\mathcal{Q}}^2((s, a_1, a_2); \mu_h) := \sup_{Q_1, Q_2 \in \mathcal{Q}} \frac{(Q_1(s, a_1, a_2) - Q_2(s, a_1, a_2))^2}{\mathbb{E}_{(s', a'_1, a'_2) \sim \mu_h} [(Q_1 - Q_2)^2]},$$

with the extended-real conventions $c/0 := +\infty$ for $c > 0$ and $0/0 := 0$. It enters the analysis only through the transfer inequality (72).

Throughout, equilibria, best responses, deviations and duality gaps are taken over the reference-supported classes $\Pi_1^{\text{ref}} \times \Pi_2^{\text{ref}}$ of (2), and action maxima over the reference supports $\mathcal{A}_{h,i}^{\text{ref}}(s)$.

Equipped with the D^2 -divergence, we adopt the *Unilateral Concentrability* condition proposed by Cui & Du (2022a),

Algorithm 2 Sequential Offline Self-Play Mirror Descent (SOS-MD)

Require: Regularization $\eta > 0$, Reference policies $\pi^{\text{ref}} = \{\pi_{h,1}^{\text{ref}}, \pi_{h,2}^{\text{ref}}\}_{h=1}^H$, Offline dataset $\mathcal{D}$, Stagewise function classes $\{\mathcal{Q}_h\}_{h=1}^H$, Number of self-play iterations $T \geq 1$, Stepsize schedule γ_t .

- 1: Initialize terminal value function $\hat{V}_{H+1}(s) = 0$ for all $s \in \mathcal{S}$.
- 2: **for** $h = H, \dots, 1$ **do**
- 3: **Stagewise Q-Function Regression:**
- 4: Construct empirical regression targets using the next-step value estimate:

$$y_{i,h} = r_{i,h} + \hat{V}_{h+1}(s'_{i,h}) \quad \forall i \in [n]$$

- 5: Estimate $\hat{Q}_h$ via stagewise least-squares minimization over function class $\mathcal{Q}_h$:

$$\hat{Q}_h \leftarrow \operatorname{argmin}_{f \in \mathcal{Q}_h} \sum_{i=1}^{|\mathcal{D}_h|} (f(s_{i,h}, a_{i,h,1}, a_{i,h,2}) - y_{i,h})^2$$

- 6: **Self-Play Optimization (Dual Mirror Descent):**
- 7: Initialize policy iterates: $\pi_{h,1}^{(0)} = \pi_{h,1}^{\text{ref}}$ and $\pi_{h,2}^{(0)} = \pi_{h,2}^{\text{ref}}$.
- 8: **for** $t = 0, \dots, T-1$ **do**
- 9: **Marginal Payoff Evaluation:**
- 10: Compute Player 1's expected payoff against current Player 2 iterate:

$$\hat{q}_{h,1}^{(t)}(s, a_1) = \mathbb{E}_{a_2 \sim \pi_{h,2}^{(t)}(\cdot|s)}[\hat{Q}_h(s, a_1, a_2)]$$

- 11: Compute Player 2's expected payoff against current Player 1 iterate:

$$\hat{q}_{h,2}^{(t)}(s, a_2) = \mathbb{E}_{a_1 \sim \pi_{h,1}^{(t)}(\cdot|s)}[\hat{Q}_h(s, a_1, a_2)]$$

- 12: **Policy Update:** all updates are on the reference support $\mathcal{A}_{h,i}^{\text{ref}}(s) = \operatorname{supp}(\pi_{h,i}^{\text{ref}}(\cdot|s))$, with $\pi_{h,i}^{(t+1)}(a|s) = 0$ for $a \notin \mathcal{A}_{h,i}^{\text{ref}}(s)$ and each $\propto$ normalized over $\mathcal{A}_{h,i}^{\text{ref}}(s)$.
- 13: Update Player 1 (Ascent - Maximizing $\hat{Q}_h$) using stepsize γ_t , for all $s \in \mathcal{S}$ and $a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)$:

$$\pi_{h,1}^{(t+1)}(a_1|s) \propto \pi_{h,1}^{(t)}(a_1|s)^{1-\gamma_t\eta^{-1}} \cdot \exp\left(\gamma_t \cdot \hat{q}_{h,1}^{(t)}(s, a_1)\right) \cdot \pi_{h,1}^{\text{ref}}(a_1|s)^{\gamma_t\eta^{-1}} \quad (5)$$

- 14: Update Player 2 (Descent - Minimizing $\hat{Q}_h$) using stepsize γ_t , for all $s \in \mathcal{S}$ and $a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)$:

$$\pi_{h,2}^{(t+1)}(a_2|s) \propto \pi_{h,2}^{(t)}(a_2|s)^{1-\gamma_t\eta^{-1}} \cdot \exp\left(-\gamma_t \cdot \hat{q}_{h,2}^{(t)}(s, a_2)\right) \cdot \pi_{h,2}^{\text{ref}}(a_2|s)^{\gamma_t\eta^{-1}} \quad (6)$$

- 15: **end for**
- 16: **Update Value Function:**
- 17: Compute stage h value using the last iterate $\pi_h^{(T)}$:

$$\hat{V}_h(s) \leftarrow \mathbb{E}_{\substack{a_1 \sim \pi_{h,1}^{(T)} \\ a_2 \sim \pi_{h,2}^{(T)}}} \left[\hat{Q}_h(s, a_1, a_2) \right] - \eta^{-1} \operatorname{KL}(\pi_{h,1}^{(T)} \| \pi_{h,1}^{\text{ref}}) + \eta^{-1} \operatorname{KL}(\pi_{h,2}^{(T)} \| \pi_{h,2}^{\text{ref}})$$

- 18: **end for**
- 19: **Return:** Learned policy sequence $\pi^{(T)} = (\{\pi_{h,1}^{(T)}\}_{h=1}^H, \{\pi_{h,2}^{(T)}\}_{h=1}^H)$.

which is recognized as the minimal coverage requirement for offline learning in Markov games in the worst case.

Assumption 6.2 (Unilateral Concentrability). Let $\Pi :=$

$\Pi_1^{\text{ref}} \times \Pi_2^{\text{ref}}$ be the class of joint reference-supported policies. We define the set of *unilateral deviation policies* $\Pi_{\text{uni}} \subset \Pi$ as the union of policy pairs where at least one player adheres

to the Nash equilibrium π^* :

$$\Pi_{\text{uni}} := (\{\pi_1^*\} \times \Pi_2^{\text{ref}}) \cup (\Pi_1^{\text{ref}} \times \{\pi_2^*\}).$$

There exists a constant $C_{\text{uni}} \geq 1$ such that, for all $h \in [H]$ and every player $i \in \{1, 2\}$,

$$\sup_{\pi_i \in \Pi_i^{\text{ref}}} \mathbb{E}_{(s, a_1, a_2) \sim d_h^{\pi_i \times \pi_{-i}^*}} [D_Q^2((s, a_1, a_2); \mu_h)] \leq C_{\text{uni}},$$

where $\pi_i \times \pi_{-i}^* \in \Pi_{\text{uni}}$ denotes the joint policy in which Player i plays π_i and Player $-i$ plays π_{-i}^* , and $d_h^{\pi'}$ denotes the state-action visitation distribution at step h induced by π' .

Let $\alpha := \inf\{\pi_{\text{ref},h,i}(a|s) \mid \pi_{\text{ref},h,i}(a|s) > 0\}$, the minimum positive mass of the reference policy, taken over its support so that α is defined for arbitrary, including deterministic, references; we assume $\alpha > 0$. Write $\lambda := 1 + \eta^{-1} \log(1/\alpha)$ and $C_{\text{trans}} := e$, the absolute trajectory-level transfer constant obtained under the side condition of Theorem 6.3 (Lemma C.5).

Theorem 6.3. *Let $\hat{\pi}$ be the policy output by Algorithm 1. Suppose Assumptions 6.2 and C.1 hold with $\alpha > 0$, together with the function-class conditions of Section 3 for Algorithm 1, and that*

$$\eta \leq \frac{1}{4H^2(1 + \eta^{-1} \log(1/\alpha))} = \frac{1}{4H^2\lambda}. \quad (7)$$

Then with probability at least $1 - \delta$, the duality gap of $\hat{\pi}$ satisfies:

$$\text{Gap}(\hat{\pi}) \leq \tilde{\mathcal{O}}\left(\frac{\eta H^6 \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n}\right).$$

Notably, this result establishes a fast $\tilde{\mathcal{O}}(1/n)$ rate in the sample size, improving upon the standard $\tilde{\mathcal{O}}(1/\sqrt{n})$ rate in offline Markov games (Cui & Du, 2022a; Zhong et al., 2022; Zhang et al., 2023, see also Table 1). The bound scales linearly with the unilateral concentrability C_{uni} and logarithmically with $|\mathcal{Q}|$, so ordinary unilateral coverage in the function-class-aware D^2 -divergence form is the only coverage condition we impose.

Proof Sketch. The proof of Theorem 6.3 relies on a novel decomposition of the regularized duality gap that bypasses pessimism. We analyze the exploitability of Player 1 — the Player-2 half is obtained not by an appeal to symmetry of the original game, whose transitions and reward need not be symmetric, but by the explicit reflected-game reduction of Appendix C — by splitting the error relative to the true Nash equilibrium π^* :

$$\begin{aligned} & \underbrace{J(\pi_1^\dagger, \hat{\pi}_2) - J(\pi_1^*, \pi_2^*)}_{\text{Total Error}} \\ &= \underbrace{(J(\pi_1^\dagger, \hat{\pi}_2) - J(\pi_1^*, \hat{\pi}_2))}_{\text{Gap 1: Optimization}} + \underbrace{(J(\pi_1^*, \hat{\pi}_2) - J(\pi_1^*, \pi_2^*))}_{\text{Gap 2: Evaluation}}, \end{aligned}$$

where $\pi_1^\dagger$ is the best response to $\hat{\pi}_2$. The two gaps are bounded *separately* (Lemmas 6.6 and 6.5 below) and each ends at the same cumulative squared Q -estimation error term.

Lemma 6.4. *For any state $s \in \mathcal{S}$ and step $h \in [H]$, the L_1 -distance between the learned policy $\hat{\pi}_h$ and the regularized Nash equilibrium π_h^* is controlled by the estimation error of the Q -function:*

$$\|\hat{\pi}_h(\cdot|s) - \pi_h^*(\cdot|s)\|_1 \leq 2\eta \|\hat{Q}_h(s, \cdot) - Q_h^*(s, \cdot)\|_\infty.$$

Its proof is in Appendix C.4 and follows from the strong monotonicity of the KL-regularized gradient map. The appendix also establishes the squared form $\|\hat{\pi}_h(\cdot|s) - \pi_h^*(\cdot|s)\|_1^2 \leq 4\eta^2 \cdot \xi_h^2(s)$ (see (64)), where $\xi_h^2(s)$ is the *single-player Nash-averaged squared Q -error*

$$\xi_h^2(s) := \max \left(\max_{a_1} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot|s)} [(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)], \right.$$

$$\left. \max_{a_2} \mathbb{E}_{a_1 \sim \pi_{h,1}^*(\cdot|s)} [(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)] \right). \quad (8)$$

Each branch of the max is a Player- i deterministic stage- h deviation in Π_i^{ref} paired with the opponent at Nash, so ξ_h^2 is compatible with Assumption 6.2; the cross-branch case is reduced to a Nash-Nash prefix in Appendix C at an absolute C_{trans}^2 cost under (7).

Lemma 6.5. *Under the side condition (7), the optimization gap is bounded by the cumulative single-player Nash-averaged squared Q -estimation error:*

$$\begin{aligned} & \underbrace{J(\pi_1^\dagger, \hat{\pi}_2) - J(\pi_1^*, \hat{\pi}_2)}_{\text{Gap 1}} \leq \mathcal{O}(C_{\text{trans}}^2 \eta^3 H^4 \lambda^2) \\ & \cdot \sum_{h=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\xi_h^2(s)]. \end{aligned}$$

Lemma 6.6. *Under the side condition (7), the evaluation gap is bounded linearly in η by the cumulative single-player Nash-averaged squared Q -estimation error ξ_h^2 (defined in (8)):*

$$\begin{aligned} & \underbrace{J(\pi_1^*, \hat{\pi}_2) - J(\pi_1^*, \pi_2^*)}_{\text{Gap 2}} \\ & \leq \mathcal{O}(\eta (C_{\text{trans}} + \lambda^2)) \cdot \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} [\xi_h^2(s)]. \end{aligned}$$

The proofs of Lemmas 6.5 and 6.6 are deferred to Appendices C.2 and C.5, respectively. Both bounds end at the same cumulative squared Q -estimation error, which is the key feature of the bandit-style analysis: it enables Gap 1 and Gap 2 to be combined and controlled jointly via Bellman unrolling without circular dependencies. The *squared* error scaling here is the mathematical source of the fast $\tilde{\mathcal{O}}(1/n)$

rate, as opposed to the standard $\tilde{O}(1/\sqrt{n})$ rates associated with unregularized gaps.

Combining Lemmas 6.5 and 6.6 with a recursive Bellman error unrolling completes the proof of Theorem 6.3 (detailed in Appendix C).

6.2. Optimization Convergence

Next, we analyze the practical instantiation, SOS-MD (Algorithm 2), where the equilibrium is approximated by T steps of mirror descent. Throughout this subsection $\hat{Q}_h$ denotes the Q -function fitted by Algorithm 2 and $\hat{\pi}_h$ its exact stagewise regularized equilibrium. The two algorithms fit different Q -sequences, so $\hat{\pi}_h$ is not the output of Algorithm 1; Theorem 6.3 is re-derived for it in Appendix D, at the cost of the $\tilde{O}(1/T)$ term of Corollary 6.9.

We control the optimization error via mirror-descent KL convergence (Lemma 6.7) combined with Pinsker’s inequality. Two supporting regularity facts—that all SOS-MD iterates remain in a bounded log-density-ratio class relative to π^{ref} (Lemma D.4), and that the regularized duality gap is Lipschitz in the L_1 policy distance over this class (Lemma D.7)—are deferred to Appendices D.4 and D.7.

Lemma 6.7. *Let $\hat{\pi}_h$ be the unique regularized Nash equilibrium of the stage game with payoff $\hat{Q}_h$ fitted by Algorithm 2. With the time-varying stepsize $\gamma_t = \frac{2\eta}{t+2}$ in Algorithm 2, the optimization error satisfies, $\forall h \in [H], \forall T \geq 1$,*

$$\begin{aligned} & \sup_{s \in \mathcal{S}} \text{KL}(\hat{\pi}_h(\cdot|s) \parallel \pi_h^{(T)}(\cdot|s)) \\ & \leq \frac{36\eta^2 H^2 (1 + \eta^{-1} \log(1/\alpha))^2}{T + 1}. \end{aligned}$$

The proof, in Appendix D.6, is a state-wise online mirror descent argument with the Hoeffding-type oscillation bound (Lemma D.5) applied within the log-linear-bounded class.

Theorem 6.8. *Let $\pi_h^{(T)}$ be the policy generated by Algorithm 2 at step h with stepsize $\gamma_t = \frac{2\eta}{t+2}$, and let $T \geq 1$. The state-uniform L_1 -distance to this exact equilibrium $\hat{\pi}_h$ of Lemma 6.7 satisfies*

$$\sup_{s \in \mathcal{S}} \|\hat{\pi}_h(\cdot|s) - \pi_h^{(T)}(\cdot|s)\|_1 \leq \tilde{O}\left(\frac{\eta H \lambda}{\sqrt{T}}\right),$$

where $\lambda = 1 + \eta^{-1} \log(1/\alpha)$ is the regularized value-magnitude factor from Lemma C.2.

Proof Sketch. Combine the KL bound of Lemma 6.7 with Pinsker’s inequality $\|\pi - \pi'\|_1 \leq \sqrt{2 \text{KL}(\pi \parallel \pi')}$. Detailed in Appendix D.

6.3. Total Sample Complexity

By combining the statistical bound of Theorem 6.3, re-derived in Appendix D for the exact equilibrium of the

SOS-MD fitted Q , with the Lipschitz transfer to the SOS-MD last iterate (Lemmas D.7, 6.7, and Theorem 6.8), we derive the final error bound for Algorithm 2.

Corollary 6.9. *Let $\pi^{(T)}$ be the policy learned by Algorithm 2 with $T \geq 1$ iterations per stage, n offline samples and stepsize $\gamma_t = \frac{2\eta}{t+2}$. Assume the hypotheses of Theorem 6.3, with Bellman completeness for the value class induced by Algorithm 2, and the condition (7). Then with probability at least $1 - \delta$, the duality gap satisfies*

$$\begin{aligned} & \text{Gap}(\pi^{(T)}) \\ & \leq \underbrace{\tilde{O}\left(\frac{\eta H^4 \lambda^2}{\sqrt{T}}\right)}_{\text{Optimization Error}} + \underbrace{\tilde{O}\left(\frac{\eta H^6 \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n}\right)}_{\text{Statistical Error}} \\ & \quad + \underbrace{\tilde{O}\left(\frac{\eta^3 H^7 \lambda^6}{T}\right)}_{\text{Last-iterate residual}}. \end{aligned}$$

Its proof is in Appendix D.10. The residual is of lower order in T , so the headline $\tilde{O}(1/n) + \tilde{O}(1/\sqrt{T})$ is unchanged. We keep λ explicit because it carries an η^{-1} . The last iterate of SOS-MD therefore attains the fast $\tilde{O}(1/n)$ rate under the same coupled reference-policy condition (7), against the standard $\tilde{O}(1/\sqrt{n})$ of offline RL (Xie et al., 2021; Rashidinejad et al., 2021) and offline zero-sum games (Cui & Du, 2022a).

7. Conclusion

In this paper, we study offline learning of Nash equilibria in two-player zero-sum Markov games under KL regularization. We first introduce Regularized Offline Sequential Equilibrium (ROSE), an idealized framework for offline learning with KL regularization. We show that KL regularization alone stabilizes learning and yields a fast $\tilde{O}(1/n)$ convergence rate under *unilateral concentrability* together with the coupled regularization/reference-policy condition in Theorem 6.3. We then propose Sequential Offline Self-Play Mirror Descent (SOS-MD), a practical model-free algorithm that replaces the exact equilibrium solver with iterative mirror-descent self-play. Our analysis shows that the last iterate of SOS-MD inherits the same $\tilde{O}(1/n)$ statistical guarantee, with optimization error vanishing at rate $\tilde{O}(1/\sqrt{T})$ as the number of self-play iterations T grows. Future work includes extending our analysis to partially observable games and developing KL-regularized self-play methods for aligning large language models via Nash Learning from Human Feedback.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal

consequences of our work, none of which we feel must be specifically highlighted here.

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A. Notation

Table 2. Summary of Notation

Symbol	Description
<i>Markov Games & Reinforcement Learning</i>	
$\mathcal{M}$	Episodic two-player zero-sum Markov game tuple $(\mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r, H)$
$\mathcal{S}$	State space
$\mathcal{A}_1, \mathcal{A}_2$	Finite action spaces for Player 1 and Player 2
H	Episode horizon length
$P_h(\cdot \mid s, a_1, a_2)$	Transition probability kernel at step h
$r_h^*(s, a_1, a_2)$	Deterministic reward function at step h (Player 1 maximizes, Player 2 minimizes)
ρ	Initial state distribution
$\pi = (\pi_1, \pi_2)$	Joint policy, where $\pi_k = \{\pi_{h,k}\}_{h=1}^H$
d_h^π	State-action visitation law at step h induced by joint policy π : $d_h^\pi = \mathcal{L}_\pi(s_h, a_{h,1}, a_{h,2})$
$d_{h,\mathcal{S}}^\pi$	State marginal of d_h^π : $d_{h,\mathcal{S}}^\pi = \mathcal{L}_\pi(s_h)$
V_h^π, Q_h^π	Regularized Value function and Q-function under policy π
$\pi^\dagger, \pi^\ddagger$	Best response policy for Player 1 (maximizer) and Player 2 (minimizer)
<i>Regularization & Offline Learning</i>	
η	KL regularization coefficient
π^{ref}	Fixed reference policy pair $(\pi_1^{\text{ref}}, \pi_2^{\text{ref}})$
α	Minimum positive probability of the reference policy within its support
$\mathcal{D}$	Offline dataset consisting of n trajectories
n	Number of samples/trajectories in the offline dataset
$\mathcal{Q}_h$	Function class for Q -value estimation at step h (abbreviated $\mathcal{Q}$ when the step is clear); $ \mathcal{Q} := \max_{h \in [H]} \mathcal{Q}_h $ (Assumption C.1)
Π_i^{ref}	Reference-supported policies of player i : $\text{supp}(\pi_{h,i}(\cdot s)) \subseteq \{a : \pi_{h,i}^{\text{ref}}(a s) > 0\}$ for all (h, s)
Π_{uni}	Set of unilateral deviation policies (where at least one player plays Nash), the deviating player ranging over Π_i^{ref}
C_{uni}	Unilateral concentrability coefficient (Assumption 6.2)
$\text{Gap}(\pi)$	Duality gap of joint policy π (sum of exploitability)
<i>Algorithms (ROSE & SOS-MD)</i>	
$\hat{Q}_h, \hat{V}_h$	Estimated Q-function and Value function at step h
$\hat{\pi}$	Policy returned by the exact equilibrium solver (ROSE)
$\pi^{(T)}$	Last iterate policy returned by SOS-MD after T iterations
γ	Learning rate for Mirror Descent

continued on next page

Table 2 – continued from previous page

Symbol	Description
T	Number of iterations for SOS-MD
$\mathcal{T}_h$	Bellman backup operator at step h : $(\mathcal{T}_h f)(s, a) = r_h^*(s, a) + \mathbb{E}_{s' \sim P_h(\cdot s, a)}[f(s')]$, with $a = (a_1, a_2)$

B. Discussion

In this section, we discuss the practical implications of our theoretical framework, particularly for Large Language Model (LLM) alignment, and highlight how our results improve upon the existing literature.

Implications for LLM Alignment. Recently, formulating LLM alignment as a two-player zero-sum game has attracted significant research attention (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025a;b). The primary motivation lies in the limitations of standard Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Xiong et al., 2024; Rafailov et al., 2023), which typically relies on the Bradley-Terry (BT) modeling of human preferences. The BT model inherently assumes the *transitivity* of preferences (i.e., if $A \succ B$ and $B \succ C$, then $A \succ C$). However, this assumption contradicts empirical evidence in human decision-making, where preferences can be intransitive (May, 1954).

To bypass the restrictive BT assumption and capture general preferences, recent works (Azar et al., 2024; Munos et al., 2024) propose a KL-regularized game-theoretic framework. In this setting, the objective is to identify a policy that maximizes the win rate against an adversarial opponent while staying close to a reference policy. However, the majority of this literature focuses on the *online* setting, where the agent can actively query preference signals. Furthermore, existing theoretical analyses are predominantly asymptotic, failing to provide finite-sample complexity bounds.

Comparison with Offline Nash Learning. While Ye et al. (2024) provide sample complexity results for the offline setting, their approach largely mirrors the algorithmic design and analysis of unregularized offline Markov games (Cui & Du, 2022a; Zhang et al., 2023). Consequently, they do not fully leverage the geometry of KL regularization, resulting in a standard $\tilde{O}(1/\sqrt{n})$ statistical rate. Our work advances beyond this result in three key aspects: (1) Multi-step Dynamics: We study the multi-step setting, generalizing the single-step (bandit) formulation in Ye et al. (2024). This formulation captures more complex LLM applications, such as multi-turn user interactions and tool-use scenarios involving intermediate observations (Chen et al., 2026; Xie et al., 2026). (2) No Explicit Pessimism: Unlike prior approaches, our algorithm requires no explicit pessimistic bonuses added to the value function, making it align better with practical post-training algorithms. (3) Fast Rates: We develop a novel analysis distinct from the standard pessimistic framework used in offline Markov games, unlocking the strong convexity of the objective to achieve a fast rate of $\tilde{O}(1/n)$.

C. Proof of Statistical Efficiency (Theorem 6.3)

In this section, we provide the detailed proofs for the statistical efficiency of Algorithm 1. The main results (Theorem 6.3 and Corollary 6.9) and all appendix proofs are stated under the following standing assumption on the function class of Section 3.

Assumption C.1 (Finite, uniformly bounded stagewise function classes). For each $h \in [H]$, the learner uses a function class $\mathcal{Q}_h \subset (\mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathbb{R})$ that is finite and uniformly bounded:

$$|\mathcal{Q}_h| < \infty, \quad \mathcal{Q}_h \subseteq \{f : \|f\|_\infty \leq C_{\text{val}}\}, \quad C_{\text{val}} := H(1 + \eta^{-1} \log(1/\alpha)) = H\lambda,$$

and we write $|\mathcal{Q}| := \max_{h \in [H]} |\mathcal{Q}_h|$ for the cardinality appearing in Theorem 6.3. When the stage index is immaterial we abbreviate $\mathcal{Q}_h$ by $\mathcal{Q}$, as in Definition 6.1 and Assumption 6.2, where $\mathcal{Q}$ always denotes the stage- h class matching the index of μ_h and $d_h^{\pi'}$.

This is consistent with Bellman completeness. Write $B_h := \sup_{f \in \mathcal{Q}_h} \|f\|_\infty$, take the terminal induced value class to be the singleton $\{0\}$, and build the minimal complete hierarchy backwards by $\mathcal{Q}_h := \{\mathcal{T}_h V : V \in \mathcal{V}_{h+1}^{\text{alg}}\}$, where $\mathcal{V}_{h+1}^{\text{alg}}$ denotes the value class induced at stage $h+1$ by the algorithm under consideration from stage payoffs in $\mathcal{Q}_{h+1}$ (for Algorithm 1, the class of Lemma C.9). The construction proceeds from $h = H$ downwards, with $\mathcal{Q}_H = \{\mathcal{T}_H 0\}$, so $B_H \leq 1$.

(a) *Algorithm 1.* The equilibrium step does not inflate the payoff scale: by Corollary C.3(iii), the regularized *saddle* value of a stage payoff $\tilde{Q}_{h+1} \in \mathcal{Q}_{h+1}$ obeys $\|V\|_\infty \leq \|\tilde{Q}_{h+1}\|_\infty \leq B_{h+1}$. Hence $\|\mathcal{T}_h V\|_\infty \leq 1 + B_{h+1}$ and $B_h \leq 1 + B_{h+1}$, so that $B_h \leq H - h + 1 \leq H \leq H\lambda = C_{\text{val}}$, using $\lambda \geq 1$.

(b) *Algorithm 2.* The value update evaluates the same stage objective at the last iterate, which need not be a saddle point, so only the first bound of Corollary C.3(iii) is available, namely $\|V\|_\infty \leq \|\tilde{Q}_{h+1}\|_\infty + \eta^{-1} \log(1/\alpha) \leq B_{h+1} + \eta^{-1} \log(1/\alpha)$. This gives $B_h \leq 1 + B_{h+1} + \eta^{-1} \log(1/\alpha) = B_{h+1} + \lambda$, and therefore $B_h \leq (H - h + 1)\lambda \leq H\lambda = C_{\text{val}}$.

In both cases a complete class can be taken inside the C_{val} -ball at every stage. The slack $C_{\text{val}} - H = H\eta^{-1} \log(1/\alpha)$ is exactly what case (b) consumes, and it also accommodates the regularized value functions of the true game, whose magnitude is $(H - h + 1)\lambda$ by Lemma C.2. Such bounded function classes are standard in offline RL (Xie et al., 2021) and KL-regularized learning (Xiong et al., 2024; Ye et al., 2024; Zhao et al., 2025).

C.1. Proof of Lemma C.2

Lemma C.2 (Uniform Value Bound). *Write $\pi_{h,i}^{\text{ref}} \equiv \pi_{\text{ref},h,i}$ for the stage- h reference policy of player $i \in \{1, 2\}$, and set*

$$\mathcal{A}_{h,i}^{\text{ref}}(s) := \{a \in \mathcal{A}_i : \pi_{h,i}^{\text{ref}}(a | s) > 0\}, \quad \Pi_i^{\text{ref}} := \{\pi_i : \text{supp}(\pi_{h,i}(\cdot | s)) \subseteq \mathcal{A}_{h,i}^{\text{ref}}(s) \text{ for all } (h, s)\}.$$

Let $\nu = \{\nu_h\}_{h=1}^H$ be any reference-supported opponent policy sequence, i.e. $\nu \in \Pi_2^{\text{ref}}$. Then the magnitude of the regularized best-response value function is bounded uniformly over $\nu \in \Pi_2^{\text{ref}}$. Specifically, $\forall h \in [H]$:

$$\|V_h^{\dagger,\nu}\|_\infty \leq (H - h + 1) (1 + \eta^{-1} \log(1/\alpha)). \quad (9)$$

A policy that places mass outside the reference support at a state reached with positive probability incurs an infinite forward-KL term, so its value is $-\infty$ for the maximizing Player 1 and $+\infty$ for the minimizing Player 2; no finite envelope holds over the unrestricted classes, and such a policy is not a best response. Off-support behavior at states reached with probability zero may be replaced by π^{ref} without changing any value. Accordingly, all policy classes appearing below—best responses, the deviation classes of Assumption 6.2, and the policies generated by Algorithms 1 and 2, which are of Gibbs form and hence reference-supported—are understood to lie in $\Pi_1^{\text{ref}} \times \Pi_2^{\text{ref}}$.

We will also prove that, in particular, for all $h \in [H]$,

$$1 + \|V_{h+1}^{\dagger,\nu}\|_\infty \leq (H - h + 1) (1 + \eta^{-1} \log(1/\alpha)). \quad (10)$$

Proof. We prove (9) by backward induction on h . We adopt the standard convention that the terminal value satisfies $V_{H+1}^{\dagger,\nu} \equiv 0$, so (9) holds for $h = H + 1$.

Fix $h \in [H]$ and assume (9) holds for $h + 1$. Fix any state $s \in \mathcal{S}$. Since Player 2's policy ν is fixed, its regularization term acts as a state-dependent constant shift:

$$C_{h,s}^\nu := \eta^{-1} \text{KL}(\nu_h(\cdot | s) \| \pi_{\text{ref},h,2}(\cdot | s)).$$

Then

$$V_h^{\dagger,\nu}(s) = \max_{\mu \in \Delta(\mathcal{A}_1)} \left\{ \left\langle \mu, Q_{h,\nu}^\dagger(s, \cdot) \right\rangle - \eta^{-1} \text{KL}(\mu \| \pi_{\text{ref},h,1}(\cdot | s)) \right\} + C_{h,s}^\nu, \quad (11)$$

where $Q_{h,\nu}^\dagger(s, a_1) := \mathbb{E}_{a_2 \sim \nu_h(\cdot | s)} [Q_h^{\dagger,\nu}(s, a_1, a_2)]$ and

$$Q_h^{\dagger,\nu}(s, a_1, a_2) = r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger,\nu}(s')].$$

Using $|r_h^*| \leq 1$ and Jensen's inequality, for all (s, a_1, a_2) ,

$$|Q_h^{\dagger,\nu}(s, a_1, a_2)| \leq 1 + \|V_{h+1}^{\dagger,\nu}\|_\infty \quad \Rightarrow \quad \|Q_{h,\nu}^\dagger(s, \cdot)\|_\infty \leq 1 + \|V_{h+1}^{\dagger,\nu}\|_\infty. \quad (12)$$

Denote the maximization term in (11) by $V_{h,s}^{\max}$. By the variational identity $\max_{\mu} \{\langle \mu, q \rangle - \eta^{-1} \text{KL}(\mu \| \pi)\} = \eta^{-1} \log \sum_a \pi(a) \exp(\eta q(a))$ with $q = Q_{h,\nu}^{\dagger}(s, \cdot)$ and $\pi = \pi_{\text{ref},h,1}(\cdot | s)$,

$$V_{h,s}^{\max} = \eta^{-1} \log \sum_{a \in \mathcal{A}_1} \pi_{\text{ref},h,1}(a | s) \exp(\eta Q_{h,\nu}^{\dagger}(s, a)). \quad (13)$$

Using the property that $\eta^{-1} \log \sum \pi_i e^{\eta q_i} \in [\min q_i, \max q_i]$, we bound the magnitude:

$$|V_{h,s}^{\max}| \leq \|Q_{h,\nu}^{\dagger}(s, \cdot)\|_{\infty}.$$

Substituting this back into (11):

$$|V_h^{\dagger,\nu}(s)| \leq |V_{h,s}^{\max}| + |C_{h,s}^{\nu}| \leq \|Q_{h,\nu}^{\dagger}(s, \cdot)\|_{\infty} + C_{h,s}^{\nu}. \quad (14)$$

(Note that KL divergence is non-negative, so $|C_{h,s}^{\nu}| = C_{h,s}^{\nu}$).

By the definition of α , we have $\pi_{h,i}^{\text{ref}}(a | s) \geq \alpha$ for every a in the reference support $\mathcal{A}_{h,i}^{\text{ref}}(s)$. Hence for any distribution p supported in $\mathcal{A}_{h,i}^{\text{ref}}(s)$,

$$\text{KL}(p \| \pi_{h,i}^{\text{ref}}(\cdot | s)) = \sum_{a \in \mathcal{A}_{h,i}^{\text{ref}}(s)} p(a) \log \frac{p(a)}{\pi_{h,i}^{\text{ref}}(a | s)} \leq \sum_a p(a) \log \frac{1}{\alpha} = \log(1/\alpha),$$

using $p(a) \leq 1$ and $\pi_{h,i}^{\text{ref}}(a | s) \geq \alpha$ on the support. The maximization in (11), although written over $\Delta(\mathcal{A}_1)$, is effectively over $\Delta(\mathcal{A}_{h,1}^{\text{ref}}(s))$: any μ charging an unsupported action has objective value $-\infty$. Therefore, $0 \leq C_{h,s}^{\nu} \leq \eta^{-1} \log(1/\alpha)$ for all (h, s) .

Taking the supremum over s in (14) and using (12), we obtain the recursion:

$$\|V_h^{\dagger,\nu}\|_{\infty} \leq \left(1 + \|V_{h+1}^{\dagger,\nu}\|_{\infty}\right) + \eta^{-1} \log(1/\alpha). \quad (15)$$

Applying the induction hypothesis $\|V_{h+1}^{\dagger,\nu}\|_{\infty} \leq (H - h)(1 + \eta^{-1} \log(1/\alpha))$ to (15) gives

$$\begin{aligned} \|V_h^{\dagger,\nu}\|_{\infty} &\leq 1 + \eta^{-1} \log(1/\alpha) + (H - h)(1 + \eta^{-1} \log(1/\alpha)) \\ &= (H - h + 1)(1 + \eta^{-1} \log(1/\alpha)), \end{aligned}$$

which establishes (9). The inequality (10) follows immediately: since $\lambda = 1 + \eta^{-1} \log(1/\alpha) \geq 1$, we have $1 + \|V_{h+1}^{\dagger,\nu}\|_{\infty} \leq 1 + (H - h)\lambda \leq (H - h + 1)\lambda$. $\square$

The same envelope holds for every value function used in the sequel, in either player's role, and uniformly over the fixed opponent's policy. No restriction on η is needed; the side condition (7) is not used here.

Corollary C.3 (Uniform value and payoff envelopes). *Recall $\lambda = 1 + \eta^{-1} \log(1/\alpha) \geq 1$ and $C_{\text{val}} = H\lambda$. Fix $h \in [H]$.*

(i) (Policy values.) *For every reference-supported joint policy $\pi = (\pi_1, \pi_2) \in \Pi_1^{\text{ref}} \times \Pi_2^{\text{ref}}$,*

$$\|V_h^{\pi}\|_{\infty} \leq (H - h + 1)\lambda, \quad \|Q_h^{\pi}\|_{\infty} \leq 1 + (H - h)\lambda \leq (H - h + 1)\lambda.$$

(ii) (Best-response values, uniformly over the fixed opponent.) *For every $\nu \in \Pi_2^{\text{ref}}$ and every $\pi_1 \in \Pi_1^{\text{ref}}$,*

$$\max\{\|V_h^{\dagger,\nu}\|_{\infty}, \|V_h^{\pi_1, \dagger}\|_{\infty}\} \leq (H - h + 1)\lambda, \quad \max\{\|Q_h^{\dagger,\nu}\|_{\infty}, \|Q_h^{\pi_1, \dagger}\|_{\infty}\} \leq 1 + (H - h)\lambda \leq (H - h + 1)\lambda,$$

and the same bounds hold for the opponent-marginalized payoffs $Q_{h,\nu}^{\dagger}(s, a_1) = \mathbb{E}_{a_2 \sim \nu_h(\cdot | s)}[Q_h^{\dagger,\nu}(s, a_1, a_2)]$ and $Q_{h,\pi_1}^{\dagger}(s, a_2) = \mathbb{E}_{a_1 \sim \pi_{h,1}(\cdot | s)}[Q_h^{\pi_1, \dagger}(s, a_1, a_2)]$. In particular every quantity above is at most $H\lambda = C_{\text{val}}$, uniformly over the fixed opponent policy, and the regularized Nash quantities $V_h^{\pi^} = V_h^{\dagger, \pi_2^*} = V_h^{\pi_1^*, \dagger}$ and Q_h^* obey the same bounds (take $\nu = \pi_2^*$, resp. $\pi_1 = \pi_1^*$).*

(iii) (Stage envelope; no inflation at a saddle point.) Fix s , let $q : \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathbb{R}$ be any payoff, and for p_i supported in $\mathcal{A}_{h,i}^{\text{ref}}(s)$ define

$$F(p_1, p_2) := \langle p_1 \otimes p_2, q \rangle - \eta^{-1} \text{KL}(p_1 \| \pi_{h,1}^{\text{ref}}(\cdot | s)) + \eta^{-1} \text{KL}(p_2 \| \pi_{h,2}^{\text{ref}}(\cdot | s)).$$

Then $|F(p_1, p_2)| \leq \|q\|_\infty + \eta^{-1} \log(1/\alpha)$ for every such pair, and $|F(p_1^*, p_2^*)| \leq \|q\|_\infty$ whenever (p_1^*, p_2^*) is a saddle point of F .

(iv) (Algorithm-induced values.) Consequently the value produced by the equilibrium step of Algorithm 1 from a stage payoff $\tilde{Q}_h$ satisfies $\|V\|_\infty \leq \|\tilde{Q}_h\|_\infty$, and every $V \in \mathcal{V}_{h+1}^{\text{alg}}$ (Lemma C.9) satisfies $\|V\|_\infty \leq \sup_{f \in \mathcal{Q}_{h+1}} \|f\|_\infty \leq C_{\text{val}}$. For a stage value evaluated at a pair that is not a saddle point—as in the value update of Algorithm 2, which uses the last iterate—only the first bound of (iii) is available, namely $\|V\|_\infty \leq \|\tilde{Q}_h\|_\infty + \eta^{-1} \log(1/\alpha)$.

Proof. (iii) Both KL terms lie in $[0, \eta^{-1} \log(1/\alpha)]$ for reference-supported arguments (shown in the proof of Lemma C.2) and enter F with opposite signs, so $|F(p_1, p_2) - \langle p_1 \otimes p_2, q \rangle| \leq \eta^{-1} \log(1/\alpha)$, giving the first claim. For the second, test the saddle-point inequalities $F(p_1, p_2^*) \leq F(p_1^*, p_2^*) \leq F(p_1^*, p_2)$ against the reference policies, which are themselves admissible and satisfy $\text{KL}(\pi^{\text{ref}} \| \pi^{\text{ref}}) = 0$:

$$\begin{aligned} F(p_1^*, p_2^*) &\leq F(p_1^*, \pi_{h,2}^{\text{ref}}(\cdot | s)) = \langle p_1^* \otimes \pi_{h,2}^{\text{ref}}, q \rangle - \eta^{-1} \text{KL}(p_1^* \| \pi_{h,1}^{\text{ref}}) \leq \|q\|_\infty, \\ F(p_1^*, p_2^*) &\geq F(\pi_{h,1}^{\text{ref}}(\cdot | s), p_2^*) = \langle \pi_{h,1}^{\text{ref}} \otimes p_2^*, q \rangle + \eta^{-1} \text{KL}(p_2^* \| \pi_{h,2}^{\text{ref}}) \geq -\|q\|_\infty, \end{aligned}$$

both using $\text{KL} \geq 0$.

(i)–(ii) Repeat the backward induction of Lemma C.2 verbatim, replacing the maximization in (11) by the expectation $\langle \pi_{h,1}(\cdot | s), \cdot \rangle$ for (i), and by Player 2’s minimization for the $V^{\pi_1, \dagger}$ half of (ii). In each case $|V_h(s)| \leq \|Q_h\|_\infty + \eta^{-1} \log(1/\alpha)$: for (i) because the two stagewise KL terms lie in $[0, \eta^{-1} \log(1/\alpha)]$ and enter with opposite signs, and for (ii) because $\min_p \{ \langle p, \bar{q} \rangle + \eta^{-1} \text{KL}(p \| \pi_{h,2}^{\text{ref}}) \} = -\eta^{-1} \log \sum_a \pi_{h,2}^{\text{ref}}(a | s) e^{-\eta \bar{q}(a)} \in [\min_a \bar{q}(a), \max_a \bar{q}(a)]$ by the same range property applied to $-\bar{q}$, while the frozen Player-1 KL term contributes at most $\eta^{-1} \log(1/\alpha)$ in magnitude. Combining with $\|Q_h\|_\infty \leq 1 + \|V_{h+1}\|_\infty$ gives the recursion $\|V_h\|_\infty \leq \lambda + \|V_{h+1}\|_\infty$ with $V_{H+1} \equiv 0$, hence $\|V_h\|_\infty \leq (H - h + 1)\lambda$; the Q -bounds and their opponent-marginalized versions follow from $\|Q_h\|_\infty \leq 1 + \|V_{h+1}\|_\infty \leq 1 + (H - h)\lambda$ and $|\mathbb{E}_{a_2 \sim \nu_h}[\cdot]| \leq \|\cdot\|_\infty$.

(iv) At each s , the equilibrium step of Algorithm 1 computes a saddle point of F with $q = \tilde{Q}_h(s, \cdot, \cdot)$ and sets $\hat{V}_h(s)$ to the value of F there, so the first claim is exactly the second part of (iii); applying it at stage $h + 1$ with $\tilde{Q}_{h+1} \in \mathcal{Q}_{h+1}$ and invoking Assumption C.1 gives the second. The value update of Algorithm 2 evaluates the same F at $(\pi_{h,1}^{(T)}, \pi_{h,2}^{(T)})$, which need not be a saddle point, so only the first part of (iii) applies. $\square$

C.2. Proof of Lemma 6.5 (Optimization Gap Bound)

Duality Gap Decomposition. Let $\hat{\pi} = (\hat{\pi}_1, \hat{\pi}_2)$ denote the learned policy pair. Throughout this proof, $J(r, \pi_1, \pi_2)$ denotes the KL-regularized expected return defined in Section 3: $J(r, \pi_1, \pi_2) = \mathbb{E}_{s_1 \sim \rho}[V_1^{\pi_1, \pi_2}(s_1)]$ where $V_1^{\pi_1, \pi_2}$ is the regularized value function with reward r and KL penalties for both players. Fix $\hat{\pi}_2$; the regularized best response of Player 1 under the true game reward r^* is the maximizer of the KL-regularized objective:

$$\pi^\dagger \in \arg \max_{\pi_1} J(r^*, \pi_1, \hat{\pi}_2),$$

where the argmax is taken over the KL-regularized objective. Because that objective carries the penalty $-\eta^{-1} \text{KL}(\pi_{h,1}(\cdot | s) \| \pi_{h,1}^{\text{ref}}(\cdot | s))$ at every step, mass placed outside $\mathcal{A}_{h,1}^{\text{ref}}(s) := \{a_1 \in \mathcal{A}_1 : \pi_{h,1}^{\text{ref}}(a_1 | s) > 0\}$ at a state reached with positive probability incurs an infinite forward-KL term and cannot be optimal; the maximizer is therefore reference supported, and at every such (h, s) it is the unique Gibbs-form softmax policy with logits $\eta Q_{h,1}^{\dagger, \hat{\pi}_2}(s, \cdot) + \log \pi_{h,1}^{\text{ref}}(\cdot | s)$ on $\mathcal{A}_{h,1}^{\text{ref}}(s)$. (At states unreachable under $(\pi^\dagger, \hat{\pi}_2)$ the maximizer is not unique; we fix the Gibbs representative, which changes no value and no occupancy.) Similarly, let $\pi^\ddagger$ be the regularized best response of Player 2 against $\hat{\pi}_1$.

The duality gap of the learned policy pair $\hat{\pi}$ is defined as the sum of exploitabilities:

$$\text{DualGap}(\hat{\pi}) := \underbrace{\left(\max_{\pi'_1} J(r^*, \pi'_1, \hat{\pi}_2) - J(r^*, \hat{\pi}_1, \hat{\pi}_2) \right)}_{\text{Player 1 Exploitability}} + \underbrace{\left(J(r^*, \hat{\pi}_1, \hat{\pi}_2) - \min_{\pi'_2} J(r^*, \hat{\pi}_1, \pi'_2) \right)}_{\text{Player 2 Exploitability}}.$$

Equivalently, using the best response notation and the value of the true Nash equilibrium (π_1^*, π_2^*) :

$$\text{DualGap}(\hat{\pi}) = \underbrace{(J(r^*, \pi^\dagger, \hat{\pi}_2) - J(r^*, \pi_1^*, \pi_2^*))}_{\text{Term A}} + \underbrace{(J(r^*, \pi_1^*, \pi_2^*) - J(r^*, \hat{\pi}_1, \pi^\dagger))}_{\text{Term B}}.$$

Our goal is to bound Term A. Term B is not disposed of by an appeal to symmetry: the game has asymmetric dynamics, reference policies and behavior distribution. Instead, as shown in *Term B via the reflected game* below, Term B is Term A of an explicitly constructed auxiliary game to which every hypothesis (Assumption 6.2, Assumption C.1, Bellman completeness, the reference-support constant α) transfers. We decompose Term A as follows:

$$J(r^*, \pi^\dagger, \hat{\pi}_2) - J(r^*, \pi_1^*, \pi_2^*) = \underbrace{J(r^*, \pi^\dagger, \hat{\pi}_2) - J(r^*, \pi_1^*, \hat{\pi}_2)}_{\text{Gap 1: Optimization / Stability}} + \underbrace{J(r^*, \pi_1^*, \hat{\pi}_2) - J(r^*, \pi_1^*, \pi_2^*)}_{\text{Gap 2: Evaluation / Fixed-Opponent}}. \quad (16)$$

Term B via the reflected game (no appeal to symmetry). Let $\sigma(s, a_1, a_2) := (s, a_2, a_1)$ denote the action-coordinate swap and define the *reflected game* $\tilde{\mathcal{M}} = (\mathcal{S}, \tilde{\mathcal{A}}_1, \tilde{\mathcal{A}}_2, \tilde{P}, \tilde{r}, H)$ by

$$\tilde{\mathcal{A}}_1 := \mathcal{A}_2, \quad \tilde{\mathcal{A}}_2 := \mathcal{A}_1, \quad \tilde{P}_h(\cdot \mid s, b_1, b_2) := P_h(\cdot \mid s, b_2, b_1), \quad \tilde{r}_h(s, b_1, b_2) := 1 - r_h^*(s, b_2, b_1),$$

with reference policies $\tilde{\pi}_{h,1}^{\text{ref}} := \pi_{h,1}^{\text{ref}}$ and $\tilde{\pi}_{h,2}^{\text{ref}} := \pi_{h,2}^{\text{ref}}$, the same η and the same initial law ρ , and with the policy relabelling $\tilde{\pi} := (\pi_2, \pi_1)$. Since $r_h^* \in [0, 1]$ we have $\tilde{r}_h \in [0, 1]$, so $\tilde{\mathcal{M}}$ is again an instance of the model of Section 3, with the *same* α and hence the same λ and C_{trans} , and the side condition (7) is unaffected. Write $c_h := H - h + 1$ (so $c_{H+1} = 0$ and $c_h - 1 - c_{h+1} = 0$). The following correspondences hold; each is verified by direct substitution in (1).

- (a) *Values.* $\tilde{Q}_h^{\tilde{\pi}}(s, b_1, b_2) = c_h - Q_h^\pi(s, b_2, b_1)$ and $\tilde{V}_h^{\tilde{\pi}}(s) = c_h - V_h^\pi(s)$, by backward induction from $\tilde{V}_{H+1} \equiv V_{H+1} \equiv 0$. The KL signs match: the maximizer of $\tilde{\mathcal{M}}$ is $\tilde{\pi}_1 = \pi_2$, whose KL is *subtracted* in $\tilde{V}$ and *added* in V , and symmetrically for π_1 .
- (b) *Equilibria, best responses and Term B.* $\tilde{\pi}^* = (\pi_2^*, \pi_1^*)$ is the regularized Nash equilibrium of $\tilde{\mathcal{M}}$, and the regularized best response of $\tilde{\mathcal{M}}$'s Player 1 against $\tilde{\pi}_2 = \hat{\pi}_1$ is $\pi_1^\dagger$, because $\tilde{J}(\tilde{\pi}_1, \hat{\pi}_1) = H - J(r^*, \hat{\pi}_1, \tilde{\pi}_1)$. Hence

$$\underbrace{\tilde{J}(\pi_1^\dagger, \hat{\pi}_1) - \tilde{J}(\pi_2^*, \pi_1^*)}_{\text{Term A of } \tilde{\mathcal{M}}} = J(r^*, \pi_1^*, \pi_2^*) - J(r^*, \hat{\pi}_1, \pi^\dagger) = \text{Term B}.$$

- (c) *Occupancies.* Occupancy measures transform as *pushforwards*, not as identities: $\tilde{d}_h^{\tilde{\pi}} = \sigma_{\#} d_h^\pi$. In the countable case this reads $\tilde{d}_h^{\tilde{\pi}}(s, b_1, b_2) = d_h^\pi(s, b_2, b_1)$. The state marginals coincide.
- (d) *Data, function class and completeness.* Set $\tilde{\mu}_h := \sigma_{\#} \mu_h$ and, for each $h \in [H]$, $\tilde{\mathcal{Q}}_h := \{R_h q : q \in \mathcal{Q}\}$ with $(R_h q)(s, b_1, b_2) := c_h - q(s, b_2, b_1)$. Then $|\tilde{\mathcal{Q}}_h| = |\mathcal{Q}|$, so the log-cardinality entering the concentration bounds is unchanged, and $\|R_h q\|_\infty \leq c_h + C_{\text{val}} \leq 2C_{\text{val}}$ (using $\lambda \geq 1$), so Assumption C.1 holds for each $\tilde{\mathcal{Q}}_h$ with C_{val} inflated by an absolute factor. This factor-two inflation affects only absolute constants in the regression bounds, not the density-ratio constant: the term c_h is action independent and cancels from every Gibbs normalization and every logit difference. Thus the reflected policy-ratio bounds are still governed by the original envelope $C_{\text{val}} = H\lambda$, and $\tilde{C}_{\text{trans}} = C_{\text{trans}} = e$. (The class is h -indexed because R_h carries c_h ; all later steps are stagewise.) Bellman completeness transfers exactly, since $\tilde{T}_h \tilde{V}_{h+1} = R_h(\mathcal{T}_h V_{h+1})$ whenever $\tilde{V}_{h+1} = c_{h+1} - V_{h+1}$.
- (e) *Coverage.* R_h is an involution that negates differences, and σ is an involution, so both the numerator and the denominator in Definition 6.1 are preserved and $D_{\tilde{\mathcal{Q}}_h}^2(\sigma(z); \tilde{\mu}_h) = D_{\mathcal{Q}}^2(z; \mu_h)$ for every $z = (s, a_1, a_2)$. Combined with (b), (c) and $\tilde{\Pi}_{\text{uni}} = \{(\pi_2, \pi_1) : (\pi_1, \pi_2) \in \Pi_{\text{uni}}\}$, Assumption 6.2 holds for $\tilde{\mathcal{M}}$ with $\tilde{C}_{\text{uni}} = C_{\text{uni}}$.
- (f) *Algorithm.* Run Algorithm 1 on the reflected dataset $\tilde{\mathcal{D}} = \{(s_{i,h}, a_{i,h,2}, a_{i,h,1}, 1 - r_{i,h}, s'_{i,h})\}$ with the classes $\{\tilde{\mathcal{Q}}_h\}_{h=1}^H$. The reflected noise is $\tilde{\xi}_{i,h} = -\xi_{i,h}$, again zero-mean 1-sub-Gaussian. Each least-squares residual satisfies $(R_h q)(\sigma(z_{i,h})) - (1 - r_{i,h}) - \tilde{V}_{h+1}(s'_{i,h}) = -(q(z_{i,h}) - r_{i,h} - \hat{V}_{h+1}(s'_{i,h}))$ because $c_h - 1 - c_{h+1} = 0$, so the two squared objectives agree term by term and $\tilde{Q}_h = R_h \hat{Q}_h$. Adding a constant to, negating and transposing a stage payoff exchanges the roles of the maximizing and minimizing players, so the regularized equilibrium solve returns $\tilde{\pi}_h = (\hat{\pi}_{h,2}, \hat{\pi}_{h,1})$ and the value update returns $\tilde{V}_h = c_h - \hat{V}_h$, closing the induction.

Consequently the bound proved below for Term A, applied to $\tilde{\mathcal{M}}$, bounds Term B with the same H -, η - and α -dependence and the same C_{uni} , up to an absolute constant factor. The same swap applies to Algorithm 2: under R_h the fitted stage payoff becomes c_h minus its transpose and the two players' KL signs exchange, so the two mirror-descent updates correspond.

Support and ℓ_∞ conventions. Throughout this appendix we write $\mathcal{A}_{h,i}^{\text{ref}}(s) := \{a \in \mathcal{A}_i : \pi_{h,i}^{\text{ref}}(a | s) > 0\}$ for the reference support of player i at (h, s) ; it is nonempty and finite. Every policy appearing below — π^* , $\hat{\pi}$, $\pi^\dagger$, $\pi^\ddagger$ and all stagewise best responses — is of Gibbs form relative to π^{ref} , hence is supported on $\mathcal{A}_{h,i}^{\text{ref}}(s)$ and *strictly positive* there. Accordingly, all logit vectors, Softmax maps and log-partition functions are understood as indexed by $\mathcal{A}_{h,i}^{\text{ref}}(s)$: off the reference support the logit $\log \pi_{h,i}^{\text{ref}}(\cdot | s)$ equals $-\infty$, so logit *differences* are only defined on $\mathcal{A}_{h,i}^{\text{ref}}(s)$. Every optimization over $\Delta(\mathcal{A}_i)$ below is likewise confined to $\Delta(\mathcal{A}_{h,i}^{\text{ref}}(s))$, since the KL penalty is $+\infty$ otherwise and enters each player's own objective unfavourably. For a function g of the Player-1 action we write

$$\|g\|_{\infty, \text{ref}} := \max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} |g(a_1)| \leq \|g\|_\infty := \max_{a_1 \in \mathcal{A}_1} |g(a_1)|,$$

with (h, s) clear from context; norms of functions of a_2 , or of the state, remain unrestricted ℓ_∞ norms. The restricted form keeps logit differences finite, and the deterministic maximizing selectors it produces are *reference-supported* Player-1 deviations, as the routing into Π_{uni} in Step 4.2 requires.

In the following, we bound **Gap 1** in terms of the cumulative squared Q -estimation error, combining the KL identity of Step 1 (which also serves Lemma 6.6), the logits/ Q -difference decomposition of Steps 2–3, and the stability bound of Lemma 6.4 (Step 4.4).

Step 1: KL Identity for the Evaluation Gap (Gap 2). We analyze the second term in the decomposition (16), namely

$$\text{Gap 2} := J(r^*, \pi_1^*, \hat{\pi}_2) - J(r^*, \pi_1^*, \pi_2^*) = \mathbb{E}_{s_1 \sim \rho} [V_1^{\pi_1^*, \hat{\pi}_2}(s_1) - V_1^{\pi_1^*, \pi_2^*}(s_1)].$$

Fix (h, s) and fix Player 1 to π_1^* . For any Player 2 policy $\nu = \{\nu_t\}_{t=1}^H$, define the (regularized) stagewise objective of Player 2 at (h, s) as a function of the stage- h mixed action $\nu_h(\cdot | s)$:

$$L_{h,s}^\nu(\nu_h(\cdot | s)) := \sum_{a_2 \in \mathcal{A}_2} \nu_h(a_2 | s) \underbrace{\mathbb{E}_{a_1 \sim \pi_{h,1}^*(\cdot | s)} [Q_h^{\pi_1^*, \nu}(s, a_1, a_2)]}_{=: h_h^\nu(s, a_2)} + \eta^{-1} \sum_{a_2 \in \mathcal{A}_2} \nu_h(a_2 | s) \log \frac{\nu_h(a_2 | s)}{\pi_{h,2}^{\text{ref}}(a_2 | s)}.$$

Here $Q_h^{\pi_1^*, \nu}$ is defined using the *true continuation* under (π_1^*, ν) :

$$Q_h^{\pi_1^*, \nu}(s, a_1, a_2) := r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\pi_1^*, \nu}(s')], \quad V_{H+1}^{\pi_1^*, \nu} \equiv 0.$$

Since Player 2 minimizes the value, $L_{h,s}^\nu$ is a strictly convex objective in $\nu_h(\cdot | s)$.

Local optimality of π_2^* . Because (π_1^*, π_2^*) is a (regularized) Nash equilibrium, for every (h, s) we have

$$\pi_{h,2}^*(\cdot | s) \in \arg \min_{\mu \in \Delta(\mathcal{A}_2)} L_{h,s}^{\pi_2^*}(\mu).$$

By the conventions above, the minimization is confined to $\Delta(\mathcal{A}_{h,2}^{\text{ref}}(s))$, and by the variational identity used in (13) the minimizer is the Gibbs law

$$\pi_{h,2}^*(a_2 | s) = \frac{\pi_{h,2}^{\text{ref}}(a_2 | s) \exp(-\eta h_h^{\pi_2^*}(s, a_2))}{Z_{h,s}}, \quad Z_{h,s} := \sum_{a \in \mathcal{A}_{h,2}^{\text{ref}}(s)} \pi_{h,2}^{\text{ref}}(a | s) e^{-\eta h_h^{\pi_2^*}(s, a)}, \quad (17)$$

which is strictly positive on $\mathcal{A}_{h,2}^{\text{ref}}(s)$ and zero off it, hence lies in the relative interior of $\Delta(\mathcal{A}_{h,2}^{\text{ref}}(s))$, where $L_{h,s}^{\pi_2^*}$ is differentiable and stationarity of the Lagrangian for the single equality constraint $\sum_{a'} \mu(a') = 1$ is legitimate: with a

multiplier $\varsigma_{h,s} \in \mathbb{R}$, the first-order optimality condition at $\mu = \pi_{h,2}^*(\cdot | s)$ reads, for every $a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)$,

$$\begin{aligned} 0 &= \frac{\partial}{\partial \mu(a_2)} \left(L_{h,s}^{\pi_2^*}(\mu) + \varsigma_{h,s} \left(\sum_{a' \in \mathcal{A}_{h,2}^{\text{ref}}(s)} \mu(a') - 1 \right) \right) \Big|_{\mu=\pi_{h,2}^*(\cdot | s)} \\ &= h_h^{\pi_2^*}(s, a_2) + \eta^{-1} \left(1 + \log \frac{\pi_{h,2}^*(a_2 | s)}{\pi_{h,2}^{\text{ref}}(a_2 | s)} \right) + \varsigma_{h,s}. \end{aligned} \quad (18)$$

Alternatively, (18) follows by substituting (17), with $\varsigma_{h,s} = \eta^{-1} \log Z_{h,s} - \eta^{-1}$. Rearranging (18) gives, for every $a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)$,

$$h_h^{\pi_2^*}(s, a_2) = -\eta^{-1} \log \frac{\pi_{h,2}^*(a_2 | s)}{\pi_{h,2}^{\text{ref}}(a_2 | s)} - c_{h,s}, \quad (19)$$

where $c_{h,s} := \varsigma_{h,s} + \eta^{-1} = \eta^{-1} \log Z_{h,s}$ does not depend on a_2 ; it is unrelated to the $C_{h,s}$ of Step 3 below.

Evaluating the local gap. Now fix (h, s) and compare $\hat{\pi}_{h,2}(\cdot | s)$ to $\pi_{h,2}^*(\cdot | s)$ *under the same continuation*. Both are supported on $\mathcal{A}_{h,2}^{\text{ref}}(s)$ and strictly positive there, so (19) may be paired with $\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)$; since that difference sums to zero the constant $c_{h,s}$ drops out, and the remaining terms collapse to $\sum_{a_2} \hat{\pi}_{h,2}(a_2 | s) \log(\hat{\pi}_{h,2}(a_2 | s) / \pi_{h,2}^*(a_2 | s))$. We therefore obtain the exact identity

$$L_{h,s}^{\pi_2^*}(\hat{\pi}_{h,2}(\cdot | s)) - L_{h,s}^{\pi_2^*}(\pi_{h,2}^*(\cdot | s)) = \eta^{-1} \text{KL}(\hat{\pi}_{h,2}(\cdot | s) \parallel \pi_{h,2}^*(\cdot | s)). \quad (20)$$

Telescoping to Gap 2. Define $\Delta_h(s) := V_h^{\pi_1^*, \hat{\pi}_2}(s) - V_h^{\pi_1^*, \pi_2^*}(s)$. Using the Bellman equations for $V_h^{\pi_1^*, \hat{\pi}_2}$ and $V_h^{\pi_1^*, \pi_2^*}$ and the definition of $L_{h,s}^{\pi_2^*}$ (which uses the continuation $V_{h+1}^{\pi_1^*, \pi_2^*}$), one obtains the recursion

$$\Delta_h(s) = \left(L_{h,s}^{\pi_2^*}(\hat{\pi}_{h,2}(\cdot | s)) - L_{h,s}^{\pi_2^*}(\pi_{h,2}^*(\cdot | s)) \right) + \mathbb{E}_{s' \sim P_h(\cdot | s, \pi_1^*, \hat{\pi}_2)} [\Delta_{h+1}(s')], \quad (21)$$

with terminal condition $\Delta_{H+1} \equiv 0$. Unrolling (21) yields

$$\text{Gap } 2 = \mathbb{E}_{s_1 \sim \rho} [\Delta_1(s_1)] = \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[L_{h,s}^{\pi_2^*}(\hat{\pi}_{h,2}(\cdot | s)) - L_{h,s}^{\pi_2^*}(\pi_{h,2}^*(\cdot | s)) \right]. \quad (22)$$

Combining (22) with (20) gives

$$\text{Gap } 2 = \frac{1}{\eta} \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\text{KL}(\hat{\pi}_{h,2}(\cdot | s) \parallel \pi_{h,2}^*(\cdot | s)) \right]. \quad (23)$$

Equation (23) is the exact KL identity for Gap 2 invoked by Lemma 6.6 (Appendix C.5). The rest of this subsection bounds Gap 1.

Step 2: Lipschitz Continuity of the Best Response. Throughout this step, all value and Q -functions are defined with respect to the true reward function r^* .

Fix a time–state pair (h, s) and a Player 2 policy $\nu = \{\nu_{h'}\}_{h'=1}^H$. Define the best-response value of Player 1 against ν by

$$V_h^{\dagger, \nu}(s) := \max_{\pi_1} V_h^{\pi_1, \nu}(s),$$

and the associated best-response Q -function

$$Q_h^{\dagger, \nu}(s, a_1, a_2) := r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \nu}(s')].$$

We also write the opponent-averaged Q -vector

$$Q_{h,\nu}^\dagger(s, a_1) := \mathbb{E}_{a_2 \sim \nu_h(\cdot | s)} [Q_h^{\dagger, \nu}(s, a_1, a_2)].$$

Under KL regularization with Player 1 reference policy $\pi_{\text{ref}, h, 1}(\cdot | s)$, the stagewise best response of Player 1 to ν admits the Gibbs form

$$\pi_{h,1}^{\text{br}}(\cdot | s; \nu) = \text{Softmax}\left(\eta Q_{h,\nu}^\dagger(s, \cdot) + \log \pi_{\text{ref}, h, 1}(\cdot | s)\right). \quad (24)$$

Define the corresponding logits, indexed by $\mathcal{A}_{h,1}^{\text{ref}}(s)$ as in the conventions above, for the best responses to $\hat{\pi}_2$ and π_2^* :

$$z_h^\dagger := \eta Q_{h,\hat{\pi}_2}^\dagger(s, \cdot) + \log \pi_{\text{ref}, h, 1}(\cdot | s), \quad z_h^* := \eta Q_{h,\pi_2^*}^\dagger(s, \cdot) + \log \pi_{\text{ref}, h, 1}(\cdot | s).$$

By (24), $\pi_{h,1}^\dagger(\cdot | s) = \text{Softmax}(z_h^\dagger)$. Taking $\nu = \pi_2^*$ in (24) and using the saddle-point property $V_h^{\dagger, \pi_2^*} = V_h^{\pi_2^*}$ of the regularized Nash equilibrium (Section 3), which makes π_1^* the regularized best response to π_2^* and hence $Q_h^{\dagger, \pi_2^*} = Q_h^{\pi_2^*}$, we also get $\pi_{h,1}^*(\cdot | s) = \text{Softmax}(z_h^*)$. The Softmax mapping is 1-Lipschitz from $(\mathbb{R}^{\mathcal{A}_{h,1}^{\text{ref}}(s)}, \|\cdot\|_\infty)$ to $(\mathbb{R}^{\mathcal{A}_{h,1}^{\text{ref}}(s)}, \|\cdot\|_1)$ — equivalently, Φ is 1-smooth with respect to $\|\cdot\|_\infty$, as established in Step 3 below — so

$$\begin{aligned} \|\pi_{h,1}^\dagger(\cdot | s) - \pi_{h,1}^*(\cdot | s)\|_1 &\leq \|z_h^\dagger - z_h^*\|_{\infty, \text{ref}} \\ &\leq \eta \|Q_{h,\hat{\pi}_2}^\dagger(s, \cdot) - Q_{h,\pi_2^*}^\dagger(s, \cdot)\|_\infty, \end{aligned}$$

where the reference term $\log \pi_{\text{ref}, h, 1}(\cdot | s)$ is finite and cancels in the logit difference on $\mathcal{A}_{h,1}^{\text{ref}}(s)$, and $\|\cdot\|_{\infty, \text{ref}} \leq \|\cdot\|_\infty$.

Next, we bound the difference between the opponent-averaged Q -vectors. Fix (h, s) and a_1 . We decompose the difference by adding and subtracting a mixed term:

$$\begin{aligned} &Q_{h,\hat{\pi}_2}^\dagger(s, a_1) - Q_{h,\pi_2^*}^\dagger(s, a_1) \\ &= \underbrace{\left(\mathbb{E}_{a_2 \sim \hat{\pi}_{h,2}}[Q_h^{\dagger, \hat{\pi}_2}(s, a_1, a_2)] - \mathbb{E}_{a_2 \sim \pi_{h,2}^*}[Q_h^{\dagger, \hat{\pi}_2}(s, a_1, a_2)]\right)}_{T_1} \\ &\quad + \underbrace{\left(\mathbb{E}_{a_2 \sim \pi_{h,2}^*}[Q_h^{\dagger, \hat{\pi}_2}(s, a_1, a_2)] - \mathbb{E}_{a_2 \sim \pi_{h,2}^*}[Q_h^{\dagger, \pi_2^*}(s, a_1, a_2)]\right)}_{T_2}. \end{aligned}$$

Bounding the first term. Applying Hölder's inequality yields

$$\begin{aligned} |T_1| &= \left| \sum_{a_2 \in \mathcal{A}_2} (\hat{\pi}_{h,2}(a_2 | s) - \pi_{h,2}^*(a_2 | s)) Q_h^{\dagger, \hat{\pi}_2}(s, a_1, a_2) \right| \\ &\leq \|\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)\|_1 \|Q_h^{\dagger, \hat{\pi}_2}(s, a_1, \cdot)\|_\infty. \end{aligned}$$

Next, using the Bellman form

$$Q_h^{\dagger, \hat{\pi}_2}(s, a_1, a_2) = r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \hat{\pi}_2}(s')],$$

we have

$$\begin{aligned} \|Q_h^{\dagger, \hat{\pi}_2}(s, a_1, \cdot)\|_\infty &\leq \|r_h^*(s, a_1, \cdot)\|_\infty + \sup_{a_2 \in \mathcal{A}_2} \left| \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \hat{\pi}_2}(s')] \right| \\ &\leq 1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_\infty, \end{aligned}$$

where we used $|r_h^*| \leq 1$ and $|\mathbb{E}[V_{h+1}^{\dagger, \hat{\pi}_2}(s')]| \leq \mathbb{E}[|V_{h+1}^{\dagger, \hat{\pi}_2}(s')|] \leq \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_\infty$. Therefore,

$$|T_1| \leq \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_\infty\right) \|\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)\|_1. \quad (25)$$

Bounding the second term. For the second term, substituting the Bellman form of $Q_h^{\dagger, \nu}$ gives

$$\begin{aligned} T_2 &= \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \left[r_h^*(s, a_1, a_2) + \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \hat{\pi}_2}(s')] \right. \\ &\quad \left. - r_h^*(s, a_1, a_2) - \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \pi_2^*}(s')] \right] \\ &= \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \left[\mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [V_{h+1}^{\dagger, \hat{\pi}_2}(s') - V_{h+1}^{\dagger, \pi_2^*}(s')] \right]. \end{aligned}$$

Now denote the next-step value difference function

$$\Delta_{h+1}(s') := V_{h+1}^{\dagger, \hat{\pi}_2}(s') - V_{h+1}^{\dagger, \pi_2^*}(s').$$

Then the above becomes

$$T_2 = \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')].$$

Using $(a + b)^2 \leq 2a^2 + 2b^2$ on the decomposition $Q - Q^* = T_1 + T_2$, followed by $\max_{a_1} (u + v) \leq \max_{a_1} u + \max_{a_1} v$:

$$\|Q_{h, \hat{\pi}_2}^{\dagger}(s, \cdot) - Q_{h, \pi_2^*}^{\dagger}(s, \cdot)\|_{\infty, \text{ref}}^2 \leq \max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} (2|T_1(s, a_1)|^2 + 2|T_2(s, a_1)|^2).$$

This is a *single-level* application: (26) below is never re-expanded through itself, and the residual Δ_{h+1} is passed to Lemma C.4, whose unrolling uses $(x + y)^2 \leq (1 + H)x^2 + (1 + \frac{1}{H})y^2$ with $(1 + \frac{1}{H})^H \leq e$; no factor of 2 is compounded across steps h . Applying the bound for T_1 from (25) yields

$$|T_1(s, a_1)|^2 \leq \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_{\infty}\right)^2 \|\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)\|_1^2,$$

which does not depend on a_1 . For T_2 , we apply Jensen's inequality $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$:

$$\begin{aligned} |T_2(s, a_1)|^2 &= \left| \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')] \right|^2 \\ &\leq \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')^2]. \end{aligned}$$

Combining these estimates, we obtain the pointwise (in s) squared Q -difference bound:

$$\begin{aligned} \|Q_{h, \hat{\pi}_2}^{\dagger}(s, \cdot) - Q_{h, \pi_2^*}^{\dagger}(s, \cdot)\|_{\infty, \text{ref}}^2 &\leq 2 \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_{\infty}\right)^2 \|\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)\|_1^2 \\ &\quad + 2 \max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')^2]. \end{aligned} \quad (26)$$

Step 3: Upper Bounding Gap 1. We focus on **Gap 1**, the suboptimality of π^* against the fixed opponent $\hat{\pi}_2$ relative to the true best response $\pi^{\dagger}$.

$$\begin{aligned} \text{Gap 1} &:= J(r^*, \pi^{\dagger}, \hat{\pi}_2) - J(r^*, \pi_1^*, \hat{\pi}_2) \\ &= \mathbb{E}_{s_1 \sim \rho} [V_1^{\dagger, \hat{\pi}_2}(s_1) - V_1^{\pi_1^*, \hat{\pi}_2}(s_1)]. \end{aligned}$$

For each time-state pair (h, s) , define the regularized stagewise objective of Player 1 against $\hat{\pi}_2$ as

$$f_{h,s}(\mu) := \langle \mu, Q_{h, \hat{\pi}_2}^{\dagger}(s, \cdot) \rangle - \eta^{-1} \text{KL}(\mu \| \pi_{\text{ref}, h, 1}(\cdot | s)),$$

where $Q_{h, \hat{\pi}_2}^{\dagger}(s, \cdot)$ is the opponent-averaged best-response Q -vector defined above.

The stagewise best response of Player 1 is given by

$$\pi_{h,1}^{\dagger}(\cdot | s) \in \arg \max_{\mu \in \Delta(\mathcal{A}_1)} f_{h,s}(\mu).$$

Exact Characterization via KL Divergence. By the Gibbs/log-partition identity used in (13), the maximizer of $f_{h,s}$ is $\pi_{h,1}^\dagger(a \mid s) = \pi_{\text{ref},h,1}(a \mid s)e^{\eta Q_{h,\hat{\pi}_2}^\dagger(s,a)}/Z_{h,s}$ on $\mathcal{A}_{h,1}^{\text{ref}}(s)$, with $Z_{h,s} = \sum_{a \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \pi_{\text{ref},h,1}(a \mid s)e^{\eta Q_{h,\hat{\pi}_2}^\dagger(s,a)}$ and $f_{h,s}(\pi_{h,1}^\dagger) = \eta^{-1} \log Z_{h,s}$. Hence, for any μ supported on $\mathcal{A}_{h,1}^{\text{ref}}(s)$,

$$f_{h,s}(\pi_{h,1}^\dagger) - f_{h,s}(\mu) = \eta^{-1} \sum_a \mu(a) \left(\log \frac{\mu(a)}{\pi_{\text{ref},h,1}(a \mid s)} - \eta Q_{h,\hat{\pi}_2}^\dagger(s,a) + \log Z_{h,s} \right) = \eta^{-1} \text{KL}(\mu \parallel \pi_{h,1}^\dagger(\cdot \mid s)),$$

where the first equality uses $\sum_a \mu(a) = 1$. Since $\pi_{h,1}^*(\cdot \mid s)$ is supported on $\mathcal{A}_{h,1}^{\text{ref}}(s)$, where $\pi_{h,1}^\dagger(\cdot \mid s)$ is strictly positive, $\mu = \pi_{h,1}^*(\cdot \mid s)$ is admissible and the right-hand side is finite:

$$f_{h,s}(\pi_{h,1}^\dagger) - f_{h,s}(\pi_{h,1}^*) = \eta^{-1} \text{KL}(\pi_{h,1}^*(\cdot \mid s) \parallel \pi_{h,1}^\dagger(\cdot \mid s)).$$

Consequently, the overall gap can be expressed as

$$\text{Gap 1} = \eta^{-1} \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\text{KL}(\pi_{h,1}^*(\cdot \mid s) \parallel \pi_{h,1}^\dagger(\cdot \mid s)) \right], \quad (27)$$

where $d_h^{\pi_1^*, \hat{\pi}_2}$ denotes the state visitation distribution induced by $(\pi_1^*, \hat{\pi}_2)$.

Derivation of (27). We define the value difference at step h as:

$$\tilde{\Delta}_h(s) := V_h^{\dagger, \hat{\pi}_2}(s) - V_h^{\pi_1^*, \hat{\pi}_2}(s).$$

(The tilde distinguishes this from the $\Delta_{h+1} = V_{h+1}^{\dagger, \hat{\pi}_2} - V_{h+1}^{\pi_1^*, \hat{\pi}_2}$ of Step 2 and the Δ_h of Step 1 and Lemma C.4.) For a fixed opponent $\hat{\pi}_2$, the Player-2 regularization term in $V^{\pi_1^*, \hat{\pi}_2}$ is independent of Player 1's action. We denote this state-dependent term as:

$$C_{h,s} := \eta^{-1} \text{KL}(\hat{\pi}_{h,2}(\cdot \mid s) \parallel \pi_{\text{ref},h,2}(\cdot \mid s)).$$

We also define the regularization penalty for Player 1 as:

$$\text{Reg}_{1,h,s}(\mu) := \eta^{-1} \text{KL}(\mu \parallel \pi_{\text{ref},h,1}(\cdot \mid s)).$$

Using these definitions, the stagewise objective $f_{h,s}(\mu)$ can be written as:

$$f_{h,s}(\mu) = \langle \mu, Q_{h,\hat{\pi}_2}^\dagger(s, \cdot) \rangle - \text{Reg}_{1,h,s}(\mu).$$

The value of the best response $V_h^{\dagger, \hat{\pi}_2}(s)$ is the maximum of the total regularized objective:

$$\begin{aligned} V_h^{\dagger, \hat{\pi}_2}(s) &= \max_{\mu} \left(\langle \mu, Q_{h,\hat{\pi}_2}^\dagger(s, \cdot) \rangle - \text{Reg}_{1,h,s}(\mu) + C_{h,s} \right) \\ &= \max_{\mu} f_{h,s}(\mu) + C_{h,s} \\ &= f_{h,s}(\pi_{h,1}^\dagger) + C_{h,s}. \end{aligned}$$

Similarly, the value of the Nash policy π^* satisfies the Bellman equation, which we can decompose using the same terms:

$$\begin{aligned} V_h^{\pi_1^*, \hat{\pi}_2}(s) &= \mathbb{E}_{a_1 \sim \pi_{h,1}^*} \left[r_h^*(s, a_1, \hat{\pi}_2) + \mathbb{E}_{s'} [V_{h+1}^{\pi_1^*, \hat{\pi}_2}(s')] \right] - \text{Reg}_{1,h,s}(\pi_{h,1}^*) + C_{h,s} \\ &= \underbrace{\mathbb{E}_{a_1 \sim \pi_{h,1}^*} \left[r_h^*(s, a_1, \hat{\pi}_2) + \mathbb{E}_{s'} [V_{h+1}^{\dagger, \hat{\pi}_2}(s')] \right]}_{= f_{h,s}(\pi_{h,1}^*)} - \text{Reg}_{1,h,s}(\pi_{h,1}^*) + C_{h,s} \\ &\quad - \mathbb{E}_{s' \sim P_h(\cdot \mid s, \pi_1^*, \hat{\pi}_2)} \left[V_{h+1}^{\dagger, \hat{\pi}_2}(s') - V_{h+1}^{\pi_1^*, \hat{\pi}_2}(s') \right]. \end{aligned}$$

When we subtract $V_h^{\pi_1^*, \hat{\pi}_2}(s)$ from $V_h^{\dagger, \hat{\pi}_2}(s)$, the constant term $C_{h,s}$ cancels out:

$$\begin{aligned}\tilde{\Delta}_h(s) &= \left(f_{h,s}(\pi_{h,1}^\dagger) + C_{h,s}\right) - \left(f_{h,s}(\pi_{h,1}^*) + C_{h,s}\right) + \mathbb{E}_{s' \sim P_h}[\tilde{\Delta}_{h+1}(s')] \\ &= f_{h,s}(\pi_{h,1}^\dagger) - f_{h,s}(\pi_{h,1}^*) + \mathbb{E}_{s' \sim P_h(\cdot | s, \pi_1^*, \hat{\pi}_2)}[\tilde{\Delta}_{h+1}(s')].\end{aligned}$$

Using the exact characterization $f(\pi^\dagger) - f(\pi^*) = \eta^{-1} \text{KL}(\pi^* \| \pi^\dagger)$, we obtain the recurrence:

$$\tilde{\Delta}_h(s) = \eta^{-1} \text{KL}\left(\pi_{h,1}^*(\cdot | s) \parallel \pi_{h,1}^\dagger(\cdot | s)\right) + \mathbb{E}_{s' \sim P_h(\cdot | s, \pi_1^*, \hat{\pi}_2)}[\tilde{\Delta}_{h+1}(s')].$$

Smoothness of the Potential Function. Fix a time–state pair (h, s) . Define, for a finite index set $\mathcal{B}$, the log-partition function

$$\Phi(\theta) := \log \sum_{a \in \mathcal{B}} \exp(\theta_a), \quad \theta \in \mathbb{R}^{\mathcal{B}};$$

below we use it with $\mathcal{B} = \mathcal{A}_{h,1}^{\text{ref}}(s)$, and in the proof of Lemma 6.6 with $\mathcal{B} = \mathcal{A}_{h,2}^{\text{ref}}(s)$. Then Φ is **1-smooth** with respect to the ℓ_∞ -norm: its Hessian at θ is $\text{diag}(P_\theta) - P_\theta P_\theta^\top$ with $P_\theta = \text{Softmax}(\theta)$, so $v^\top \nabla^2 \Phi(\theta) v = \text{Var}_{P_\theta}(v) \leq \mathbb{E}_{P_\theta}[v^2] \leq \|v\|_\infty^2$ for every v . Hence, for all $\theta, \theta' \in \mathbb{R}^{\mathcal{B}}$,

$$\Phi(\theta') \leq \Phi(\theta) + \langle \nabla \Phi(\theta), \theta' - \theta \rangle + \frac{1}{2} \|\theta' - \theta\|_\infty^2.$$

Quadratic Upper Bound on Bregman Divergence. Recall the log-partition function

$$\Phi(\theta) = \log \sum_{a \in \mathcal{B}} \exp(\theta_a),$$

whose gradient map $\nabla \Phi(\theta)$ corresponds to the Softmax distribution $P_\theta = \text{Softmax}(\theta)$. The Bregman divergence generated by Φ is defined as

$$D_\Phi(\theta', \theta) := \Phi(\theta') - \Phi(\theta) - \langle \nabla \Phi(\theta), \theta' - \theta \rangle.$$

A standard result in convex analysis implies that this Bregman divergence coincides exactly with the KL divergence between the induced distributions:

$$D_\Phi(\theta', \theta) = \text{KL}(P_{\theta'} \| P_\theta).$$

Since Φ is 1-smooth with respect to the ℓ_∞ -norm, the associated Bregman divergence admits the quadratic upper bound

$$D_\Phi(\theta', \theta) \leq \frac{1}{2} \|\theta' - \theta\|_\infty^2.$$

Consequently, for any $\theta, \theta' \in \mathbb{R}^{\mathcal{B}}$,

$$\text{KL}(P_\theta \| P_{\theta'}) \leq \frac{1}{2} \|\theta' - \theta\|_\infty^2. \quad (28)$$

Application to Policy Logits. Fix a time–state pair (h, s) . Define the natural parameters (logits)

$$\begin{aligned}\theta_h^\dagger(s, \cdot) &:= \eta Q_{h, \hat{\pi}_2}^\dagger(s, \cdot) + \log \pi_{\text{ref}, h, 1}(\cdot | s), \\ \theta_h^*(s, \cdot) &:= \eta Q_{h, \pi_2^*}^\dagger(s, \cdot) + \log \pi_{\text{ref}, h, 1}(\cdot | s),\end{aligned}$$

which induce the policies

$$\pi_{h,1}^\dagger(\cdot | s) = \text{Softmax}(\theta_h^\dagger(s, \cdot)), \quad \pi_{h,1}^*(\cdot | s) = \text{Softmax}(\theta_h^*(s, \cdot)).$$

The difference between the logits is therefore

$$\theta_h^\dagger(s, \cdot) - \theta_h^*(s, \cdot) = \eta(Q_{h, \hat{\pi}_2}^\dagger(s, \cdot) - Q_{h, \pi_2^*}^\dagger(s, \cdot)). \quad (29)$$

Applying the quadratic KL bound (28) with $\mathcal{B} = \mathcal{A}_{h,1}^{\text{ref}}(s)$, $\theta = \theta_h^*(s, \cdot)$ and $\theta' = \theta_h^\dagger(s, \cdot)$ yields

$$\begin{aligned} \text{KL}(\pi_{h,1}^*(\cdot | s) \parallel \pi_{h,1}^\dagger(\cdot | s)) &\leq \frac{1}{2} \left\| \theta_h^\dagger(s, \cdot) - \theta_h^*(s, \cdot) \right\|_{\infty, \text{ref}}^2 \\ &= \frac{\eta^2}{2} \left\| Q_{h, \hat{\pi}_2}^\dagger(s, \cdot) - Q_{h, \pi_2^*}^\dagger(s, \cdot) \right\|_{\infty, \text{ref}}^2, \end{aligned}$$

the last step being (29) together with the cancellation of the (finite) reference logits on $\mathcal{A}_{h,1}^{\text{ref}}(s)$.

Substituting this bound into the representation of Gap 1, we obtain

$$\begin{aligned} \text{Gap 1} &= \eta^{-1} \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\text{KL}(\pi_{h,1}^*(\cdot | s) \parallel \pi_{h,1}^\dagger(\cdot | s)) \right] \\ &\leq \frac{\eta}{2} \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\left\| Q_{h, \hat{\pi}_2}^\dagger(s, \cdot) - Q_{h, \pi_2^*}^\dagger(s, \cdot) \right\|_{\infty, \text{ref}}^2 \right]. \end{aligned} \quad (30)$$

By (26), for every (h, s) ,

$$\begin{aligned} \left\| Q_{h, \hat{\pi}_2}^\dagger(s, \cdot) - Q_{h, \pi_2^*}^\dagger(s, \cdot) \right\|_{\infty, \text{ref}}^2 &\leq 2 \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_\infty \right)^2 \left\| \hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s) \right\|_1^2 \\ &\quad + 2 \max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')^2]. \end{aligned}$$

Substituting into (30) gives

$$\begin{aligned} \text{Gap 1} &\leq \eta \sum_{h=1}^H \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_\infty \right)^2 \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\left\| \hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s) \right\|_1^2 \right] \\ &\quad + \eta \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')^2] \right]. \end{aligned} \quad (31)$$

Here $\Delta_{h+1}(s') = V_{h+1}^{\dagger, \hat{\pi}_2}(s') - V_{h+1}^{\pi_1^*, \pi_2^*}(s')$ is the best-response value difference introduced in Step 2, *not* the quantity $\tilde{\Delta}$ used in the derivation of (27). Since $\mathcal{A}_{h,1}^{\text{ref}}(s)$ is finite and the inner integrand is measurable in s , the inner maximum is attained by a measurable deterministic selector $\nu_h^{\arg \max} : \mathcal{S} \rightarrow \mathcal{A}_1$ with $\nu_h^{\arg \max}(s) \in \mathcal{A}_{h,1}^{\text{ref}}(s)$.

A uniform value envelope for an arbitrary reference-supported pair. Lemma C.2 is stated for best-response values, but the identical envelope holds for the value of *any* pair of policies supported on the reference support. In the definition (1) the two KL penalties enter with opposite signs and each lies in $[0, \eta^{-1} \log(1/\alpha)]$ (by the computation following (14)), so their signed sum lies in $[-\eta^{-1} \log(1/\alpha), \eta^{-1} \log(1/\alpha)]$. Combined with $|r_h^*| \leq 1$ and Jensen's inequality this yields $\|V_h^{\pi_1, \pi_2}\|_\infty \leq 1 + \|V_{h+1}^{\pi_1, \pi_2}\|_\infty + \eta^{-1} \log(1/\alpha)$, and backward induction from $V_{H+1}^{\pi_1, \pi_2} \equiv 0$ gives, for every $h \in [H]$ and every such pair,

$$\|V_h^{\pi_1, \pi_2}\|_\infty \leq (H - h + 1)(1 + \eta^{-1} \log(1/\alpha)) = (H - h + 1)\lambda, \quad \lambda := 1 + \eta^{-1} \log(1/\alpha). \quad (32)$$

Lemma C.4 (Stability for value difference). *Fix an arbitrary Player 1 policy π_1 whose stage distributions satisfy $\text{supp } \pi_{h,1}(\cdot | s) \subseteq \text{supp } \pi_{h,1}^{\text{ref}}(\cdot | s)$ for all (h, s) , and let $\hat{\pi}_2, \pi_2^*$ be two Player 2 policies of Gibbs form $\pi_{h,2}(a | s) \propto \pi_{h,2}^{\text{ref}}(a | s) \exp(\eta q_h(s, a))$ with payoff vectors obeying $\|q_h\|_\infty \leq H\lambda$. ($\hat{\pi}_2$ and π_2^* are both of this form, by Assumption C.1 and Lemma C.2 respectively.) Let $d_h^{\pi_1, \hat{\pi}_2}$ denote the state visitation distribution at step h induced by $(\pi_1, \hat{\pi}_2)$. Define the value difference and policy difference as*

$$\Delta_h(s) := V_h^{\pi_1, \hat{\pi}_2}(s) - V_h^{\pi_1, \pi_2^*}(s), \quad \Delta\pi_{h,2}(\cdot | s) := \hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s).$$

Define the coefficient

$$L_{t+1} := 1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_\infty + 2H\lambda \leq 3H\lambda, \quad (33)$$

where the inequality uses (32) and $\lambda \geq 1$, and is therefore uniform in π_1 . Then for every $h \in [H]$,

$$\mathbb{E}_{s \sim d_h^{\pi_1, \hat{\pi}_2}}[\Delta_h(s)^2] \leq 6H \sum_{t=h}^H L_{t+1}^2 \mathbb{E}_{s \sim d_t^{\pi_1, \hat{\pi}_2}}[\|\Delta\pi_{t,2}(\cdot | s)\|_1^2]. \quad (34)$$

Moreover, the same estimate holds for the process started deterministically at an arbitrary state: for every $h \in [H]$ and every $s \in \mathcal{S}$,

$$\begin{aligned} \Delta_h(s)^2 &\leq 2H \sum_{t=h}^H \left(1 + \frac{1}{H}\right)^{t-h} L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{\pi_1, \hat{\pi}_2} | s_h = s}[\|\Delta\pi_{t,2}(\cdot | s_t)\|_1^2] \\ &\leq 6H \sum_{t=h}^H L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{\pi_1, \hat{\pi}_2} | s_h = s}[\|\Delta\pi_{t,2}(\cdot | s_t)\|_1^2], \end{aligned} \quad (35)$$

where $d_t^{\pi_1, \hat{\pi}_2} | s_h = s$ is the visitation law at step t conditioned on $s_h = s$.

Proof. Fix $t \in [H]$ and any state s . Since π_1 is fixed, the stage- t regularization term of Player 1 at s is identical in $V_t^{\pi_1, \hat{\pi}_2}(s)$ and in $V_t^{\pi_1, \pi_2^*}(s)$ and cancels in the difference; Player 1's regularization at the later steps is carried inside the continuation Q -functions. Let $Q_t^*(s, a) := Q_t^{\pi_1, \pi_2^*}(s, a)$.

$$\begin{aligned} \Delta_t(s) &= \mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2}[Q_t^{\pi_1, \hat{\pi}_2}(s, a)] - \mathbb{E}_{a \sim \pi_1 \times \pi_2^*}[Q_t^*(s, a)] + \eta^{-1}(\text{KL}(\hat{\pi}_{t,2} \| \pi^{\text{ref}}) - \text{KL}(\pi_{t,2}^* \| \pi^{\text{ref}})) \\ &= \underbrace{\mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2}[Q_t^{\pi_1, \hat{\pi}_2}(s, a) - Q_t^*(s, a)]}_{\text{(I)}} + \underbrace{\mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2}[Q_t^*(s, a)] - \mathbb{E}_{a \sim \pi_1 \times \pi_2^*}[Q_t^*(s, a)] + \Delta\text{Reg}_2(s)}_{\text{(II)}}, \end{aligned}$$

where $\text{Reg}_2(\pi) := \eta^{-1}\text{KL}(\pi(\cdot | s) \| \pi_{t,2}^{\text{ref}}(\cdot | s))$ and $\Delta\text{Reg}_2(s) := \text{Reg}_2(\hat{\pi}_{t,2}) - \text{Reg}_2(\pi_{t,2}^*)$. Term (I) captures the value difference propagation:

$$\text{(I)} = \mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2} \mathbb{E}_{s' \sim P_t(\cdot | s, a)}[\Delta_{t+1}(s')].$$

Term (II) captures the policy error at step t . Note that

$$\begin{aligned} \mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2}[Q_t^*] - \mathbb{E}_{a \sim \pi_1 \times \pi_2^*}[Q_t^*] &= \mathbb{E}_{a_1 \sim \pi_{t,1}} \left[\sum_{a_2} (\hat{\pi}_{t,2}(a_2 | s) - \pi_{t,2}^*(a_2 | s)) Q_t^*(s, a_1, a_2) \right] \\ &= \mathbb{E}_{a_1 \sim \pi_{t,1}} \left[\langle \Delta\pi_{t,2}(\cdot | s), Q_t^*(s, a_1, \cdot) \rangle \right], \end{aligned}$$

so by Hölder's inequality this contribution is at most $\|Q_t^*(s, \cdot, \cdot)\|_\infty \|\Delta\pi_{t,2}(\cdot | s)\|_1$ in absolute value.

The regularization difference. By hypothesis both $\hat{\pi}_{t,2}(\cdot | s)$ and $\pi_{t,2}^*(\cdot | s)$ are Gibbs policies $\pi(a) \propto \pi_{t,2}^{\text{ref}}(a | s) \exp(\eta q(s, a))$ with $\|q(s, \cdot)\|_\infty \leq H\lambda$: for $\pi_{t,2}^*$ the payoff is $q = -Q_{t, \pi_1}^{\dagger, \pi_2^*}(s, \cdot) = -\mathbb{E}_{a_1 \sim \pi_{t,1}}[Q_t^{\pi_1, \pi_2^*}(s, a_1, \cdot)]$, bounded by $1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_\infty \leq H\lambda$ by Lemma C.2; for $\hat{\pi}_{t,2}$ it is $q = -\mathbb{E}_{a_1 \sim \hat{\pi}_{t,1}}[\hat{Q}_t(s, a_1, \cdot)]$, bounded by $C_{\text{val}} = H\lambda$ by Assumption C.1. Since $\log Z := \log \sum_a \pi_{t,2}^{\text{ref}}(a | s) e^{\eta q(s, a)} \in [-\eta\|q\|_\infty, \eta\|q\|_\infty]$,

$$\log \frac{\pi(a)}{\pi_{t,2}^{\text{ref}}(a | s)} = \eta q(s, a) - \log Z, \quad \left| \log \frac{\pi(a)}{\pi_{t,2}^{\text{ref}}(a | s)} \right| \leq 2\eta H\lambda \quad (36)$$

for every a in the support of $\pi_{t,2}^{\text{ref}}(\cdot | s)$, while $\pi(a) = 0$ off that support. In particular $\pi(a) \geq \alpha e^{-2\eta H\lambda} > 0$ on the support, so every convex combination $\xi = (1 - \theta)\hat{\pi}_{t,2} + \theta\pi_{t,2}^*$, $\theta \in [0, 1]$, has exactly the support of $\pi_{t,2}^{\text{ref}}(\cdot | s)$; moreover $\xi(a)/\pi_{t,2}^{\text{ref}}(a | s)$ is a convex combination of the two ratios in (36), so $\|\log(\xi/\pi_{t,2}^{\text{ref}})\|_\infty \leq 2\eta H\lambda$ as well. Restricted to the

face of the simplex supported on $\text{supp } \pi_{t,2}^{\text{ref}}(\cdot | s)$, the map $p \mapsto \text{Reg}_2(p)$ is continuously differentiable in a neighbourhood of this segment, with

$$\nabla_p \eta^{-1} \text{KL}(p \| \pi_{t,2}^{\text{ref}}) = \eta^{-1} (\mathbf{1} + \log(p/\pi_{t,2}^{\text{ref}})),$$

so the Mean Value Theorem applies: there is a ξ on the segment joining $\pi_{t,2}^*(\cdot | s)$ and $\hat{\pi}_{t,2}(\cdot | s)$ with

$$\text{Reg}_2(\hat{\pi}_{t,2}) - \text{Reg}_2(\pi_{t,2}^*) = \langle \eta^{-1} (\mathbf{1} + \log(\xi/\pi_{t,2}^{\text{ref}})), \hat{\pi}_{t,2} - \pi_{t,2}^* \rangle.$$

The constant part of the gradient cancels *exactly*: $\hat{\pi}_{t,2}(\cdot | s)$ and $\pi_{t,2}^*(\cdot | s)$ are both probability vectors, hence $\langle \mathbf{1}, \hat{\pi}_{t,2} - \pi_{t,2}^* \rangle = 0$. Therefore, by Hölder's inequality,

$$\begin{aligned} |\text{Reg}_2(\hat{\pi}_{t,2}) - \text{Reg}_2(\pi_{t,2}^*)| &= \eta^{-1} |\langle \log(\xi/\pi_{t,2}^{\text{ref}}), \hat{\pi}_{t,2} - \pi_{t,2}^* \rangle| \\ &\leq \eta^{-1} \|\log(\xi/\pi_{t,2}^{\text{ref}})\|_{\infty} \|\hat{\pi}_{t,2} - \pi_{t,2}^*\|_1 \leq B_{\text{reg}} \|\Delta \pi_{t,2}(\cdot | s)\|_1, \quad B_{\text{reg}} := 2H\lambda. \end{aligned}$$

Combining the two contributions yields the Lipschitz bound for Term (II):

$$\begin{aligned} |(\text{II})| &\leq (\|Q_t^*(s, \cdot, \cdot)\|_{\infty} + B_{\text{reg}}) \|\Delta \pi_{t,2}(\cdot | s)\|_1 \\ &\leq (1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_{\infty} + 2H\lambda) \|\Delta \pi_{t,2}(\cdot | s)\|_1 = L_{t+1} \|\Delta \pi_{t,2}(\cdot | s)\|_1, \end{aligned}$$

where we used $\|Q_t^*\|_{\infty} \leq 1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_{\infty}$ (immediate from $|r_t^*| \leq 1$) together with $B_{\text{reg}} = 2H\lambda$ and the definition (33) of L_{t+1} .

Thus, we have the pointwise recursive bound:

$$|\Delta_t(s)| \leq L_{t+1} \|\Delta \pi_{t,2}(\cdot | s)\|_1 + |\mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2} \mathbb{E}_{s' \sim P_t(\cdot | s, a)} [\Delta_{t+1}(s')]|.$$

Square both sides and apply the weighted AM-GM inequality $(x + y)^2 \leq (1 + H)x^2 + (1 + \frac{1}{H})y^2$ (valid for all x, y and all $H > 0$, since $2xy \leq Hx^2 + y^2/H$):

$$\Delta_t(s)^2 \leq (1 + H)L_{t+1}^2 \|\Delta \pi_{t,2}(\cdot | s)\|_1^2 + \left(1 + \frac{1}{H}\right) (\mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2} \mathbb{E}_{s' \sim P_t} [\Delta_{t+1}(s')])^2.$$

Apply Jensen's inequality to the second term: $(\mathbb{E}Z)^2 \leq \mathbb{E}[Z^2]$.

$$\Delta_t(s)^2 \leq 2HL_{t+1}^2 \|\Delta \pi_{t,2}(\cdot | s)\|_1^2 + \left(1 + \frac{1}{H}\right) \mathbb{E}_{a \sim \pi_1 \times \hat{\pi}_2} \mathbb{E}_{s' \sim P_t} [\Delta_{t+1}(s')^2].$$

(Using $1 + H \leq 2H$.) Take the expectation with respect to $s \sim d_t^{\pi_1, \hat{\pi}_2}$; if $s \sim d_t^{\pi_1, \hat{\pi}_2}$, $a \sim \pi_1 \times \hat{\pi}_2$ and $s' \sim P_t$, then $s' \sim d_{t+1}^{\pi_1, \hat{\pi}_2}$. Thus:

$$\mathbb{E}_{s \sim d_t^{\pi_1, \hat{\pi}_2}} [\Delta_t(s)^2] \leq 2HL_{t+1}^2 \mathbb{E}_{s \sim d_t^{\pi_1, \hat{\pi}_2}} [\|\Delta \pi_{t,2}\|_1^2] + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim d_{t+1}^{\pi_1, \hat{\pi}_2}} [\Delta_{t+1}(s')^2].$$

Unrolling this recurrence from $t = h$ to H (with $\Delta_{H+1} \equiv 0$):

$$\mathbb{E}_{d_h} [\Delta_h^2] \leq 2H \sum_{t=h}^H \left(1 + \frac{1}{H}\right)^{t-h} L_{t+1}^2 \mathbb{E}_{d_t} [\|\Delta \pi_{t,2}\|_1^2].$$

Since $t - h \leq H$ and $(1 + \frac{1}{H})^H \leq e < 3$, the pre-factor is bounded by $6H$, yielding (34). Finally, every inequality above is pointwise in the current state and only then averaged over the visitation law, so replacing $d_t^{\pi_1, \hat{\pi}_2}$ throughout by the conditional visitation $d_t^{\pi_1, \hat{\pi}_2} | s_h = s$ gives (35). $\square$

Step 4: Combining the two terms in the Gap 1 bound. Recall from (31) that

$$\begin{aligned} \text{Gap 1} &\leq \eta \sum_{h=1}^H \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_{\infty}\right)^2 \mathbb{E}_{s \sim d_h^{\pi_1, \hat{\pi}_2}} [\|\hat{\pi}_{h,2}(\cdot | s) - \pi_{h,2}^*(\cdot | s)\|_1^2] \\ &\quad + \eta \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1, \hat{\pi}_2}} \left[\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s)} \mathbb{E}_{s' \sim P_h(\cdot | s, a_1, a_2)} [\Delta_{h+1}(s')^2] \right]. \end{aligned}$$

Step 4.1: Controlling the first term. By Lemma C.2, for every $h \in [H]$,

$$1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_{\infty} \leq H \left(1 + \eta^{-1} \log(1/\alpha)\right).$$

Plugging into the first term of (31) yields

$$\begin{aligned} & \eta \sum_{h=1}^H \left(1 + \|V_{h+1}^{\dagger, \hat{\pi}_2}\|_{\infty}\right)^2 \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1^2 \right] \\ & \leq \eta H^2 \left(1 + \eta^{-1} \log(1/\alpha)\right)^2 \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1^2 \right]. \end{aligned} \quad (37)$$

Step 4.2: Controlling the second term. Remember we defined, in (33),

$$L_{t+1} := 1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_{\infty} + 2H\lambda \leq 3H\lambda,$$

where π_1 denotes the fixed Player 1 policy with which Lemma C.4 is invoked, and where the bound $L_{t+1} \leq 3H\lambda$ holds uniformly in π_1 by the pairwise envelope (32).

Δ -decomposition (bridging (26) and Lemma C.4). The Δ_{h+1} in (31) is the best-response value difference $V_{h+1}^{\dagger, \hat{\pi}_2}(s') - V_{h+1}^{\dagger, \pi_2^*}(s')$, while Lemma C.4 bounds the value difference $V_{h+1}^{\pi_1, \hat{\pi}_2}(s') - V_{h+1}^{\pi_1, \pi_2^*}(s')$ for any fixed Player 1 policy π_1 . Let $\pi_1^{\dagger}(\hat{\pi}_2)$ denote the regularized best response of Player 1 to $\hat{\pi}_2$. Since $V_{h+1}^{\dagger, \hat{\pi}_2} = V_{h+1}^{\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2}$ and $V_{h+1}^{\dagger, \pi_2^*} = V_{h+1}^{\pi_1^*, \pi_2^*} \geq V_{h+1}^{\pi_1^{\dagger}(\hat{\pi}_2), \pi_2^*}$ by Nash optimality of π_1^* against π_2^* ,

$$\Delta_{h+1}(s') \leq R(s') := V_{h+1}^{\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2}(s') - V_{h+1}^{\pi_1^{\dagger}(\hat{\pi}_2), \pi_2^*}(s').$$

Moreover, by Player 1's best-response optimality, $V_{h+1}^{\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2}(s') \geq V_{h+1}^{\pi_1^*, \hat{\pi}_2}(s') \geq V_{h+1}^{\pi_1^*, \pi_2^*}(s')$ (the last inequality uses Nash optimality of π_2^* against π_1^*), so $\Delta_{h+1}(s') \geq 0$. Combined with $\Delta_{h+1}(s') \leq R(s')$, this gives $0 \leq \Delta_{h+1}(s') \leq R(s')$, hence $R(s') \geq 0$ and $\Delta_{h+1}(s')^2 \leq R(s')^2$.

Step 4.2.a: Pointwise V-stability and rerouting to a Π_{uni} measure. Apply (35) to $R(s')$ at $s_{h+1} = s'$ with $\pi_1 = \pi_1^{\dagger}(\hat{\pi}_2)$. The lemma is applicable: $\pi_1^{\dagger}(\hat{\pi}_2)$ is a regularized best response, hence Gibbs and in particular supported on $\text{supp } \pi_{h,1}^{\text{ref}}$, and $L_{t+1} \leq 3H\lambda$ holds uniformly in the fixed Player 1 policy by (32). Unrolling from $h+1$ to H and using $\Delta_{h+1}(s')^2 \leq R(s')^2$ from the Δ -decomposition above yields

$$\Delta_{h+1}(s')^2 \leq 2H \sum_{t=h+1}^H \left(1 + \frac{1}{H}\right)^{t-h-1} L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{(\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2) \mid s_{h+1}=s'}} \left[\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2 \right], \quad (38)$$

where $d_t^{(\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2) \mid s_{h+1}=s'}$ denotes the visitation distribution at step t under the joint policy $(\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2)$ conditioned on starting at $s_{h+1} = s'$. Applying Lemma C.5 (i) to this *conditional* measure (part (v) of that lemma; measure domination of the state-action law also implies domination of the state marginal) bridges $d_t^{(\pi_1^{\dagger}(\hat{\pi}_2), \hat{\pi}_2) \mid s'}$ to $d_t^{(\pi_1^*, \hat{\pi}_2) \mid s'}$ at constant cost C_{trans} :

$$\Delta_{h+1}(s')^2 \leq 2C_{\text{trans}} H \sum_{t=h+1}^H \left(1 + \frac{1}{H}\right)^{t-h-1} L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{(\pi_1^*, \hat{\pi}_2) \mid s_{h+1}=s'}} \left[\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2 \right]. \quad (39)$$

Step 4.2.b: Outer averaging produces a hybrid measure that we reroute to Π_{uni} by a path-measure change of measure. Since $\mathcal{A}_{h,1}^{\text{ref}}(s_h)$ is finite, the inner maximum over a_1 is attained at every s_h ; fix the selector

$$\nu_h^{\arg \max}(s_h) \in \arg \max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s_h)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot|s_h)} \mathbb{E}_{s' \sim P_h(\cdot|s_h, a_1, a_2)} [\Delta_{h+1}(s')^2],$$

with ties broken by a fixed ordering of $\mathcal{A}_1$ (so that $\nu_h^{\arg \max}$ is a measurable map $\mathcal{S} \rightarrow \mathcal{A}_1$). Replacing the reference-support maximum by this deterministic stage policy with *equality* and then substituting the pointwise bound (39) gives, for the second term of (31),

$$\begin{aligned} & \eta \sum_{h=1}^H \mathbb{E}_{s_h \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s_h)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot | s_h)} \mathbb{E}_{s' \sim P_h(\cdot | s_h, a_1, a_2)} [\Delta_{h+1}(s')^2] \right] \\ & \leq 2 C_{\text{trans}} H \eta \sum_{h=1}^H \sum_{t=h+1}^H \left(1 + \frac{1}{H}\right)^{t-h-1} L_{t+1}^2 \mathbb{E}_{s_t \sim \nu'_{t,h}} [\|\hat{\pi}_{t,2}(\cdot | s_t) - \pi_{t,2}^*(\cdot | s_t)\|_1^2], \end{aligned} \quad (40)$$

where $\nu'_{t,h}$ is the step- t state marginal of the trajectory law obtained by running the game from $s_1 \sim \rho$ under the time-varying joint policy

$$\varpi(h)_{h'} := \begin{cases} (\pi_{h',1}^*, \hat{\pi}_{h',2}), & h' \neq h, \\ (\nu_h^{\arg \max}, \pi_{h,2}^*), & h' = h. \end{cases} \quad (41)$$

Player 2 plays $\hat{\pi}_2$ at all steps except step h (where it plays $\pi_{h,2}^*$), so $\nu'_{t,h}$ is *not* of the form $d_t^{\pi_1^*, \hat{\pi}_2}$ and is *not* in Π_{uni} directly.

To bring $\nu'_{t,h}$ into a Π_{uni} -compatible form, define the Player 1 policy $\pi'_1(h)$ on *all* H steps by

$$\pi'_{h',1}(h) := \begin{cases} \pi_{h',1}^*, & h' \neq h, \\ \nu_h^{\arg \max}, & h' = h, \end{cases} \quad h' \in [H],$$

so that $\pi'_1(h)$ depends on h alone and not on t ; its step- h component is a deterministic reference-supported stage policy. Since Π_1 consists of time-varying reference-supported Markov policies with arbitrary admissible stage components, including deterministic and dataset-dependent ones, over all of which Assumption 6.2 takes a supremum, we have $\pi'_1(h) \in \Pi_1$ and $(\pi'_1(h), \pi_2^*) \in \Pi_1 \times \{\pi_2^*\} \subset \Pi_{\text{uni}}$.

Path-measure likelihood ratio. The comparison is made between *trajectory* laws rather than between step- t marginals, whose ratio is an average of path ratios and not a product of per-stage policy ratios. Write $\tau_t := (s_1, a_{1,1}, a_{1,2}, \dots, s_{t-1}, a_{t-1,1}, a_{t-1,2}, s_t)$ for the truncated trajectory through step t , and let $\mathbb{P}_{\varpi}^{(t)}$ be its law under a joint (possibly time-varying, possibly deterministic) Markov policy ϖ started from $s_1 \sim \rho$, which disintegrates as

$$\mathbb{P}_{\varpi}^{(t)}(d\tau_t) = \rho(ds_1) \prod_{h'=1}^{t-1} \varpi_{h',1}(a_{h',1} | s_{h'}) \varpi_{h',2}(a_{h',2} | s_{h'}) P_{h'}(ds_{h'+1} | s_{h'}, a_{h',1}, a_{h',2}). \quad (42)$$

Set $\varpi' := \varpi(h)$ from (41) and $\varpi'' := (\pi'_1(h), \pi_2^*)$. The two path laws share the initial distribution ρ , the transition kernels $\{P_{h'}\}_{h' < t}$, every Player-1 stage component (namely $\pi_{h',1}^*$ for $h' \neq h$ and the same deterministic $\nu_h^{\arg \max}$ at step h), and the step- h Player-2 component $\pi_{h,2}^*$; they differ *only* in Player 2's stage policy at the steps $h' \in \{1, \dots, t-1\} \setminus \{h\}$. Since $\hat{\pi}_{h',2}$ and $\pi_{h',2}^*$ are Gibbs with respect to the same reference $\pi_{h',2}^{\text{ref}}$ they have identical supports, so $\mathbb{P}_{\varpi'}^{(t)} \ll \mathbb{P}_{\varpi''}^{(t)}$, and cancelling the common factors in (42) gives, $\mathbb{P}_{\varpi''}^{(t)}$ -almost surely,

$$\frac{d\mathbb{P}_{\varpi'}^{(t)}}{d\mathbb{P}_{\varpi''}^{(t)}}(\tau_t) = \prod_{h' \in \{1, \dots, t-1\} \setminus \{h\}} \frac{\hat{\pi}_{h',2}(a_{h',2} | s_{h'})}{\pi_{h',2}^*(a_{h',2} | s_{h'})}. \quad (43)$$

Each per-step factor is a softmax-vs-softmax Player 2 ratio bounded by $\exp(4\eta H(1 + \eta^{-1} \log(1/\alpha)))$ (the per-step bound of Step D.5 of Appendix C.3, applied pointwise at every $(s_{h'}, a_{h',2})$; both policies are Gibbs for the *same* player). The product has at most $t-2 \leq H-2$ factors. Under (7), the telescoped product satisfies

$$\prod_{h' \in \{1, \dots, t-1\} \setminus \{h\}} \frac{\hat{\pi}_{h',2}(a_{h',2} | s_{h'})}{\pi_{h',2}^*(a_{h',2} | s_{h'})} \leq \exp(4\eta H^2 \lambda) \leq e = C_{\text{trans}}.$$

The step- h Player 1 factor is identical on the two sides, so it cancels in (43) and no argmax-versus-softmax comparison, and hence no additional $1/\alpha$ cost, arises at step h . Consequently, for any nonnegative measurable f ,

$$\mathbb{E}_{s_t \sim \nu'_{t,h}} [f(s_t)] = \mathbb{E}_{\tau_t \sim \mathbb{P}_{\varpi''}^{(t)}} \left[f(s_t) \frac{d\mathbb{P}_{\varpi'}^{(t)}}{d\mathbb{P}_{\varpi''}^{(t)}}(\tau_t) \right] \leq C_{\text{trans}} \cdot \mathbb{E}_{s_t \sim d_t^{(\pi'_1(h), \pi_2^*)}} [f(s_t)], \quad (44)$$

where the first equality is the change of measure at the path level combined with the push-forward $\tau_t \mapsto s_t$, which sends $\mathbb{P}_{\varpi'}^{(t)}$ to $\nu'_{t,h}$ and $\mathbb{P}_{\varpi''}^{(t)}$ to $d_t^{(\pi'_1(h), \pi_2^*)}$.

Applying (44) with $f(s_t) = \|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2$ to (40), and bounding $(1 + 1/H)^{t-h-1} \leq e$, $2e \leq 6$:

$$\begin{aligned} & \eta \sum_{h=1}^H \mathbb{E}_{s_h \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s_h)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*} \mathbb{E}_{s'} [\Delta_{h+1}(s')^2] \right] \\ & \leq 6 C_{\text{trans}}^2 H \eta \sum_{h=1}^H \sum_{t=h+1}^H L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{(\pi'_1(h), \pi_2^*)}} [\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2], \end{aligned} \quad (45)$$

where the C_{trans}^2 prefactor collects the Player-1 conditional transfer (from (39)) and the Player-2 path-measure bridge (from (44)). **Swap and summation.** Each inner term $L_{t+1}^2 \mathbb{E}_{s_t \sim d_t^{(\pi'_1(h), \pi_2^*)}} [\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2]$ depends on the outer index h only through the deterministic step- h stage policy $\nu_h^{\arg \max}$ embedded in $\pi'_1(h)$. For each fixed t , bound the inner sum over h by replacing each Π_{uni} -measure with the sup over Π_{uni} :

$$\sum_{h=1}^{t-1} \mathbb{E}_{s_t \sim d_t^{(\pi'_1(h), \pi_2^*)}} [\|\hat{\pi}_{t,2} - \pi_{t,2}^*\|_1^2] \leq (t-1) \cdot \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s_t \sim d_t^{\pi'}} [\|\hat{\pi}_{t,2} - \pi_{t,2}^*\|_1^2], \quad (46)$$

and use $(t-1) \leq H$. Substituting (46) into (45) and swapping $\sum_h \sum_t = \sum_t \sum_h$:

$$\eta \sum_{h=1}^H \mathbb{E}_{s_h \sim d_h^{\pi_1^*, \hat{\pi}_2}} \left[\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s_h)} \mathbb{E}_{a_2} \mathbb{E}_{s'} [\Delta_{h+1}(s')^2] \right] \leq 6 C_{\text{trans}}^2 \eta H^2 \sum_{t=1}^H L_{t+1}^2 \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s_t \sim d_t^{\pi'}} [\|\hat{\pi}_{t,2} - \pi_{t,2}^*\|_1^2]. \quad (47)$$

Step 4.3: Final bound on Gap 1. Putting (37) and (47) into (31), and noting that $(\pi_1^*, \hat{\pi}_2) \in \Pi_{\text{uni}}$ so $\mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} [\cdot] \leq \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\cdot]$, we conclude

$$\begin{aligned} \text{Gap 1} & \leq \eta H^2 (1 + \eta^{-1} \log(1/\alpha))^2 \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} [\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1^2] \\ & \quad + 6 C_{\text{trans}}^2 \eta H^2 \sum_{t=1}^H L_{t+1}^2 \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s_t \sim d_t^{\pi'}} [\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2], \end{aligned}$$

Recall from (33) that $L_{t+1} := 1 + \|V_{t+1}^{\pi_1, \pi_2^*}\|_{\infty} + 2H\lambda$, where $\pi_1 = \pi_1^{\dagger}(\hat{\pi}_2)$ is the Player-1 policy with which Lemma C.4 was invoked in Step 4.2.a. By the pairwise value envelope (32) we have $\|V_{t+1}^{\pi_1, \pi_2^*}\|_{\infty} \leq (H-t)\lambda$, hence

$$L_{t+1} \leq 1 + (H-t)\lambda + 2H\lambda \leq 3H\lambda \quad \text{for every } t \in [H],$$

using $1 \leq \lambda$. Bounding the first term's distribution $d_h^{\pi_1^*, \hat{\pi}_2}$ by $\sup_{\pi' \in \Pi_{\text{uni}}} d_h^{\pi'}$ (since $(\pi_1^*, \hat{\pi}_2) \in \Pi_{\text{uni}}$) and substituting,

$$\begin{aligned} \text{Gap 1} & \leq \eta H^2 (1 + \eta^{-1} \log(1/\alpha))^2 \sum_{h=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1^2] \\ & \quad + 54 C_{\text{trans}}^2 \eta H^4 (1 + \eta^{-1} \log(1/\alpha))^2 \sum_{t=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s_t \sim d_t^{\pi'}} [\|\hat{\pi}_{t,2}(\cdot|s_t) - \pi_{t,2}^*(\cdot|s_t)\|_1^2] \\ & \leq 55 C_{\text{trans}}^2 \eta H^4 (1 + \eta^{-1} \log(1/\alpha))^2 \sum_{h=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1^2]. \end{aligned} \quad (48)$$

Step 4.4: Conversion to single-player Nash-averaged squared Q -error via squared L_1 stability. By the squared stability bound (64) (Appendix C.4), for every (h, s) ,

$$\|\hat{\pi}_h(\cdot|s) - \pi_h^*(\cdot|s)\|_1^2 \leq 4\eta^2 \cdot \xi_h^2(s),$$

with $\xi_h^2(s)$ the single-player Nash-averaged squared Q -error from (65). Here, as in (64), $\hat{\pi}_h(\cdot|s) - \pi_h^*(\cdot|s)$ denotes the difference of the concatenated pairs of stage distributions, so in particular $\|\hat{\pi}_{h,2}(\cdot|s) - \pi_{h,2}^*(\cdot|s)\|_1 \leq \|\hat{\pi}_h(\cdot|s) - \pi_h^*(\cdot|s)\|_1$. Substituting into (48),

$$\begin{aligned} \text{Gap 1} &\leq 55 C_{\text{trans}}^2 \eta H^4 (1 + \eta^{-1} \log(1/\alpha))^2 \cdot 4\eta^2 \sum_{h=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\xi_h^2(s)] \\ &= 220 C_{\text{trans}}^2 \eta^3 H^4 (1 + \eta^{-1} \log(1/\alpha))^2 \sum_{h=1}^H \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}} [\xi_h^2(s)]. \end{aligned} \quad (49)$$

This proves Lemma 6.5 with the $\mathcal{O}(C_{\text{trans}}^2 \eta^3 H^4 (1 + \eta^{-1} \log(1/\alpha))^2)$ leading constant; under (7), $C_{\text{trans}}^2 = e^2$ is absorbed into the universal constant.

C.3. Proof of Lemma C.5 (Density-Ratio Stability under Small η)

Lemma C.5 (Density-Ratio Stability under Small η). *Under the condition (7) stated with Theorem 6.3, namely $4\eta H^2 \lambda \leq 1$, let $\pi_1^\dagger(\hat{\pi}_2)$ denote the regularized best response of Player 1 against $\hat{\pi}_2$, and set $C_{\text{trans}} := e$. For finite measures ν and μ , write $\nu \preceq C\mu$ if $\nu(B) \leq C\mu(B)$ for every measurable set B ; equivalently, $\nu \ll \mu$ and $d\nu/d\mu \leq C$ μ -almost surely. Then for every $h \in [H]$,*

- (i) (Player 1 transfer) $d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2} \preceq C_{\text{trans}} d_h^{\pi_1^*, \hat{\pi}_2}$.
- (ii) (Player 2 transfer) $d_h^{\pi_1^*, \hat{\pi}_2} \preceq C_{\text{trans}} d_h^{\pi_1^*, \pi_2^*}$.

In the countable case this is exactly the corresponding pointwise density-ratio statement, with $0/0 = 0$. In the general standard-Borel setting the measure-domination form above is used.

Moreover, the same one-sided transfer remains valid when the deviating component is an arbitrary reference-supported policy rather than a regularized one. Write

$$\mathcal{A}_{h,i}^{\text{ref}}(s) := \{a \in \mathcal{A}_i : \pi_{\text{ref},h,i}(a | s) > 0\}, \quad \Pi_i^{\text{ref}} := \{\pi_i : \text{supp } \pi_{h,i}(\cdot | s) \subseteq \mathcal{A}_{h,i}^{\text{ref}}(s) \ \forall (h, s)\}$$

for the reference support and the corresponding support-restricted policy class. Then for every $\varpi_1 \in \Pi_1^{\text{ref}}$, every $\varpi_2 \in \Pi_2^{\text{ref}}$, and every $h \in [H]$,

$$d_h^{\varpi_1, \hat{\pi}_2} \preceq C_{\text{trans}} d_h^{\pi_1^*, \hat{\pi}_2}, \quad d_h^{\pi_1^*, \varpi_2} \preceq C_{\text{trans}} d_h^{\pi_1^*, \pi_2^*}, \quad d_h^{\varpi_1, \varpi_2} \preceq C_{\text{trans}}^2 d_h^{\pi_1^*, \hat{\pi}_2}, \quad (50)$$

More generally, when the dominating law is the visitation law of a joint policy whose two components are regularized, the transfer costs one factor C_{trans} for each player component that differs: a one-player transfer is bounded by C_{trans} , while a two-player transfer is bounded by C_{trans}^2 . Only these directions are asserted: reverse domination need not hold when the transferred law contains a deterministic component, and is never used. This is proved in Step D.7 below. Specializing (50) to $\varpi_1 = \pi_1^\dagger(\hat{\pi}_2)$ and $\varpi_2 = \hat{\pi}_2$ recovers parts (i) and (ii). Consequently, for any nonnegative function f and any $h \in [H]$,

$$\mathbb{E}_{(s,a) \sim d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}}[f] \leq C_{\text{trans}} \cdot \mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \hat{\pi}_2}}[f] \leq C_{\text{trans}}^2 \cdot \mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \pi_2^*}}[f]. \quad (51)$$

The proof leverages the softmax representation of regularized best responses together with the worst-case bound $\|Q_h^{\dagger, \hat{\pi}_2} - Q_h^*\|_\infty \leq 2H(1 + \eta^{-1} \log(1/\alpha))$ from Lemma C.2. The remaining parts (i')–(v), which combine the same per-stage argument with the reference-support lower bound (56) for non-Gibbs deviations, are proved in Steps D.7–D.8.

We first establish domination of the BR-prefix law $d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}$ by the Nash-prefix law $d_h^{\pi_1^*, \hat{\pi}_2}$, which share $\hat{\pi}_2$ and the kernels and differ only in Player 1's policy.

Step D.1: Per-step softmax-ratio bound. At each step $h' \in [H]$ and each state s , both $\pi_1^\dagger(\hat{\pi}_2)$ and π_1^* are KL-regularized best responses (the former to $\hat{\pi}_2$, the latter to π_2^* , which is itself the regularized Nash). By the Gibbs form (24),

$$\pi_{h',1}^\dagger(\hat{\pi}_2)(\cdot | s) = \text{Softmax}(\eta Q_{h',\hat{\pi}_2}^{\dagger, \hat{\pi}_2}(s, \cdot) + \log \pi_{\text{ref},h',1}(\cdot | s)), \quad \pi_{h',1}^*(\cdot | s) = \text{Softmax}(\eta Q_{h',\pi_2^*}^{\dagger, \pi_2^*}(s, \cdot) + \log \pi_{\text{ref},h',1}(\cdot | s)).$$

Both Gibbs distributions put zero mass outside the reference support $\mathcal{A}_{h',1}^{\text{ref}}(s) = \{a_1 : \pi_{\text{ref},h',1}(a_1 | s) > 0\}$ and strictly positive mass on it, so every action ratio below is 0/0 off $\mathcal{A}_{h',1}^{\text{ref}}(s)$ (read as 0) and finite on it. Throughout this appendix all action suprema and ℓ_∞ -norms for player i at (h', s) are taken over $\mathcal{A}_{h',i}^{\text{ref}}(s)$, on which the logits $\pm\eta Q + \log \pi_{\text{ref},h',i}(\cdot | s)$ are finite. There the reference logit cancels in any ratio, so for every $a_1 \in \mathcal{A}_{h',1}^{\text{ref}}(s)$,

$$\frac{\pi_{h',1}^\dagger(\hat{\pi}_2)(a_1 | s)}{\pi_{h',1}^*(a_1 | s)} = \frac{\exp(\eta Q_{h',\hat{\pi}_2}^\dagger(s, a_1))}{\exp(\eta Q_{h',\pi_2^*}^\dagger(s, a_1))} \cdot \frac{Z_{h',s}^*}{Z_{h',s}^\dagger},$$

where $Z_{h',s}^\dagger$ and $Z_{h',s}^*$ are the corresponding partition functions. By a standard softmax-ratio bound, for any finite index set $\mathcal{B}$ and any two *finite* logit vectors $z, z' \in \mathbb{R}^{\mathcal{B}}$, with normalizers $Z = \sum_{a \in \mathcal{B}} e^{z_a}$ and $Z' = \sum_{a \in \mathcal{B}} e^{z'_a}$,

$$\frac{\text{Softmax}(z)(a)}{\text{Softmax}(z')(a)} \leq \exp(\|z - z'\|_\infty + \log \frac{Z'}{Z}) \leq \exp(2\|z - z'\|_\infty), \quad (52)$$

where the second inequality uses $|\log(Z'/Z)| \leq \|z - z'\|_\infty$ (LSE 1-Lipschitz in ℓ_∞). Every application of (52) below takes $\mathcal{B} = \mathcal{A}_{h',i}^{\text{ref}}(s)$.

Step D.2: Worst-case logit-difference bound. The logit difference at step h' is

$$\eta (Q_{h',\hat{\pi}_2}^\dagger(s, a_1) - Q_{h',\pi_2^*}^\dagger(s, a_1)).$$

By Lemma C.2, both $Q_{h',\hat{\pi}_2}^\dagger$ and $Q_{h',\pi_2^*}^\dagger$ are bounded by $H(1 + \eta^{-1} \log(1/\alpha))$ in absolute value, uniformly over Player 2 policies. Hence the worst-case (TV) logit-difference bound is

$$\|\eta (Q_{h',\hat{\pi}_2}^\dagger(s, \cdot) - Q_{h',\pi_2^*}^\dagger(s, \cdot))\|_\infty \leq 2\eta H(1 + \eta^{-1} \log(1/\alpha)).$$

Combining with (52),

$$\frac{\pi_{h',1}^\dagger(\hat{\pi}_2)(a_1 | s)}{\pi_{h',1}^*(a_1 | s)} \leq \exp(4\eta H(1 + \eta^{-1} \log(1/\alpha))). \quad (53)$$

Step D.3: Telescoping over the horizon (path-measure form). Since d_h^π is the marginal of the trajectory law, it does not factorize into per-stage policy ratios; the change of measure must therefore be performed at the level of the *trajectory law*. Write

$$\tau_h := (s_1, a_{1,1}, a_{1,2}, \dots, s_h, a_{h,1}, a_{h,2})$$

for the trajectory truncated at stage h , and let $\mathbb{P}_{\pi_1, \pi_2}^{(h)}$ denote its law under the joint policy (π_1, π_2) . In iterated-kernel notation,

$$\mathbb{P}_{\pi_1, \pi_2}^{(h)}(d\tau_h) = \rho(ds_1) \prod_{h'=1}^{h-1} \left[\pi_{h',1}(da_{h',1} | s_{h'}) \pi_{h',2}(da_{h',2} | s_{h'}) P_{h'}(ds_{h'+1} | s_{h'}, a_{h',1}, a_{h',2}) \right] \pi_{h,1}(da_{h,1} | s_h) \pi_{h,2}(da_{h,2} | s_h). \quad (54)$$

The two path laws $\mathbb{P}_{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}^{(h)}$ and $\mathbb{P}_{\pi_1^*, \pi_2^*}^{(h)}$ share ρ , the kernels $\{P_{h'}\}_{h' < h}$ and Player 2's policy, and their Player-1 components are Gibbs with respect to the same reference, hence have identical support at every (h', s) ; they are therefore mutually absolutely continuous, and by the chain rule the Radon–Nikodym derivative is the product of the differing (Player-1) per-stage policy ratios: $\mathbb{P}_{\pi_1^*, \hat{\pi}_2}^{(h)}$ -almost surely,

$$\frac{d\mathbb{P}_{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}^{(h)}(\tau_h)}{d\mathbb{P}_{\pi_1^*, \pi_2^*}^{(h)}(\tau_h)} = \prod_{h'=1}^h \frac{\pi_{h',1}^\dagger(\hat{\pi}_2)(a_{h',1} | s_{h'})}{\pi_{h',1}^*(a_{h',1} | s_{h'})} \leq \exp(4\eta H \cdot h \cdot (1 + \eta^{-1} \log(1/\alpha))),$$

by (53) applied to each factor. Thus the Radon–Nikodym derivative has a version uniformly bounded by the displayed right-hand side. Since $h \leq H$, that bound is at most

$$c_h := \exp(4\eta H^2(1 + \eta^{-1} \log(1/\alpha))).$$

Let $\Phi_h(\tau_h) := (s_h, a_{h,1}, a_{h,2})$, so that $d_h^{\pi_1, \pi_2} = (\Phi_h)_\# \mathbb{P}_{\pi_1, \pi_2}^{(h)}$. Push-forward preserves measure domination: for every measurable $B \subseteq \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2$,

$$d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}(B) = \mathbb{P}_{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}^{(h)}(\Phi_h^{-1}(B)) \leq c_h \mathbb{P}_{\pi_1^*, \hat{\pi}_2}^{(h)}(\Phi_h^{-1}(B)) = c_h d_h^{\pi_1^*, \hat{\pi}_2}(B).$$

This path-measure-then-push-forward device is left implicit in the remaining transfers.

Step D.4: Closing the Player 1 transfer under the coupled side condition. The condition (7) is exactly $4\eta H^2 \lambda \leq 1$. Hence

$$c_h = \exp(4\eta H^2 \lambda) \leq \exp(1) = e = C_{\text{trans}}.$$

This proves part (i) of Lemma C.5.

Step D.5: Symmetric argument for the Player 2 transfer. We now prove part (ii): under the same coupled condition $4\eta H^2 \lambda \leq 1$,

$$d_h^{\pi_1^*, \hat{\pi}_2} \preceq C_{\text{trans}} d_h^{\pi_1^*, \pi_2^*}.$$

The two distributions differ only in Player 2's policy ($\hat{\pi}_2$ vs. π_2^*), and the argument is symmetric to Steps D.1–D.4. Player 2's regularized stage policies admit Gibbs forms

$$\hat{\pi}_{h',2}(\cdot | s) = \text{Softmax}(-\eta \hat{Q}_{h',\hat{\pi}_1}(s, \cdot) + \log \pi_{\text{ref},h',2}(\cdot | s)), \quad \pi_{h',2}^*(\cdot | s) = \text{Softmax}(-\eta Q_{h',\pi_1^*}^*(s, \cdot) + \log \pi_{\text{ref},h',2}(\cdot | s)).$$

Both opponent-marginalized Q -vectors satisfy the same uniform bound $\|\hat{Q}_{h',\hat{\pi}_1}\|_\infty, \|Q_{h',\pi_1^*}^*\|_\infty \leq H(1 + \eta^{-1} \log(1/\alpha)) =: C_{\text{val}}$: Lemma C.2 gives this bound for Q^* uniformly over opponent policies, and the empirical $\hat{Q}_h$ inherits it by Assumption C.1. The logit difference is therefore bounded in ℓ_∞ by $2\eta H(1 + \eta^{-1} \log(1/\alpha))$, and (52) gives a per-step Player 2 ratio of at most $\exp(4\eta H(1 + \eta^{-1} \log(1/\alpha)))$. Telescoping over $h \leq H$ steps and applying Step D.4 yields $C_{\text{trans}} = e$ for part (ii).

Step D.6: Composited transfer (51). For any nonnegative f , importance reweighting using part (i) gives

$$\mathbb{E}_{(s,a) \sim d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}}[f(s, a)] = \mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \hat{\pi}_2}}[f(s, a) \cdot \frac{d d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}}{d d_h^{\pi_1^*, \hat{\pi}_2}}(s, a)] \leq C_{\text{trans}} \cdot \mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \hat{\pi}_2}}[f(s, a)],$$

and a second importance reweighting using part (ii) gives

$$\mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \hat{\pi}_2}}[f(s, a)] \leq C_{\text{trans}} \cdot \mathbb{E}_{(s,a) \sim d_h^{\pi_1^*, \pi_2^*}}[f(s, a)].$$

Composing yields the bound in (51).

Step D.7: Reverse directions, arbitrary reference-supported deviations, and cross-branch transfers. Every transfer in this appendix rests on one of two per-stage action-policy ratio bounds.

(a) *Regularized vs. regularized.* The per-step softmax-ratio bound (52), combined with the worst-case logit-difference bound of Step D.2, depends only on the uniform ℓ_∞ -bound $\|q\|_\infty \leq H(1 + \eta^{-1} \log(1/\alpha)) = H\lambda$ on the payoff vector q entering the Gibbs form of a regularized stage policy. It holds for $Q_{h,\hat{\pi}_2}^\dagger, Q_{h,\pi_2^*}^\dagger, \hat{Q}_{h,\hat{\pi}_1}$ and $Q_{h,\pi_1^*}^*$ and their Player-2-marginalized counterparts: Lemma C.2 gives $\|Q_{h,\nu}^\dagger\|_\infty \leq 1 + \|V_{h+1}^{\dagger,\nu}\|_\infty \leq H\lambda$ uniformly over opponent policies ν , and $\hat{Q}_h$ inherits the bound from Assumption C.1. Consequently

$$\frac{\pi_{h,i}(a | s)}{\pi'_{h,i}(a | s)} \leq \exp(4\eta H\lambda), \quad a \in \mathcal{A}_{h,i}^{\text{ref}}(s), \quad (55)$$

holds in *both* directions for any two regularized stage policies $\pi_{h,i}, \pi'_{h,i}$ of the same player i , built from the common reference $\pi_{\text{ref},h,i}(\cdot | s)$ and from payoff vectors of ℓ_∞ -norm at most $H\lambda$.

(b) *Arbitrary reference-supported deviation vs. regularized.* Bound (55) is *not* available when the deviating stage policy fails to be of Gibbs form, as in part (iv) below: a deterministic $\arg \max$ or branch selector produced by the $\max_{a_i}$ operators

of (81) and (82), or an arbitrary Player- i policy as quantified in Assumption 6.2. For these we invoke the reference-policy lower bound α directly. Let $\pi_{h,i}(\cdot | s) = \text{Softmax}(\eta q + \log \pi_{\text{ref},h,i}(\cdot | s))$ be a regularized stage policy with $\|q\|_\infty \leq H\lambda$ and normalizer $Z = \sum_a \pi_{\text{ref},h,i}(a | s) e^{\eta q(a)}$. Since $\pi_{\text{ref},h,i}(\cdot | s)$ is a probability vector, $Z \in [e^{-\eta H\lambda}, e^{\eta H\lambda}]$, so for every $a \in \mathcal{A}_{h,i}^{\text{ref}}(s)$,

$$\pi_{h,i}(a | s) = \frac{\pi_{\text{ref},h,i}(a | s) e^{\eta q(a)}}{Z} \geq \alpha e^{-2\eta H\lambda}.$$

Hence for *any* stage distribution $\varpi_{h,i}(\cdot | s)$ supported in $\mathcal{A}_{h,i}^{\text{ref}}(s)$ — Dirac masses included —

$$\frac{\varpi_{h,i}(a | s)}{\pi_{h,i}(a | s)} \leq \frac{e^{2\eta H\lambda}}{\alpha} = e^{2\eta H} \alpha^{-(2H+1)} \leq e^{4\eta H} \alpha^{-4H} = \exp(4\eta H\lambda), \quad (56)$$

where we used $\eta\lambda = \eta + \log(1/\alpha)$ and, for the last inequality, $2H + 1 \leq 4H$ together with $\alpha \leq 1$ (so $\alpha^{-(2H+1)} \leq \alpha^{-4H}$) and $H \geq 1$. The deviation-vs-regularized per-stage ratio therefore obeys the same bound (53) as the regularized-vs-regularized one, so Steps D.3 and D.4 go through with $\pi_1^\dagger(\hat{\pi}_2)$ replaced by an arbitrary $\varpi_1 \in \Pi_1^{\text{ref}}$ (resp. $\hat{\pi}_2$ by an arbitrary $\varpi_2 \in \Pi_2^{\text{ref}}$): the deviating law is supported inside the support of the regularized one, so absolute continuity holds in that direction, and the telescoped constant is again $\exp(4\eta H^2\lambda) \leq e = C_{\text{trans}}$ under (7). This proves the one-player transfer bounds in the “moreover” clause (50); composing at most one such transfer for each player gives the C_{trans}^2 bound for joint and cross-branch transfers.

Combining (a) and (b), telescoping as in Step D.3 and pushing forward to the step- h marginal yields the following measure-domination bounds:

- (i') (Reverse of (i)) $d_h^{\pi_1^*, \hat{\pi}_2} \preceq C_{\text{trans}} d_h^{\pi_1^\dagger(\hat{\pi}_2), \hat{\pi}_2}$.
- (ii') (Reverse of (ii)) $d_h^{\pi_1^*, \pi_2^*} \preceq C_{\text{trans}} d_h^{\pi_1^*, \hat{\pi}_2}$.
- (iii) (Player 1 transfer with π_2^* shared) $d_h^{\hat{\pi}_1, \pi_2^*} \preceq C_{\text{trans}} d_h^{\pi_1^*, \pi_2^*}$ and conversely $d_h^{\pi_1^*, \pi_2^*} \preceq C_{\text{trans}} d_h^{\hat{\pi}_1, \pi_2^*}$.
- (iv) (*Cross-branch trajectory transferred to a Π_{uni} proxy*) A trajectory generated by a *cross-branch* policy schedule — one in which Player 1 plays Nash on some segments and Player 2 plays Nash on the complementary segments, the remaining stage components being arbitrary elements of Π_1^{ref} or Π_2^{ref} , deterministic selectors included — is *not* natively a single multi-step policy in Π_{uni} . Choose as proxy a policy $\pi^{\text{prox}} \in \Pi_{\text{uni}}$ *both* of whose stage components are regularized, e.g. $\pi^{\text{prox}} = (\pi_1^*, \hat{\pi}_2) \in \{\pi_1^*\} \times \Pi_2$ or $\pi^{\text{prox}} = (\hat{\pi}_1, \pi_2^*) \in \Pi_1 \times \{\pi_2^*\}$; every stage component is then Gibbs with a payoff vector of ℓ_∞ -norm at most $H\lambda$, so every per-stage ratio below has a *regularized denominator*. The laws $\mathbb{P}_{\text{cross}}^{(h)}$ and $\mathbb{P}_{\pi^{\text{prox}}}^{(h)}$ of (54) share ρ and the kernels, so their Radon–Nikodym derivative is the product over the differing stages of the two players' per-stage policy ratios, each bounded by $\exp(4\eta H\lambda)$ by (55) (regularized component) or (56) (arbitrary reference-supported deviation). Each player contributes at most $h \leq H$ such factors, so by Step D.4 each player's cumulative product is at most C_{trans} ; pushing forward to the step- h marginal,

$$d_h^{\text{cross-branch}} \preceq C_{\text{trans}}^2 d_h^{\pi^{\text{prox}}}.$$

This shifts expectations from the cross-branch measure into the Π_{uni} -compatible concentrability class of Assumption 6.2 at cost C_{trans}^2 , absorbed into the prefactor of (87).

Throughout (b) the deviating stage components are supported in $\mathcal{A}_{h,i}^{\text{ref}}(s)$: the deviation classes Π_i in Assumption 6.2 are read as Π_i^{ref} , and the action maxima defining ξ_h^2 in (65) as maxima over $\mathcal{A}_{h,i}^{\text{ref}}(s)$. Neither reading loses anything: the maxima in ξ_h^2 arise from the Hölder step $\langle \Delta M_{h,s} \pi_{h,s}^*, \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle \leq \|\Delta M_{h,s} \pi_{h,s}^*\|_\infty \|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1$ producing (62) in the proof of Lemma 6.4. Because both policies are Gibbs with reference π_{ref} (Step D.1), the policy-difference vector $\hat{\pi}_{h,s} - \pi_{h,s}^*$ vanishes outside $\mathcal{A}_{h,i}^{\text{ref}}(s)$; hence the pairing uses only supported coordinates, and the ℓ_∞ -norm may be restricted without weakening the inequality. Replacing Π_i by Π_i^{ref} also shrinks the supremum in Assumption 6.2 and hence only *weakens* that assumption.

Step D.8: Extension to conditional measures (part (v)). The per-stage bounds (53), (55) and (56) make no reference to the law of the trajectory prefix. Running the Step D.3 argument on the law of the sub-trajectory $(s_{h'+1}, a_{h'+1,1}, a_{h'+1,2}, \dots, s_t, a_{t,1}, a_{t,2})$ — that is, on (54) with ρ replaced by $\delta_{s'}$ and the stage index shifted — and pushing forward to the step- t marginal produces the same C_{trans} -bound (or C_{trans}^2 for joint two-player transfers, as in (iv)), for every starting state s' and every sub-horizon of length at most H :

- (v) (Conditional-measure version of (i)–(iv)) For any state $s' \in \mathcal{S}$ and any $h' < t \leq H$, the conditional measure $d_t^{(\pi_1, \pi_2) | s_{h'+1}=s'}$ satisfies the same measure-domination bounds (i)–(iv) above; this version is invoked in Lemma 6.5, eqs. (38)–(39).

□

C.4. Proof of Lemma 6.4

Step 5: Local VI Stability. The proof proceeds by studying a *local variational inequality* (VI) at each time–state pair (h, s) , with the continuation value held fixed, so that the resulting operator is purely stagewise.

Local VI operator. Fix $(h, s) \in [H] \times \mathcal{S}$ and recall the reference support $\mathcal{A}_{h,i}^{\text{ref}}(s) := \{a \in \mathcal{A}_i : \pi_{\text{ref},h,i}(a | s) > 0\}$ together with the reference-supported policy classes Π_i^{ref} introduced in the preliminaries. All stagewise objects below live on the corresponding face of the product simplex,

$$\mathcal{X}_{h,s} := \Delta(\mathcal{A}_{h,1}^{\text{ref}}(s)) \times \Delta(\mathcal{A}_{h,2}^{\text{ref}}(s)),$$

and we let $\pi_{h,s} = (\pi_{h,s,1}, \pi_{h,s,2}) \in \mathcal{X}_{h,s}$ denote a pair of independent stage action distributions. Restricting to $\mathcal{X}_{h,s}$ is without loss: for any p charging an action outside $\mathcal{A}_{h,1}^{\text{ref}}(s)$ one has $\text{KL}(p \| \pi_{\text{ref},h,1}(\cdot | s)) = +\infty$, which drives the maximized objective (57) to $-\infty$ for Player 1, and symmetrically the Player 2 regularizer enters (57) with a $+$ sign in a minimization; so the stagewise optima over $\Delta(\mathcal{A}_1) \times \Delta(\mathcal{A}_2)$ and over $\mathcal{X}_{h,s}$ coincide. Throughout this subsection the product space carries the norm $\|\pi_{h,s}\|_1 := \|\pi_{h,s,1}\|_1 + \|\pi_{h,s,2}\|_1$, whose dual norm is $\|y\|_\infty = \max(\|y_1\|_\infty, \|y_2\|_\infty)$, each blockwise ℓ_∞ -norm being taken over the corresponding reference support $\mathcal{A}_{h,i}^{\text{ref}}(s)$. Two elementary consequences are used below: $\|\pi_{h,s,i} - \pi'_{h,s,i}\|_1 \leq \|\pi_{h,s} - \pi'_{h,s}\|_1$ for each single player i (the form invoked in Appendix C.5), and $\|\pi_{h,s,1} \otimes \pi_{h,s,2} - \pi'_{h,s,1} \otimes \pi'_{h,s,2}\|_1 \leq \|\pi_{h,s} - \pi'_{h,s}\|_1$, so the bound stated in Lemma 6.4 is valid whether $\|\hat{\pi}_h(\cdot | s) - \pi_h^*(\cdot | s)\|_1$ is read blockwise or as the ℓ_1 -distance between the induced joint action distributions. Given any payoff matrix $G \in \mathbb{R}^{|\mathcal{A}_1| \times |\mathcal{A}_2|}$, define the stagewise KL-regularized objective

$$J_{G,h,s}(\pi_{h,s,1}, \pi_{h,s,2}) := \mathbb{E}_{a \sim \pi_{h,s}}[G(a_1, a_2)] - \eta^{-1} \text{KL}(\pi_{h,s,1} \| \pi_{\text{ref},h,1}(\cdot | s)) + \eta^{-1} \text{KL}(\pi_{h,s,2} \| \pi_{\text{ref},h,2}(\cdot | s)). \quad (57)$$

We define the associated local VI operator (the gradient map of the regularized zero-sum game) by

$$F_{G,h,s}(\pi_{h,s}) := \begin{pmatrix} -\nabla_{\pi_1} J_{G,h,s}(\pi_{h,s}) \\ \nabla_{\pi_2} J_{G,h,s}(\pi_{h,s}) \end{pmatrix} = M_G \pi_{h,s} + \eta^{-1} \nabla \text{Reg}_{h,s}(\pi_{h,s}), \quad (58)$$

where M_G is the skew-symmetric matrix composed of blocks $-G$ and $G^\top$, so that $M_G \pi_{h,s} = (-G \pi_{h,s,2}, G^\top \pi_{h,s,1})$, and $\text{Reg}_{h,s}(\pi_{h,s}) := \text{KL}(\pi_{h,s,1} \| \pi_{\text{ref},h,1}(\cdot | s)) + \text{KL}(\pi_{h,s,2} \| \pi_{\text{ref},h,2}(\cdot | s))$ collects the KL regularization terms, so that

$$\nabla \text{Reg}_{h,s}(\pi_{h,s}) = \left(\log \frac{\pi_{h,s,1}}{\pi_{\text{ref},h,1}(\cdot | s)} + \mathbf{1}, \log \frac{\pi_{h,s,2}}{\pi_{\text{ref},h,2}(\cdot | s)} + \mathbf{1} \right).$$

Because $\nabla \text{Reg}_{h,s}$ diverges wherever a coordinate vanishes, $F_{G,h,s}$ is finite only on the relative interior $\text{ri } \mathcal{X}_{h,s}$; every policy at which we evaluate $F_{G,h,s}$ below is a Gibbs distribution with reference $\pi_{\text{ref},h,i}(\cdot | s)$, hence strictly positive on $\mathcal{A}_{h,i}^{\text{ref}}(s)$ and therefore an element of $\text{ri } \mathcal{X}_{h,s}$.

Define the continuation-augmented payoff matrix for the true game and the estimated game respectively:

$$\begin{aligned} G_{h,s}^*(a_1, a_2) &:= Q_h^{\pi^*, \pi^*}(s, a_1, a_2), \\ \hat{G}_{h,s}(a_1, a_2) &:= \hat{Q}_h(s, a_1, a_2). \end{aligned}$$

Both $\pi_{h,s}^*$ and $\hat{\pi}_{h,s}$ are saddle points of the corresponding stage objective *constrained* to $\mathcal{X}_{h,s}$. Because the normalization constraints $\langle \mathbf{1}, \pi_{h,s,i} \rangle = 1$ are active, each block of F is only guaranteed to be constant (a Lagrange multiplier times 1) at a solution, so the root equations $F_{G_{h,s}^*,h,s}(\pi_{h,s}^*) = 0$ and $F_{\hat{G}_{h,s},h,s}(\hat{\pi}_{h,s}) = 0$ are *false* in general. The correct stagewise first-order conditions are the two *variational inequalities*: for every $x = (x_1, x_2) \in \mathcal{X}_{h,s}$,

$$\langle F_{G_{h,s}^*,h,s}(\pi_{h,s}^*), x - \pi_{h,s}^* \rangle \geq 0, \quad (59)$$

$$\langle F_{\hat{G}_{h,s},h,s}(\hat{\pi}_{h,s}), x - \hat{\pi}_{h,s} \rangle \geq 0. \quad (60)$$

Indeed, $\pi_{h,s,1}^*$ maximizes the concave map $p \mapsto J_{G_{h,s}^*,h,s}(p, \pi_{h,s,2}^*)$ over $\Delta(\mathcal{A}_{h,1}^{\text{ref}}(s))$, whence $\langle \nabla_{\pi_1} J_{G_{h,s}^*,h,s}(\pi_{h,s}^*), x_1 - \pi_{h,s,1}^* \rangle \leq 0$. Symmetrically, $\pi_{h,s,2}^*$ minimizes the convex map $p \mapsto J_{G_{h,s}^*,h,s}(\pi_{h,s,1}^*, p)$, giving $\langle \nabla_{\pi_2} J_{G_{h,s}^*,h,s}(\pi_{h,s}^*), x_2 - \pi_{h,s,2}^* \rangle \geq 0$. Adding the two and using the sign pattern of (58) yields (59); the identical argument applied to $\hat{G}_{h,s}$ yields (60).

Strong Monotonicity and Stability. The KL-regularized operator satisfies a strong monotonicity property with respect to the ℓ_1 -norm fixed above. Specifically, for any G and any $\pi, \pi' \in \text{ri } \mathcal{X}_{h,s}$:

$$\langle F_{G,h,s}(\pi) - F_{G,h,s}(\pi'), \pi - \pi' \rangle \geq \frac{\eta^{-1}}{2} \|\pi - \pi'\|_1^2. \quad (61)$$

To verify (61), note that the bilinear part contributes $\langle M_G(\pi - \pi'), \pi - \pi' \rangle = 0$ by skew-symmetry of M_G , while the constant 1 terms of $\nabla \text{Reg}_{h,s}$ cancel in the difference. Hence

$$\langle F_{G,h,s}(\pi) - F_{G,h,s}(\pi'), \pi - \pi' \rangle = \eta^{-1} \sum_{i \in \{1,2\}} \langle \log(\pi_i / \pi'_i), \pi_i - \pi'_i \rangle = \eta^{-1} \sum_{i \in \{1,2\}} [\text{KL}(\pi_i \| \pi'_i) + \text{KL}(\pi'_i \| \pi_i)].$$

Two applications of Pinsker's inequality bound the right-hand side below by $\eta^{-1} \sum_i \|\pi_i - \pi'_i\|_1^2$, and $x^2 + y^2 \geq \frac{1}{2}(x + y)^2$ then gives $\frac{\eta^{-1}}{2} \|\pi - \pi'\|_1^2$, as claimed.

Applying (61) to $\pi = \hat{\pi}_{h,s}$ and $\pi' = \pi_{h,s}^*$ with the true operator F_{G^*} :

$$\begin{aligned} \frac{\eta^{-1}}{2} \|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1^2 &\leq \langle F_{G_{h,s}^*,h,s}(\hat{\pi}_{h,s}) - F_{G_{h,s}^*,h,s}(\pi_{h,s}^*), \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle \\ &\leq \langle F_{G_{h,s}^*,h,s}(\hat{\pi}_{h,s}), \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle \\ &\leq \langle F_{G_{h,s}^*,h,s}(\hat{\pi}_{h,s}) - F_{\hat{G}_{h,s},h,s}(\hat{\pi}_{h,s}), \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle, \end{aligned}$$

where the second inequality drops $-\langle F_{G_{h,s}^*,h,s}(\pi_{h,s}^*), \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle \leq 0$, which is (59) at the test point $x = \hat{\pi}_{h,s}$, and the third adds $\langle F_{\hat{G}_{h,s},h,s}(\hat{\pi}_{h,s}), \pi_{h,s}^* - \hat{\pi}_{h,s} \rangle \geq 0$, which is (60) at the test point $x = \pi_{h,s}^*$. The difference in operators is linear in the difference of payoff matrices. Let $\Delta M_{h,s} = M_{G^*} - M_{\hat{G}}$. Since $\Delta M_{h,s}$ is skew-symmetric, $\langle \Delta M_{h,s} v, v \rangle = 0$. Thus:

$$\text{RHS} = \langle \Delta M_{h,s} \hat{\pi}_{h,s}, \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle = \langle \Delta M_{h,s} \pi_{h,s}^*, \hat{\pi}_{h,s} - \pi_{h,s}^* \rangle.$$

Let $\mathcal{E}_{h,s} := \|\Delta M_{h,s} \pi_{h,s}^*\|_\infty$, the dual norm being the one fixed above. Recall that M_G has the skew-symmetric block form $\begin{pmatrix} 0 & -G \\ G^\top & 0 \end{pmatrix}$, so $\Delta M_{h,s} \pi_{h,s}^* = \begin{pmatrix} -\Delta G_{h,s} \pi_{h,s,2}^* \\ \Delta G_{h,s}^\top \pi_{h,s,1}^* \end{pmatrix}$, where $\Delta G_{h,s}(a_1, a_2) = Q_h^*(s, a_1, a_2) - \hat{Q}_h(s, a_1, a_2)$ is the payoff difference matching $\Delta M_{h,s} = M_{G^*} - M_{\hat{G}}$ (the overall sign is immaterial below, since only absolute values of the blocks enter). Hence

$$\mathcal{E}_{h,s} = \max \left(\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} |\mathbb{E}_{a_2 \sim \pi_{h,s,2}^*} [\hat{Q}_h(s, a_1, a_2) - Q_h^*(s, a_1, a_2)]|, \max_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} |\mathbb{E}_{a_1 \sim \pi_{h,s,1}^*} [\hat{Q}_h(s, a_1, a_2) - Q_h^*(s, a_1, a_2)]| \right). \quad (62)$$

The vector $\hat{\pi}_{h,s} - \pi_{h,s}^*$ is supported on $\mathcal{A}_{h,1}^{\text{ref}}(s) \times \mathcal{A}_{h,2}^{\text{ref}}(s)$, so no coordinate outside the reference support is ever paired against it in the Hölder step below.

Using Hölder's inequality with (62),

$$\frac{\eta^{-1}}{2} \|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1^2 \leq \mathcal{E}_{h,s} \|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1,$$

and dividing by the norm yields the pointwise stability bound

$$\|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1 \leq 2\eta \mathcal{E}_{h,s}. \quad (63)$$

Squaring (63) and using the identity $(\max(x, y))^2 = \max(x^2, y^2)$ for $x, y \geq 0$ together with Jensen's inequality on each branch of (62) yields the *squared stability bound*

$$\|\hat{\pi}_{h,s} - \pi_{h,s}^*\|_1^2 \leq 4\eta^2 \cdot \xi_h^2(s), \quad (64)$$

where the *single-player Nash-averaged squared Q-error*

$$\xi_h^2(s) := \max \left(\max_{a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)} \mathbb{E}_{a_2 \sim \pi_{h,2}^*(\cdot|s)} [(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)], \max_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} \mathbb{E}_{a_1 \sim \pi_{h,1}^*(\cdot|s)} [(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)] \right) \quad (65)$$

is bounded above by $\|\hat{Q}_h(s) - Q_h^*(s)\|_\infty^2$ but lies in Π_{uni} -form when the outer state expectation is taken: each branch of the max corresponds to the worst Player- i deterministic deviation in Π_i^{ref} paired with the opponent's Nash policy, so $\xi_h^2(s)$ is exactly the quantity controlled by Assumption 6.2 via (72). This is the same quantity as (8), whose action maxima are likewise read over the reference supports.

C.5. Proof of Lemma 6.6 (Evaluation Gap via Squared Error)

Bounding Gap 2 via Logits Decomposition. Recall the exact identity established in Step 1 (Eq. (23)):

$$\text{Gap 2} = \eta^{-1} \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}} [\text{KL}(\hat{\pi}_{h,2}(\cdot|s) \parallel \pi_{h,2}^*(\cdot|s))]. \quad (66)$$

The Player-2 stage policies are the regularized best responses in the empirical and true games, respectively. By the saddle-point first-order condition with payoff Q and Player-1 marginalization,

$$\hat{\pi}_{h,2}(a_2|s) \propto \pi_{h,2}^{\text{ref}}(a_2|s) \exp(-\eta \hat{Q}_{h,\hat{\pi}_1}(s, a_2)), \quad \pi_{h,2}^*(a_2|s) \propto \pi_{h,2}^{\text{ref}}(a_2|s) \exp(-\eta Q_{h,\pi_1^*}^\dagger(s, a_2)),$$

where $\hat{Q}_{h,\hat{\pi}_1}(s, a_2) := \mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}}[\hat{Q}_h(s, a_1, a_2)]$ and $Q_{h,\pi_1^*}^\dagger(s, a_2) := \mathbb{E}_{a_1 \sim \pi_{h,1}^*}[Q_h^*(s, a_1, a_2)]$. The corresponding logits are

$$\hat{z}_h(a_2) := -\eta \hat{Q}_{h,\hat{\pi}_1}(s, a_2) + \log \pi_{h,2}^{\text{ref}}(a_2|s), \quad z_h^*(a_2) := -\eta Q_{h,\pi_1^*}^\dagger(s, a_2) + \log \pi_{h,2}^{\text{ref}}(a_2|s), \quad a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s).$$

Both $\hat{\pi}_{h,2}(\cdot|s)$ and $\pi_{h,2}^*(\cdot|s)$ are supported on $\mathcal{A}_{h,2}^{\text{ref}}(s)$ and strictly positive there, where $\log \pi_{h,2}^{\text{ref}}(a_2|s)$ is finite; every ℓ_∞ -norm of a logit vector in this subsection is taken over $\mathcal{A}_{h,2}^{\text{ref}}(s)$. By the smoothness bound (28),

$$\text{KL}(\hat{\pi}_{h,2}(\cdot|s) \parallel \pi_{h,2}^*(\cdot|s)) \leq \frac{1}{2} \|\hat{z}_h - z_h^*\|_\infty^2.$$

Decomposing the logit difference (reference-policy terms cancel),

$$\begin{aligned} \hat{z}_h(a_2) - z_h^*(a_2) &= -\eta \cdot \mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}}[\hat{Q}_h(s, a_1, a_2) - Q_h^*(s, a_1, a_2)] \quad (\text{Estimation term}) \\ &\quad - \eta \cdot \mathbb{E}_{a_1 \sim (\hat{\pi}_{h,1} - \pi_{h,1}^*)}[\hat{Q}_h(s, a_1, a_2)] \quad (\text{Policy deviation term}). \end{aligned} \quad (67)$$

Per-state Player-1 action-transfer step (one-step instance of Lemma C.5). For every fixed (h, s) , both $\hat{\pi}_{h,1}(\cdot|s)$ and $\pi_{h,1}^*(\cdot|s)$ are Gibbs softmax distributions on $\mathcal{A}_{h,1}^{\text{ref}}(s)$ sharing the reference logit $\log \pi_{h,1}^{\text{ref}}(\cdot|s)$:

$$\hat{\pi}_{h,1}(a_1|s) \propto \pi_{h,1}^{\text{ref}}(a_1|s) \exp(\eta \hat{Q}_{h,\hat{\pi}_2}(s, a_1)), \quad \pi_{h,1}^*(a_1|s) \propto \pi_{h,1}^{\text{ref}}(a_1|s) \exp(\eta Q_{h,\pi_2^*}^*(s, a_1)),$$

where $\hat{Q}_{h,\hat{\pi}_2}(s, a_1) := \mathbb{E}_{a_2 \sim \hat{\pi}_{h,2}}[\hat{Q}_h(s, a_1, a_2)]$ and $Q_{h,\pi_2^*}^*(s, a_1) := \mathbb{E}_{a_2 \sim \pi_{h,2}^*}[Q_h^*(s, a_1, a_2)]$. The reference logit cancels in the difference, which is therefore bounded in ℓ_∞ over $\mathcal{A}_{h,1}^{\text{ref}}(s)$ by $\eta(\|\hat{Q}_h\|_\infty + \|Q_h^*\|_\infty) \leq 2\eta C_{\text{val}} = 2\eta H\lambda$, using $\|Q_h^*\|_\infty \leq H\lambda$ from Lemma C.2 and $\|\hat{Q}_h\|_\infty \leq C_{\text{val}} = H\lambda$ from Assumption C.1, with $\lambda := 1 + \eta^{-1} \log(1/\alpha)$. By the

per-step softmax-ratio bound (52) (Steps D.1–D.2 of Appendix C.3 applied at a single state-step), the per-state action ratio satisfies

$$\frac{\hat{\pi}_{h,1}(a_1|s)}{\pi_{h,1}^*(a_1|s)} \leq \exp(4\eta H\lambda) \leq e = C_{\text{trans}} \quad \text{under (7),}$$

for every $a_1 \in \mathcal{A}_{h,1}^{\text{ref}}(s)$. Indeed, (7) gives $4\eta H\lambda \leq 1/H \leq 1$. Consequently, for any nonnegative function $g(a_1, a_2)$ on $\mathcal{A}_1 \times \mathcal{A}_2$ and every a_2 ,

$$\mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}(\cdot|s)}[g(a_1, a_2)] \leq C_{\text{trans}} \cdot \mathbb{E}_{a_1 \sim \pi_{h,1}^*(\cdot|s)}[g(a_1, a_2)], \quad (68)$$

both expectations being over $\mathcal{A}_{h,1}^{\text{ref}}(s)$.

Bounding the estimation term via Jensen + per-state transfer (direct to ξ^2). Applying Jensen to the squared estimation term, then the per-state transfer (68) with $g(a_1, a_2) = (\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)$,

$$\begin{aligned} \sup_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} (\mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}}[\hat{Q}_h - Q_h^*](s, a_1, a_2))^2 &\leq \sup_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} \mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}}[(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)] \\ &\leq C_{\text{trans}} \cdot \sup_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} \mathbb{E}_{a_1 \sim \pi_{h,1}^*}[(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)] \\ &\leq C_{\text{trans}} \cdot \xi_h^2(s), \end{aligned} \quad (69)$$

where the last inequality uses the second branch of the definition of $\xi_h^2(s)$ (65).

Bounding the policy-deviation term via squared stability. For the policy-deviation term, Hölder's inequality with $\|Q_h^*\|_\infty \leq H(1 + \eta^{-1} \log(1/\alpha)) = H\lambda =: L$ (Lemma C.2) gives

$$\sup_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} (\mathbb{E}_{a_1 \sim (\hat{\pi}_{h,1} - \pi_{h,1}^*)}[Q_h^*(s, a_1, a_2)])^2 \leq L^2 \cdot \|\hat{\pi}_{h,1}(\cdot|s) - \pi_{h,1}^*(\cdot|s)\|_1^2,$$

the signed measure $\hat{\pi}_{h,1} - \pi_{h,1}^*$ being supported on $\mathcal{A}_{h,1}^{\text{ref}}(s)$. Applying the squared stability bound (64), which by the blockwise domination of Appendix C.4 also controls $\|\hat{\pi}_{h,1}(\cdot|s) - \pi_{h,1}^*(\cdot|s)\|_1$, we obtain

$$\sup_{a_2 \in \mathcal{A}_{h,2}^{\text{ref}}(s)} (\mathbb{E}_{a_1 \sim (\hat{\pi}_{h,1} - \pi_{h,1}^*)}[Q_h^*(s, a_1, a_2)])^2 \leq L^2 \cdot 4\eta^2 \xi_h^2(s) = 4\eta^2 L^2 \xi_h^2(s). \quad (70)$$

Combining. By $(a + b)^2 \leq 2a^2 + 2b^2$ applied to the logit decomposition (67), then (69) and (70),

$$\|\hat{z}_h - z_h^*\|_\infty^2 \leq 2\eta^2 \cdot C_{\text{trans}} \cdot \xi_h^2(s) + 2\eta^2 \cdot 4\eta^2 L^2 \cdot \xi_h^2(s) = (2C_{\text{trans}} \eta^2 + 8\eta^4 L^2) \xi_h^2(s).$$

Under (7) we have $\eta^2 H^2 \leq 1/16$, so $8\eta^4 L^2 = 8\eta^4 H^2 \lambda^2 \leq \eta^2 \lambda^2 / 2$. Hence

$$\text{KL}(\hat{\pi}_{h,2}(\cdot|s) \parallel \pi_{h,2}^*(\cdot|s)) \leq \frac{1}{2} \|\hat{z}_h - z_h^*\|_\infty^2 \leq (C_{\text{trans}} + \frac{1}{4} \lambda^2) \eta^2 \xi_h^2(s).$$

Plugging into (66),

$$\text{Gap 2} \leq \mathcal{O}(\eta(C_{\text{trans}} + \lambda^2)) \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \hat{\pi}_2}}[\xi_h^2(s)]. \quad (71)$$

This proves Lemma 6.6, with the $\mathcal{O}(\eta)$ leading constant.

C.6. Proof of Theorem 6.3 (Main Statistical Bound)

We start from the bounds on Gap 1 and Gap 2 derived in Lemma 6.5 (Eq. (49)) and Lemma 6.6 (Eq. (71)).

Unilateral Deviation and Concentrability Definitions. In the analyses of Gap 1 and Gap 2, we encounter trajectory distributions in which one player follows the Nash equilibrium strategy π^* throughout while the other follows a deviation policy. The set of *Unilateral Deviation Policies* (Assumption 6.2) is

$$\Pi_{\text{uni}} := (\{\pi_1^*\} \times \Pi_2) \cup (\Pi_1 \times \{\pi_2^*\}).$$

Here, as everywhere in the paper, Π_i denotes the reference-supported class Π_i^{ref} of (2), and the maxima over actions in ξ_h^2 (8) are read over the reference supports $\mathcal{A}_{h,i}^{\text{ref}}(s)$; the restriction only shrinks the supremum in Assumption 6.2, so C_{uni} cannot increase. The distributions encountered in our analysis (e.g., $d_h^{\pi_1^*, \hat{\pi}_2}$ and $d_h^{\hat{\pi}_1, \pi_2^*}$) are induced by policies in Π_{uni} . By the unilateral D^2 -divergence concentrability of Assumption 6.2, for every $h \in [H]$, every $i \in \{1, 2\}$, and every $\pi_i \in \Pi_i$,

$$\mathbb{E}_{(s, a_1, a_2) \sim d_h^{\pi_i \times \pi_{-i}^*}} [D_{\mathcal{Q}}^2((s, a_1, a_2); \mu_h)] \leq C_{\text{uni}}.$$

As a direct consequence of Definition 6.1 and Assumption 6.2, for any unilateral policy $\pi' \in \Pi_{\text{uni}}$ of the form (π_i, π_{-i}^*) and any $Q_1, Q_2 \in \mathcal{Q}$,

$$\mathbb{E}_{(s, a_1, a_2) \sim d_h^{\pi'}} [(Q_1 - Q_2)^2] \leq \mathbb{E}_{(s, a_1, a_2) \sim d_h^{\pi'}} [D_{\mathcal{Q}}^2] \cdot \mathbb{E}_{\mu_h} [(Q_1 - Q_2)^2] \leq C_{\text{uni}} \cdot \mathbb{E}_{\mu_h} [(Q_1 - Q_2)^2]. \quad (72)$$

Combining Gap 1 and Gap 2 into a single single-player Nash-averaged Q -error sum. By Lemmas 6.5 and 6.6, Gap 1 and Gap 2 are bounded by the same cumulative single-player Nash-averaged squared Q -estimation error $\xi_h^2(s)$ of (65), with prefactors respectively cubic and linear in η (throughout, $\lambda := 1 + \eta^{-1} \log(1/\alpha)$):

$$\text{Term A}(\hat{\pi}) = \text{Gap 1} + \text{Gap 2} \leq \mathcal{O}(\eta(C_{\text{trans}} + \lambda^2) + C_{\text{trans}}^2 \eta^3 H^4 \lambda^2) \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\text{uni}}} [\xi_h^2(s)], \quad (73)$$

where $\text{Term A}(\hat{\pi}) = J(\pi_1^\dagger, \hat{\pi}_2) - J(\pi_1^*, \pi_2^*)$ is the Player-1 half of the duality gap. The sum Gap 1 + Gap 2 is Term A alone and *not* the full gap: the Player-2 half $\text{Term B}(\hat{\pi}) = J(\pi_1^*, \pi_2^*) - J(\hat{\pi}_1, \pi_2^\dagger)$ is Term A of the reflected game of Appendix D.9 and therefore obeys the identical bound with the same constants. Consequently

$$\text{Gap}(\hat{\pi}) = \text{Term A}(\hat{\pi}) + \text{Term B}(\hat{\pi}) \leq 2 \times (\text{the Term-A bound of (73)}),$$

and the factor 2 is absorbed into the universal constant in everything that follows. Here $\mathbb{E}_{s \sim d_h^{\text{uni}}}[\cdot] := \sup_{\pi' \in \Pi_{\text{uni}}} \mathbb{E}_{s \sim d_h^{\pi'}}[\cdot]$ is the worst unilateral state-marginal at step h ; it dominates both concrete marginals occurring in the two lemmas, $d_h^{\pi_1^*, \hat{\pi}_2}$ and $d_h^{\hat{\pi}_1, \pi_2^*}$, as well as the $\sup_{\pi' \in \Pi_{\text{uni}}}$ already present in Lemma 6.5.

Final Bound via Recursive Bellman Error Unrolling. We now control the cumulative error $\sum_h \mathbb{E}_{s \sim d_h^{\text{uni}}} [\xi_h^2(s)]$ of (73) via Bellman unrolling and unilateral D^2 -divergence concentrability.

Let $\Delta Q_h(s) := \|\hat{Q}_h(s, \cdot, \cdot) - Q_h^*(s, \cdot, \cdot)\|_\infty$ denote the maximal joint-action Q -difference at state s ; by definition $\xi_h^2(s) \leq \Delta Q_h(s)^2$ pointwise.

Step 1: Recursive Unrolling via Bellman Error. We decompose the Q -value difference using the Bellman backup operator. Let $\mathcal{T}_h$ be the operator such that $(\mathcal{T}_h f)(s, a) = r_h^*(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [f(s')]$; it is affine and involves no maximization. Note that $Q_h^* = \mathcal{T}_h V_{h+1}^*$. We define the pointwise *Bellman regression error* against the estimated next-step value $\hat{V}_{h+1}$ as:

$$\mathcal{E}_h(s, a) := |\hat{Q}_h(s, a) - (\mathcal{T}_h \hat{V}_{h+1})(s, a)|.$$

Using the triangle inequality, we can bound the total estimation error:

$$\begin{aligned} |\hat{Q}_h(s, a) - Q_h^*(s, a)| &= |\hat{Q}_h(s, a) - \mathcal{T}_h \hat{V}_{h+1}(s, a) + \mathcal{T}_h \hat{V}_{h+1}(s, a) - \mathcal{T}_h V_{h+1}^*(s, a)| \\ &\leq \underbrace{|\hat{Q}_h(s, a) - (\mathcal{T}_h \hat{V}_{h+1})(s, a)|}_{\text{Regression Error } \mathcal{E}_h(s, a)} + \underbrace{|(\mathcal{T}_h \hat{V}_{h+1})(s, a) - (\mathcal{T}_h V_{h+1}^*)(s, a)|}_{\text{Propagation Error}}. \end{aligned}$$

The propagation term is bounded by the expected value difference:

$$\begin{aligned} |(\mathcal{T}_h \hat{V}_{h+1})(s, a) - (\mathcal{T}_h V_{h+1}^*)(s, a)| &= |\mathbb{E}_{s' \sim P_h(\cdot|s, a)}[\hat{V}_{h+1}(s') - V_{h+1}^*(s')]| \\ &\leq \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[|\hat{V}_{h+1}(s') - V_{h+1}^*(s')|]. \end{aligned}$$

For $0 \leq h \leq H-1$ define

$$W_{h+1}^2(s) := \max \left\{ \mathbb{E}_{(a_1, a_2) \sim \pi_1^* \times \pi_2}[(\hat{Q}_{h+1} - Q_{h+1}^*)^2(s, a_1, a_2)], \mathbb{E}_{(a_1, a_2) \sim \hat{\pi}_1 \times \pi_2^*}[(\hat{Q}_{h+1} - Q_{h+1}^*)^2(s, a_1, a_2)] \right\}, \quad (74)$$

where both action laws use the stage policies at step $h+1$, together with the terminal convention

$$W_{H+1}^2 \equiv 0, \quad (75)$$

which is consistent with $\hat{V}_{H+1} = V_{H+1}^* \equiv 0$. Each term in the max has *exactly one* player at Nash, with the other player at the corresponding learned policy; both joint distributions therefore lie in Π_{uni} .

Lemma C.6 (Saddle-point V-non-expansion). *For every $s \in \mathcal{S}$ and $h \in [H]$,*

$$(\hat{V}_{h+1}(s) - V_{h+1}^*(s))^2 \leq W_{h+1}^2(s). \quad (76)$$

Proof. Let $J_s(\pi_1, \pi_2; Q) := \mathbb{E}_{a \sim \pi_1 \times \pi_2}[Q(s, a)] - \eta^{-1} \text{KL}(\pi_1 \| \pi_1^{\text{ref}}) + \eta^{-1} \text{KL}(\pi_2 \| \pi_2^{\text{ref}})$ denote the stage-game objective. The regularized minimax equilibrium $(\hat{\pi}_1, \hat{\pi}_2)$ for $\hat{Q}$ and (π_1^*, π_2^*) for Q^* satisfy the saddle-point inequalities

$$J_s(\pi_1, \hat{\pi}_2; \hat{Q}) \leq J_s(\hat{\pi}_1, \hat{\pi}_2; \hat{Q}) \leq J_s(\hat{\pi}_1, \pi_2; \hat{Q}), \quad J_s(\pi_1, \pi_2^*; Q^*) \leq J_s(\pi_1^*, \pi_2^*; Q^*) \leq J_s(\pi_1^*, \pi_2; Q^*),$$

for all (π_1, π_2) . Setting $\pi_1 = \pi_1^*$ in the first chain and $\pi_2 = \hat{\pi}_2$ in the second, and noting $\hat{V}_{h+1}(s) = J_s(\hat{\pi}_1, \hat{\pi}_2; \hat{Q})$, $V_{h+1}^*(s) = J_s(\pi_1^*, \pi_2^*; Q^*)$, we obtain

$$\hat{V}_{h+1}(s) - V_{h+1}^*(s) \geq J_s(\pi_1^*, \hat{\pi}_2; \hat{Q}) - J_s(\pi_1^*, \hat{\pi}_2; Q^*) = \mathbb{E}_{(a_1, a_2) \sim \pi_1^* \times \hat{\pi}_2}[\hat{Q}_{h+1}(s, a_1, a_2) - Q_{h+1}^*(s, a_1, a_2)],$$

where the regularization terms cancel because the policy pair $(\pi_1^*, \hat{\pi}_2)$ is identical on both sides. Symmetrically, taking $\pi_1 = \hat{\pi}_1$ and $\pi_2 = \pi_2^*$ yields

$$\hat{V}_{h+1}(s) - V_{h+1}^*(s) \leq J_s(\hat{\pi}_1, \pi_2^*; \hat{Q}) - J_s(\hat{\pi}_1, \pi_2^*; Q^*) = \mathbb{E}_{(a_1, a_2) \sim \hat{\pi}_1 \times \pi_2^*}[\hat{Q}_{h+1}(s, a_1, a_2) - Q_{h+1}^*(s, a_1, a_2)].$$

Combining the two and squaring,

$$(\hat{V}_{h+1}(s) - V_{h+1}^*(s))^2 \leq \max \left((\mathbb{E}_{\pi_1^* \times \hat{\pi}_2}[\hat{Q} - Q^*])^2, (\mathbb{E}_{\hat{\pi}_1 \times \pi_2^*}[\hat{Q} - Q^*])^2 \right).$$

Jensen's inequality applied to each term gives the bound (76). $\square$

Scope of Lemma C.6. The proof uses that $\hat{V}_{h+1}(s)$ is the *exact saddle value* of the stage game with payoff $\hat{Q}_{h+1}(s, \cdot, \cdot)$, which is how Algorithm 1 defines it. It does *not* apply to the value backed up by Algorithm 2, whose stage value is the regularized objective at the last mirror-descent iterate rather than at a saddle point; that algorithm is analysed in Appendix D.

The branch selector. The maximum in (74) is attained at a state-dependent index. For each $h \in [H]$ let

$$i_h(s) \in \arg \max_{i \in \{1, 2\}} \mathbb{E}_{(a_1, a_2) \sim \pi_{h,i}^* \times \hat{\pi}_{h,-i}}[(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)], \quad (77)$$

with ties broken in favour of $i = 1$. Both arguments of the $\arg \max$ are measurable functions of s (the action sets $\mathcal{A}_1, \mathcal{A}_2$ are finite), so $i_h : \mathcal{S} \rightarrow \{1, 2\}$ is measurable. Define the associated *switching* stage policy pair $\sigma_h(\cdot|s) := \sigma_{h,1}(\cdot|s) \times \sigma_{h,2}(\cdot|s)$ by

$$\sigma_{h, i_h(s)}(\cdot|s) := \pi_{h, i_h(s)}^*(\cdot|s), \quad \sigma_{h, -i_h(s)}(\cdot|s) := \hat{\pi}_{h, -i_h(s)}(\cdot|s).$$

Then $\sigma = (\sigma_1, \sigma_2)$ is a bona fide product-form Markov joint policy and, by construction,

$$W_h^2(s) = \mathbb{E}_{(a_1, a_2) \sim \sigma_h(\cdot|s)}[(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2)] \quad \text{for every } s \in \mathcal{S}. \quad (78)$$

In general $\sigma \notin \Pi_{\text{uni}}$: the *identity* of the player who is at Nash may change from state to state. We therefore introduce the *switching* analysis class

$$\Pi^{\text{sw}} := \{ \varpi = (\varpi_1, \varpi_2) : \text{for all } (h, s) \text{ there is } i \in \{1, 2\} \text{ with } \varpi_{h,i}(\cdot|s) = \pi_{h,i}^*(\cdot|s) \text{ and } \text{supp } \varpi_{h,-i}(\cdot|s) \subseteq \text{supp } \pi_{h,-i}^{\text{ref}}(\cdot|s) \}, \quad (79)$$

which contains σ and contains Π_{uni} (choose the same i at every (h, s)). Π^{sw} is an intermediate object generated by the proof: it is never assumed to be covered by the offline dataset, and Assumption 6.2 is never invoked on it; Lemma C.7 transfers it back to Π_{uni} once, at the constant cost C_{trans} .

The pointwise squared recursion. Substituting Lemma C.6 into the propagation-error bound and using Jensen's inequality $(\mathbb{E}_{s'}[\|\hat{V} - V^*\|])^2 \leq \mathbb{E}_{s'}[(\hat{V} - V^*)^2] \leq \mathbb{E}_{s'}[W_{h+1}^2(s')]$, we obtain a pointwise squared inequality. It is applied H times in succession, so the coefficient of the propagated term must be $1 + \mathcal{O}(1/H)$ rather than 2: iterating a coefficient 2 compounds to 2^H , and combining it with a max-to-sum expansion to 4^H . We therefore use the weighted Young inequality

$$(x + y)^2 \leq (1 + H)x^2 + (1 + \frac{1}{H})y^2, \quad x, y \geq 0,$$

the case $\beta = 1/H$ of $(x + y)^2 \leq (1 + \beta^{-1})x^2 + (1 + \beta)y^2$, whose propagation coefficient obeys $(1 + 1/H)^H \leq e$. This gives

$$(\hat{Q}_h - Q_h^*)^2(s, a_1, a_2) \leq (H + 1)\mathcal{E}_h^2(s, a_1, a_2) + (1 + \frac{1}{H})\mathbb{E}_{s' \sim P_h(\cdot|s, a_1, a_2)}[W_{h+1}^2(s')], \quad (80)$$

valid for every (s, a_1, a_2) and every $h \in [H]$ (at $h = H$ the second term vanishes by (75)). To obtain a state-only bound, we apply $\max_{a_i} \mathbb{E}_{a_{-i} \sim \pi_{h,-i}^*(\cdot|s)}[\cdot]$ to both sides of (80) (for each $i \in \{1, 2\}$) and take the max over i , using that the maximum of a sum is at most the sum of the maxima. This yields, for the single-player Nash-averaged squared error $\xi_h^2(s)$ defined in (65),

$$\xi_h^2(s) \leq (H + 1)\mathcal{Z}_{h,s}^2 + (1 + \frac{1}{H}) \max_{i \in \{1, 2\}} \sup_{\nu_i \in \Pi_i} \mathbb{E}_{(a_1, a_2) \sim \nu_i \times \pi_{h,-i}^*}[\mathbb{E}_{s' \sim P_h(\cdot|s, a_1, a_2)}[W_{h+1}^2(s')]], \quad (81)$$

where $\mathcal{Z}_{h,s}^2 := \max_i \sup_{\nu_i \in \Pi_i} \mathbb{E}_{\nu_i \times \pi_{h,-i}^*}[\mathcal{E}_h^2(s, a_1, a_2)]$ is the single-player Nash-averaged squared Bellman regression error of (85) and $\Pi_i = \Delta(\text{supp } \pi_{h,i}^{\text{ref}}(\cdot|s))$. Each branch of the max in (81) pairs a single-player deviation ν_i with the opponent at Nash, so the step- h joint distribution $\nu_i \times \pi_{h,-i}^*$ lies in Π_{uni} .

Taking instead the joint-action supremum in (80) and setting $\zeta_{h,s}^2 := \sup_a \mathcal{E}_h^2(s, a)$ gives the joint-action form

$$\Delta Q_h(s)^2 \leq (H + 1)\zeta_{h,s}^2 + (1 + \frac{1}{H}) \sup_a \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[W_{h+1}^2(s')], \quad (82)$$

which satisfies $\xi_h^2 \leq \Delta Q_h^2$ and $\mathcal{Z}_{h,s}^2 \leq \zeta_{h,s}^2$ pointwise. The supremum in (82) is over the *joint* action, i.e. a simultaneous deterministic deviation by both players; it is therefore *not* of Π_{uni} form, and in particular it is not dominated by $\mathcal{Z}^2$. We record (82) for completeness only and route the final bound through the unilateral recursion (81).

Step 2: Recursive unrolling with W^2 propagation (weighted Young, single trajectory measure). We unroll the propagation term by iterating (80) inside W^2 . By (78), W_h^2 is exactly an expectation under the branch-selector stage policy σ_h , so one level of the recursion reads

$$\begin{aligned} W_h^2(s) &= \mathbb{E}_{a \sim \sigma_h(\cdot|s)}[(\hat{Q}_h - Q_h^*)^2(s, a)] \\ &\leq (H + 1)\mathbb{E}_{a \sim \sigma_h(\cdot|s)}[\mathcal{E}_h^2(s, a)] + (1 + \frac{1}{H})\mathbb{E}_{a \sim \sigma_h(\cdot|s)}\mathbb{E}_{s' \sim P_h(\cdot|s, a)}[W_{h+1}^2(s')] \\ &\leq (H + 1)\mathcal{Z}_{h,s}^2 + (1 + \frac{1}{H})\mathbb{E}_{a \sim \sigma_h(\cdot|s)}\mathbb{E}_{s' \sim P_h(\cdot|s, a)}[W_{h+1}^2(s')]. \end{aligned} \quad (83)$$

The last step holds by the definition (85) of $\mathcal{Z}_{h,s}^2$: writing $i = i_h(s)$ we have $\sigma_h(\cdot|s) = \pi_{h,i}^* \times \hat{\pi}_{h,-i}$, i.e. Player i at Nash and Player $-i$ at the reference-supported stage policy $\hat{\pi}_{h,-i}$, which is an admissible choice of the deviating ν_{-i} in (85). Iterating (83) from level h down to H with the terminal convention (75), and using $(1 + 1/H)^{t-h} \leq (1 + 1/H)^H \leq e$ for $h \leq t \leq H$, we obtain the unrolled cumulative bound

$$W_h^2(s) \leq e(H + 1) \sum_{t=h}^H \mathbb{E}_{s_t \sim d_t^\sigma(\cdot|s_h=s)}[\mathcal{Z}_{t,s_t}^2], \quad (84)$$

where $d_t^\sigma(\cdot \mid s_h = s)$ denotes the law of s_t in the Markov chain started at $s_h = s$ and driven by σ and P .

Define the *single-player squared Bellman regression error*

$$\mathcal{Z}_{t,s}^2 := \max_{i \in \{1,2\}} \sup_{\nu_i \in \Pi_i} \mathbb{E}_{(a_1, a_2) \sim \nu_i \times \pi_{t,-i}^*} [(\hat{Q}_t(s, a_1, a_2) - \mathcal{T}_t \hat{V}_{t+1}(s, a_1, a_2))^2]. \quad (85)$$

Combining (81) with (84) applied at level $h+1$, and using $\max\{H+1, (1 + \frac{1}{H})e(H+1)\} \leq 2e(H+1)$, we obtain the *single-player unrolled bound*

$$\xi_h^2(s) \leq 2e(H+1) \sup_{\varpi \in \Pi^{\text{sw}}} \mathbb{E}_{\tau \sim \varpi(\cdot \mid s_h = s)} \left[\sum_{t=h}^H \mathcal{Z}_{t,s_t}^2 \right], \quad (86)$$

and, in the same way from (82), the joint-action bound

$$\Delta Q_h(s)^2 \leq 2e(H+1) \left(\zeta_{h,s}^2 + \sup_a \sup_{\varpi \in \Pi^{\text{sw}}} \mathbb{E}_{s_{h+1} \sim P_h(\cdot \mid s, a)} \mathbb{E}_{\tau \sim \varpi(\cdot \mid s_{h+1})} \left[\sum_{t=h+1}^H \mathcal{Z}_{t,s_t}^2 \right] \right). \quad (87)$$

In (86) the trajectory $\tau = (s_h, a_h, s_{h+1}, \dots)$ is generated by the schedule that plays, at step h , the unilateral deviation $\nu_i \times \pi_{h,-i}^*$ realising the outer maximum of (81), and, at steps $h+1, h+2, \dots$, the branch selector σ . At every step exactly one player is at Nash, so this schedule belongs to Π^{sw} ; but because the identity of that player varies with the state, it does *not* belong to Π_{uni} . Equation (87) is recorded for reference only: its step- h deviation is two-sided and therefore falls outside Π_{uni} -compatible coverage, and the remainder of the proof uses (86) exclusively.

It remains to transfer Π^{sw} back to Π_{uni} . This is done once, and it is the only place in the unrolling where the reference-policy constant α is used.

Lemma C.7 (Switching-to-unilateral transfer). *Assume the side condition (7), equivalently $4\eta H^2 \lambda \leq 1$. Let $\varpi \in \Pi^{\text{sw}}$ be as in (79), let $t \in [H]$, and let $\mathbb{P}_\varpi^{(t)}$ denote the law of the truncated trajectory $(s_1, a_{1,1}, a_{1,2}, \dots, s_t)$ under ϖ and P with $s_1 \sim \rho$. Then $\mathbb{P}_\varpi^{(t)} \ll \mathbb{P}_{\pi^*}^{(t)}$ and, $\mathbb{P}_{\pi^*}^{(t)}$ -almost surely,*

$$\frac{d\mathbb{P}_\varpi^{(t)}}{d\mathbb{P}_{\pi^*}^{(t)}} = \prod_{h'=1}^{t-1} \prod_{i \in \{1,2\}} \frac{\varpi_{h',i}(a_{h',i} \mid s_{h'})}{\pi_{h',i}^*(a_{h',i} \mid s_{h'})} \leq C_{\text{trans}} = e. \quad (88)$$

Consequently the step- t state marginals satisfy $d_{t,S}^\varpi \preceq C_{\text{trans}} d_{t,S}^{\pi^*}$, where $\pi^* = (\pi_1^*, \pi_2^*)$; equivalently, $d_{t,S}^\varpi(B) \leq C_{\text{trans}} d_{t,S}^{\pi^*}(B)$ for every measurable $B \subseteq \mathcal{S}$.

Proof. Path level. Both laws live on the same truncated path space and share the initial distribution ρ and the kernels $\{P_{h'}\}_{h' < t}$, which cancel in the ratio; this yields the displayed product of per-stage policy ratios as a Radon–Nikodym derivative *at the path level*. Absolute continuity holds because each $\varpi_{h',i}(\cdot \mid s)$ is supported in $\text{supp } \pi_{h',i}^{\text{ref}}(\cdot \mid s)$, on which $\pi_{h',i}^*$ is strictly positive by (89) below.

Per-step bound. Fix $h' < t$ and s , and let i be an index with $\varpi_{h',i}(\cdot \mid s) = \pi_{h',i}^*(\cdot \mid s)$, which exists by (79). The i -factor of the h' -th ratio then equals 1, so only the $(-i)$ -factor is nontrivial. Player $(-i)$'s Nash stage policy is Gibbs with respect to $\pi_{h',-i}^{\text{ref}}$,

$$\pi_{h',-i}^*(a \mid s) = \frac{\pi_{h',-i}^{\text{ref}}(a \mid s) \exp(\eta q_{h'}(s, a))}{Z_{h',s}},$$

where $q_{h'}$ is the opponent-marginalized equilibrium payoff of Player $(-i)$ (for Player 2 this is the negative of the Player-1 payoff, cf. Step D.5); in either case $\|q_{h'}\|_\infty \leq H\lambda = C_{\text{val}}$ by Lemma C.2. Since $Z_{h',s} \leq \exp(\eta \|q_{h'}\|_\infty)$ and $\pi_{h',-i}^{\text{ref}}(a \mid s) \geq \alpha$ on its support,

$$\pi_{h',-i}^*(a \mid s) \geq \alpha e^{-2\eta \|q_{h'}\|_\infty} \geq \alpha e^{-2\eta H\lambda} = \alpha^{2H+1} e^{-2\eta H} \quad \text{for all } a \in \text{supp } \pi_{h',-i}^{\text{ref}}(\cdot \mid s), \quad (89)$$

using $\eta\lambda = \eta + \log(1/\alpha)$. Since $\varpi_{h',-i}(a \mid s) \leq 1$, the h' -th factor of (88) is at most $\alpha^{-(2H+1)} e^{2\eta H} \leq \exp(4\eta H\lambda)$. This bound is uniform in (s, a) and uses *no* structure of $\varpi_{h',-i}$ beyond its reference support; in particular it covers the deterministic argmax deviations produced by (81) and the branch selector (77), neither of which is of Gibbs form.

Telescoping. The product contains at most $t - 1 \leq H$ nontrivial per-step factors, so

$$\frac{d\mathbb{P}_{\varpi}^{(t)}}{d\mathbb{P}_{\pi^*}^{(t)}} \leq \exp(4\eta H^2 \lambda) \leq e = C_{\text{trans}},$$

where the last inequality is exactly (7).

Push-forward to the marginal. Because the path-level bound is uniform over histories, for every measurable $B \subseteq S$,

$$d_{t,S}^{\varpi}(B) = \mathbb{E}_{\mathbb{P}_{\pi^*}^{(t)}} \left[\mathbf{1}\{s_t \in B\} \frac{d\mathbb{P}_{\varpi}^{(t)}}{d\mathbb{P}_{\pi^*}^{(t)}} \right] \leq C_{\text{trans}} d_{t,S}^{\pi^*}(B),$$

which is legitimate precisely because the bound holds path-by-path rather than only in aggregate. $\square$

Lemma C.7 plays for the switching class the role that Lemma C.5 plays for two Gibbs policies of the same player, at the same constant C_{trans} (hence at most the C_{trans}^2 quoted in Step D.7 (iv) of Appendix C.3).

Fix $h \in [H]$ and an arbitrary $\pi' \in \Pi_{\text{uni}}$, and take the expectation of (86) under $s_h \sim d_h^{\pi'}$. The supremum in (86) sits *inside* this state expectation, so it must first be realised by a single policy: the only s_h -dependence of the maximiser is the choice of the step- h pair (i, ν_i) in (81), which is a state-dependent choice of stage policy and is measurable by the same finite-action $\arg \max$ argument used for i_h in (77), and Π^{sw} is by definition closed under state-dependent choices of which player deviates. Concatenating therefore produces a *single* Markov policy $\varpi^* \in \Pi^{\text{sw}}$: the prefix $\pi' \in \Pi_{\text{uni}} \subseteq \Pi^{\text{sw}}$ on steps $1, \dots, h-1$, the selected unilateral deviation at step h , and the branch selector σ on steps $h+1, \dots, H$. The step- t state marginal appearing after the exchange is thus $d_t^{\varpi^*}$. Writing

$$\mathcal{G}_t := \sup_{\varpi \in \Pi^{\text{sw}}} \mathbb{E}_{s_t \sim d_t^{\varpi}} [\mathcal{Z}_{t,s_t}^2], \quad (90)$$

so that $\mathbb{E}_{s_t \sim d_t^{\varpi^*}} [\mathcal{Z}_{t,s_t}^2] \leq \mathcal{G}_t$ for every t , and applying the summation swap $\sum_{h=1}^H \sum_{t=h}^H = \sum_{t=1}^H \sum_{h=1}^t$, we obtain a bound that is uniform in $\pi' \in \Pi_{\text{uni}}$ and therefore survives the supremum defining $\mathbb{E}_{d_h^{\text{uni}}}$:

$$\sum_{h=1}^H \mathbb{E}_{s_h \sim d_h^{\text{uni}}} [\xi_h^2(s_h)] \leq 2e(H+1) \sum_{t=1}^H t \mathcal{G}_t \leq 4eH^2 \sum_{t=1}^H \mathcal{G}_t, \quad (91)$$

using $H+1 \leq 2H$ and $t \leq H$ for $H \geq 1$.

Substituting (91) into the combined bound (73), we obtain

$$\text{Gap}(\hat{\pi}) \leq \mathcal{O}(\eta(C_{\text{trans}} + \lambda^2)H^2 + C_{\text{trans}}^2 \eta^3 H^6 \lambda^2) \sum_{t=1}^H \mathcal{G}_t, \quad (92)$$

where $\lambda = 1 + \eta^{-1} \log(1/\alpha)$ and the absolute constant $4e$ of (91) is absorbed into $\mathcal{O}(\cdot)$.

C.7. Final Statistical Bound via Unilateral Concentrability.

Now, we provide the following two lemmas to bound the generalization error of the Q-function estimator $\hat{Q}_h$.

Lemma C.8. *Fix a step $h \in [H]$ and let μ_h denote the stage- h state-action marginal of the behavior process, i.e. the law of $(s_h, a_{h,1}, a_{h,2})$ under the behavior policy. Because the dataset consists of n independent trajectories, the stage- h samples $\{(s_{i,h}, a_{i,h,1}, a_{i,h,2})\}_{i=1}^n$ are i.i.d. draws from μ_h ; independence holds across trajectories i only, and no independence across stages within a trajectory is used. Then, with probability at least $1 - \delta$, simultaneously for all $Q_1, Q_2 \in \mathcal{Q}_h$ we have*

$$\mathbb{E}_{\mu_h} [(Q_1(s, a_1, a_2) - Q_2(s, a_1, a_2))^2] \leq \frac{2}{n} \sum_{i=1}^n (Q_1(s_{i,h}, a_{i,h,1}, a_{i,h,2}) - Q_2(s_{i,h}, a_{i,h,1}, a_{i,h,2}))^2 + \frac{80C_{\text{val}}^2}{3n} \log(2|\mathcal{Q}|/\delta),$$

where $C_{\text{val}} = H(1 + \eta^{-1} \log(1/\alpha))$.

Proof. Let $z_{i,h} = (s_{i,h}, a_{i,h,1}, a_{i,h,2})$ be the samples drawn i.i.d. from the marginal dataset distribution μ_h . For any fixed pair of functions $Q_1, Q_2 \in \mathcal{Q}_h$, define the random variable $X(z) = (Q_1(z) - Q_2(z))^2$. By Assumption C.1, every $f \in \mathcal{Q}_h$ satisfies $\|f\|_\infty \leq C_{\text{val}} = H(1 + \eta^{-1} \log(1/\alpha))$, so $X(z) \in [0, (2C_{\text{val}})^2]$. Set $B = (2C_{\text{val}})^2$.

We apply Bernstein's inequality. For a fixed pair Q_1, Q_2 , with probability at least $1 - \delta'$, we have:

$$\mathbb{E}[X] - \frac{1}{n} \sum_{i=1}^n X(z_{i,h}) \leq \sqrt{\frac{2\text{Var}(X) \log(1/\delta')}{n}} + \frac{2B \log(1/\delta')}{3n}.$$

Since $X(z) \geq 0$ and $X(z) \leq B$, we have:

$$\text{Var}(X) \leq \mathbb{E}[X^2] \leq B\mathbb{E}[X].$$

Substituting this bound into Bernstein's inequality:

$$\mathbb{E}[X] - \frac{1}{n} \sum_{i=1}^n X(z_{i,h}) \leq \sqrt{\frac{2B\mathbb{E}[X] \log(1/\delta')}{n}} + \frac{2B \log(1/\delta')}{3n}.$$

Using the inequality $\sqrt{xy} \leq \frac{x}{2} + \frac{y}{2}$ (AM-GM inequality) with $x = \mathbb{E}[X]$ and $y = \frac{2B \log(1/\delta')}{n}$, we bound the square root term:

$$\sqrt{\frac{2B\mathbb{E}[X] \log(1/\delta')}{n}} \leq \frac{1}{2}\mathbb{E}[X] + \frac{B \log(1/\delta')}{n}.$$

Plugging this back in:

$$\begin{aligned} \mathbb{E}[X] - \frac{1}{n} \sum_{i=1}^n X(z_{i,h}) &\leq \frac{1}{2}\mathbb{E}[X] + \frac{B \log(1/\delta')}{n} + \frac{2B \log(1/\delta')}{3n} \\ \implies \frac{1}{2}\mathbb{E}[X] &\leq \frac{1}{n} \sum_{i=1}^n X(z_{i,h}) + \frac{5B \log(1/\delta')}{3n}. \end{aligned}$$

Multiplying by 2:

$$\mathbb{E}[X] \leq \frac{2}{n} \sum_{i=1}^n X(z_{i,h}) + \frac{10B \log(1/\delta')}{3n}.$$

Finally, we apply a union bound over the $|\mathcal{Q}_h|^2$ pairs $(Q_1, Q_2) \in \mathcal{Q}_h \times \mathcal{Q}_h$, setting $\delta' = \delta/|\mathcal{Q}_h|^2$. Then:

$$\log(1/\delta') = \log(|\mathcal{Q}_h|^2/\delta) = 2 \log |\mathcal{Q}_h| + \log(1/\delta) \leq 2 \log(2|\mathcal{Q}|/\delta).$$

Substituting this into the bound, we obtain:

$$\mathbb{E}_{\mu_h} [(Q_1 - Q_2)^2] \leq \frac{2}{n} \sum_{i=1}^n (Q_1(z_{i,h}) - Q_2(z_{i,h}))^2 + \frac{20B}{3n} \log(2|\mathcal{Q}|/\delta).$$

□

Lemma C.9 (Least Squares Error Bound for Finite Q-Function Classes). *Let $\{\mathcal{Q}_t\}_{t=1}^H$ be finite stagewise function classes and write $|\mathcal{Q}| := \max_{t \in [H]} |\mathcal{Q}_t|$. Define the algorithm-induced next-step value class*

$$\mathcal{V}_{h+1}^{\text{alg}} := \{V_{h+1}^{\tilde{Q}_{h+1:H}} : \tilde{Q}_t \in \mathcal{Q}_t \text{ for every } t = h+1, \dots, H\},$$

where $V_{h+1}^{\tilde{Q}_{h+1:H}}$ is the regularized value obtained by running the backward-induction step of Algorithm 1 from $\tilde{Q}_H, \tilde{Q}_{H-1}, \dots, \tilde{Q}_{h+1}$ with terminal value $V_{H+1} \equiv 0$. For $h = H$ the index set is the empty product, so $\mathcal{V}_{H+1}^{\text{alg}} = \{0\}$. By construction $\hat{V}_{h+1} \in \mathcal{V}_{h+1}^{\text{alg}}$ and $|\mathcal{V}_{h+1}^{\text{alg}}| \leq \prod_{t=h+1}^H |\mathcal{Q}_t| \leq |\mathcal{Q}|^{H-h}$.

We assume stagewise completeness relative to the induced value class: for every $h \in [H]$ and every $V \in \mathcal{V}_{h+1}^{\text{alg}}$, the Bellman backup satisfies $\mathcal{T}_h V \in \mathcal{Q}_h$. Algorithm 2 induces a different next-step value class, whose completeness assumption is stated separately where it is analysed.

Every $V \in \mathcal{V}_{h+1}^{\text{alg}}$ satisfies $\|V\|_\infty \leq C_{\text{val}}$. Indeed, with $V^{\tilde{Q}}(s) = \max_p \min_q \{ \mathbb{E}_{p \times q} [\tilde{Q}(s, \cdot, \cdot)] - \eta^{-1} \text{KL}(p \| \pi_1^{\text{ref}}(\cdot | s)) + \eta^{-1} \text{KL}(q \| \pi_2^{\text{ref}}(\cdot | s)) \}$, evaluating the inner minimum at $q = \pi_2^{\text{ref}}(\cdot | s)$ and discarding the nonnegative Player-1 KL term gives $V^{\tilde{Q}}(s) \leq \|\tilde{Q}\|_\infty$, while evaluating the outer maximum at $p = \pi_1^{\text{ref}}(\cdot | s)$ and discarding the nonnegative Player-2 KL term gives $V^{\tilde{Q}}(s) \geq -\|\tilde{Q}\|_\infty$; Assumption C.1 then gives $\|\tilde{Q}\|_\infty \leq C_{\text{val}}$.

Let $\mathcal{D}_h = \{(z_{i,h}, y_{i,h})\}_{i=1}^n$ be a dataset for step h , where the regression target is

$$y_{i,h} = r_{i,h} + \hat{V}_{h+1}(s'_{i,h}) = (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}) + \epsilon_{i,h}, \quad \epsilon_{i,h} = \xi_{i,h} + (\hat{V}_{h+1}(s'_{i,h}) - \mathbb{E}_{s' \sim P_h(\cdot | z_{i,h})} [\hat{V}_{h+1}(s')]).$$

For any fixed $V \in \mathcal{V}_{h+1}^{\text{alg}}$ chosen independently of the stage- h samples, the corresponding noise $\epsilon_i(V) := \xi_{i,h} + (V(s'_{i,h}) - \mathbb{E}_{s' \sim P_h(\cdot | z_{i,h})} [V(s')])$ is conditionally zero-mean and C_{sub} -sub-Gaussian given $z_{i,h}$, with

$$C_{\text{sub}} := \sqrt{2}(1 + C_{\text{val}}).$$

Indeed, the reward noise $\xi_{i,h}$ is 1-sub-Gaussian given $z_{i,h}$, and the centered transition term $V(s'_{i,h}) - \mathbb{E}_{s' \sim P_h(\cdot | z_{i,h})} [V(s')]$ is C_{val} -sub-Gaussian by Hoeffding's lemma, since $\|V\|_\infty \leq C_{\text{val}}$ uniformly over $V \in \mathcal{V}_{h+1}^{\text{alg}}$ and the centered version therefore has half-range C_{val} . We do not assume that the reward noise and the next-state draw are conditionally independent: for any two conditionally zero-mean sub-Gaussian variables U, W with parameters σ_U, σ_W , conditional Cauchy-Schwarz gives $\mathbb{E}[e^{s(U+W)} | z] \leq \mathbb{E}[e^{2sU} | z]^{1/2} \mathbb{E}[e^{2sW} | z]^{1/2} \leq \exp(s^2(\sigma_U^2 + \sigma_W^2))$, so $U + W$ is $\sqrt{2(\sigma_U^2 + \sigma_W^2)}$ -sub-Gaussian; with $\sigma_U = 1$ and $\sigma_W = C_{\text{val}}$ this is at most $\sqrt{2}(1 + C_{\text{val}}) = C_{\text{sub}}$.

Let $\hat{Q}_h$ be the least squares estimator

$$\hat{Q}_h \in \arg \min_{f \in \mathcal{Q}_h} \sum_{i=1}^n (f(z_{i,h}) - y_{i,h})^2.$$

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}))^2 \leq 8C_{\text{sub}}^2 \log \left(\frac{|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|}{\delta} \right) \leq 8C_{\text{sub}}^2 (H - h + 1) \log \left(\frac{|\mathcal{Q}|}{\delta} \right).$$

Proof. By the definition of the least squares estimator and the completeness assumption $(\mathcal{T}_h \hat{V}_{h+1}) \in \mathcal{Q}_h$,

$$\sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - y_{i,h})^2 \leq \sum_{i=1}^n ((\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}) - y_{i,h})^2.$$

Substituting $y_{i,h} = (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}) + \epsilon_{i,h}$ and expanding both sides yields

$$\sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}))^2 \leq 2 \sum_{i=1}^n \epsilon_{i,h} (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h})). \quad (93)$$

Because $\hat{V}_{h+1}$ is constructed from the same trajectories that produce $\{(z_{i,h}, s'_{i,h})\}_{i=1}^n$ via backward induction (recall $s'_{i,h} = s_{i,h+1}$), the noise $\epsilon_{i,h}$ is *not* automatically conditionally zero-mean given $z_{i,h}$. We bypass this dependence by chaining over the algorithm-induced value class $\mathcal{V}_{h+1}^{\text{alg}}$ defined above. Fix any pair $(f, V) \in \mathcal{Q}_h \times \mathcal{V}_{h+1}^{\text{alg}}$, and let

$$\epsilon_i(V) := r_{i,h} + V(s'_{i,h}) - (\mathcal{T}_h V)(z_{i,h}), \quad \Delta_{f,V,i} := f(z_{i,h}) - (\mathcal{T}_h V)(z_{i,h}).$$

For this fixed V , the variables $\{\epsilon_i(V)\}_{i=1}^n$ are zero-mean, C_{sub} -sub-Gaussian, and mutually independent conditional on $\{z_{i,h}\}_{i=1}^n$: each $\epsilon_i(V)$ is a measurable function of the i -th trajectory alone (through $\xi_{i,h}$ and $s'_{i,h}$), the n trajectories are independent, and the conditional mean vanishes because $\mathbb{E}[\xi_{i,h} | z_{i,h}] = 0$ and $\mathbb{E}[V(s'_{i,h}) | z_{i,h}] = (\mathbb{E}_{s' \sim P_h(\cdot | z_{i,h})} [V(s')])$.

Consider

$$M_{f,V} = \exp\left(\frac{1}{2C_{\text{sub}}^2} \sum_{i=1}^n \epsilon_i(V) \Delta_{f,V,i} - \frac{1}{8C_{\text{sub}}^2} \sum_{i=1}^n \Delta_{f,V,i}^2\right).$$

Conditioning on $z_{1:n,h}$ and applying the standard C_{sub} -sub-Gaussian moment generating function bound,

$$\mathbb{E}\left[\exp\left(\frac{1}{2C_{\text{sub}}^2} \epsilon_i(V) \Delta_{f,V,i} - \frac{1}{8C_{\text{sub}}^2} \Delta_{f,V,i}^2\right) \mid z_{i,h}\right] \leq 1.$$

Since $\epsilon_1(V), \dots, \epsilon_n(V)$ are conditionally independent given $z_{1:n,h}$ and each $\Delta_{f,V,i}$ is $z_{i,h}$ -measurable, the conditional expectation of the product factorises into the n per-sample factors above, each at most 1. Hence $\mathbb{E}[M_{f,V}] = \mathbb{E}[\mathbb{E}[M_{f,V} \mid z_{1:n,h}]] \leq 1$.

Markov's inequality gives, for any $t > 0$,

$$\mathbb{P}\left(2 \sum_{i=1}^n \epsilon_i(V) \Delta_{f,V,i} > \frac{1}{2} \sum_{i=1}^n \Delta_{f,V,i}^2 + 4C_{\text{sub}}^2 t\right) \leq e^{-t}.$$

Taking a union bound over all $(f, V) \in \mathcal{Q}_h \times \mathcal{V}_{h+1}^{\text{alg}}$ (a total of $|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|$ events) and setting $t = \log(|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|/\delta)$, with probability at least $1 - \delta$ we have simultaneously for all $f \in \mathcal{Q}_h$ and $V \in \mathcal{V}_{h+1}^{\text{alg}}$,

$$2 \sum_{i=1}^n \epsilon_i(V) (f(z_{i,h}) - (\mathcal{T}_h V)(z_{i,h})) \leq \frac{1}{2} \sum_{i=1}^n (f(z_{i,h}) - (\mathcal{T}_h V)(z_{i,h}))^2 + 4C_{\text{sub}}^2 \log\left(\frac{|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|}{\delta}\right).$$

Since $\hat{Q}_h \in \mathcal{Q}_h$ and $\hat{V}_{h+1} \in \mathcal{V}_{h+1}^{\text{alg}}$, we may instantiate this uniform bound at $(f, V) = (\hat{Q}_h, \hat{V}_{h+1})$, for which $\epsilon_i(\hat{V}_{h+1}) = \epsilon_{i,h}$. Combined with (93),

$$\sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}))^2 \leq \frac{1}{2} \sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}))^2 + 4C_{\text{sub}}^2 \log\left(\frac{|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|}{\delta}\right).$$

Rearranging gives $\sum_{i=1}^n (\hat{Q}_h(z_{i,h}) - (\mathcal{T}_h \hat{V}_{h+1})(z_{i,h}))^2 \leq 8C_{\text{sub}}^2 \log(|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|/\delta)$, and $|\mathcal{Q}_h| \leq |\mathcal{Q}|$ together with $|\mathcal{V}_{h+1}^{\text{alg}}| \leq |\mathcal{Q}|^{H-h}$ yields $\log(|\mathcal{Q}_h| \cdot |\mathcal{V}_{h+1}^{\text{alg}}|/\delta) \leq (H-h+1) \log |\mathcal{Q}| + \log(1/\delta) \leq (H-h+1) \log(|\mathcal{Q}|/\delta)$. $\square$

Final Bound Integration via D^2 -divergence and the switching-to-unilateral transfer. Combining Lemmas C.8 and C.9, the former applied to the pair $Q_1 = \hat{Q}_h \in \mathcal{Q}_h$ and $Q_2 = \mathcal{T}_h \hat{V}_{h+1} \in \mathcal{Q}_h$ (the latter membership by completeness), the expected squared Bellman regression error under the data distribution μ_h satisfies

$$\mathbb{E}_{z \sim \mu_h} \left[(\hat{Q}_h(z) - (\mathcal{T}_h \hat{V}_{h+1})(z))^2 \right] \leq \frac{16C_{\text{sub}}^2 (H-h+1) \iota + \frac{80C_{\text{val}}^2}{3} \iota}{n} =: \epsilon_{\text{stat},h}^2, \quad \iota := \log(4H|\mathcal{Q}|/\delta),$$

simultaneously for all $h \in [H]$ with probability at least $1 - \delta$: apply each of the two lemmas at every stage with confidence parameter $\delta' = \delta/(2H)$ and union bound over the resulting $2H$ events. We now relate the trajectory error $\mathcal{G}_t$ of (90) to this data-distribution error in two steps.

Step (a): switching prefix to Nash–Nash prefix. By Lemma C.7, for every $\varpi \in \Pi^{\text{sw}}$ and every $t \in [H]$ we have $d_{t,\mathcal{S}}^{\varpi} \preceq C_{\text{trans}} d_{t,\mathcal{S}}^{\pi^*}$. Since $\mathcal{Z}_{t,s}^2 \geq 0$, taking the supremum over $\varpi \in \Pi^{\text{sw}}$ gives

$$\mathcal{G}_t = \sup_{\varpi \in \Pi^{\text{sw}}} \mathbb{E}_{s_t \sim d_t^{\varpi}} [\mathcal{Z}_{t,s_t}^2] \leq C_{\text{trans}} \cdot \mathbb{E}_{s_t \sim d_t^{\pi^*}} [\mathcal{Z}_{t,s_t}^2].$$

Step (b): D^2 -divergence absorbs the state-dependent single-player deviation. The quantity $\mathcal{Z}_{t,s}^2$ of (85) is a maximum over the branch index and a supremum over the deviating stage policy, both *inside* the state expectation and not exchangeable with it; we resolve this by measurable selection. Since $\mathcal{A}_1, \mathcal{A}_2$ are finite, the supremum over $\nu_i \in \Delta(\text{supp } \pi_{t,i}^{\text{ref}}(\cdot|s))$ is

attained at a vertex, so with a fixed tie-breaking order there exist measurable maps $s \mapsto j_t(s) \in \{1, 2\}$ and $s \mapsto \nu_t(\cdot|s)$, the latter a stage policy for Player $j_t(s)$, with

$$\mathcal{Z}_{t,s}^2 = \mathbb{E}_{(a_1, a_2) \sim \nu_t(\cdot|s) \times \pi_{t, -j_t(s)}^*(\cdot|s)} [(\hat{Q}_t - \mathcal{T}_t \hat{V}_{t+1})^2(s, a_1, a_2)].$$

For $i \in \{1, 2\}$ define a genuine Player- i stage policy by $\nu_t^{(i)}(\cdot|s) := \nu_t(\cdot|s)$ if $j_t(s) = i$ and $\nu_t^{(i)}(\cdot|s) := \pi_{t,i}^*(\cdot|s)$ otherwise, and let $\pi^{(i)} \in \Pi_i$ be the Player- i policy that agrees with π_i^* on steps $1, \dots, t-1$ and equals $\nu_t^{(i)}$ at step t . Splitting on the value of $j_t(s_t)$ and then dropping the (nonnegative) indicators,

$$\begin{aligned} \mathbb{E}_{s_t \sim d_t^{\pi^*}} [\mathcal{Z}_{t,s_t}^2] &= \sum_{i \in \{1, 2\}} \mathbb{E}_{s_t \sim d_t^{\pi^*}} [\mathbf{1}\{j_t(s_t) = i\} \mathbb{E}_{\nu_t^{(i)} \times \pi_{t,-i}^*} [(\hat{Q}_t - \mathcal{T}_t \hat{V}_{t+1})^2]] \\ &\leq \sum_{i \in \{1, 2\}} \mathbb{E}_{(s_t, a_1, a_2) \sim d_t^{\pi^{(i)} \times \pi_{t,-i}^*}} [(\hat{Q}_t - \mathcal{T}_t \hat{V}_{t+1})^2], \end{aligned}$$

where we used that the Nash–Nash prefix $d_t^{\pi^*}$ composed with the step- t action law $\nu_t^{(i)} \times \pi_{t,-i}^*$ is exactly the step- t state-action law $d_t^{\pi^{(i)} \times \pi_{t,-i}^*}$ of a policy pair in Π_{uni} . Each summand is therefore covered by Assumption 6.2, and (72) applies, giving

$$\mathbb{E}_{s_t \sim d_t^{\pi^*}} [\mathcal{Z}_{t,s_t}^2] \leq 2 C_{\text{uni}} \cdot \mathbb{E}_{\mu_t} [(\hat{Q}_t - \mathcal{T}_t \hat{V}_{t+1})^2].$$

The factor 2 is the union over the two unilateral branches.

Combining steps (a) and (b) and summing over $t \in [H]$ yields

$$\sum_{t=1}^H \mathcal{G}_t \leq 2 C_{\text{trans}} \cdot C_{\text{uni}} \cdot \sum_{t=1}^H \mathbb{E}_{\mu_t} [(\hat{Q}_t - \mathcal{T}_t \hat{V}_{t+1})^2] \leq 2 C_{\text{trans}} \cdot C_{\text{uni}} \cdot \sum_{t=1}^H \epsilon_{\text{stat},t}^2, \quad (94)$$

where the second inequality invokes the per-stage regression-error concentration established above. Plugging in $\epsilon_{\text{stat},t}^2$, using $C_{\text{sub}}^2 = 2(1 + C_{\text{val}})^2 \leq 8C_{\text{val}}^2$ since $C_{\text{val}} \geq 1$, and summing the arithmetic series $\sum_{t=1}^H (H - t + 1) = H(H + 1)/2$ gives $\sum_{t=1}^H \epsilon_{\text{stat},t}^2 = \mathcal{O}(C_{\text{val}}^2 H^2 \iota/n)$, a factor of H larger than the naive $H \cdot \epsilon_{\text{stat}}^2$; this extra H is the price of the algorithm-induced-value-class union bound.

Substituting (94) into (92) and using $C_{\text{val}}^2 = H^2(1 + \eta^{-1} \log(1/\alpha))^2$ yields the final duality gap:

$$\text{Gap}(\hat{\pi}) \leq \tilde{\mathcal{O}} \left(\frac{(\eta H^6 + \eta^3 H^{10}) \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n} \right),$$

where the two η -scalings correspond to the Gap 2 and Gap 1 chains. All density-ratio multipliers are powers of $C_{\text{trans}} = e$ under (7) and are therefore absorbed into universal constants. We keep the regularization-scale factor $\lambda = 1 + \eta^{-1} \log(1/\alpha)$ explicit.

- **Gap 2 contribution (ηH^6):** From Lemma 6.6, $\text{Gap}_2 \leq \mathcal{O}(\eta(C_{\text{trans}} + \lambda^2)) \sum_{h=1}^H \mathbb{E}_{s \sim d_h^{\pi_1^*, \pi_2^*}} [\xi_h^2(s)]$ on the Term A side (the symmetric marginal $d_h^{\pi_1^*, \pi_2^*}$ arises on the Term B side; both are in Π_{uni} , and both are dominated by $\mathbb{E}_{d_h^{\text{uni}}}$). The H^6 exponent decomposes as (i) H^2 from $C_{\text{val}}^2 = H^2 \lambda^2$ in ϵ_{stat}^2 , (ii) H^2 from the Bellman unrolling of ξ^2 via (91), and (iii) H^2 from horizon summation at the concentrability step combined with the value-class union-bound factor $(H - h + 1)$ in $\epsilon_{\text{stat},h}^2$.
- **Gap 1 contribution ($\eta^3 H^{10}$):** Lemma 6.5 contributes an additional H^4 multiplier on top of the shared H^6 from $\sum_h \mathbb{E}[\xi_h^2]$. This H^4 decomposes as (iv) H from the V-stability outer factor (the $\mathcal{O}(H)$ constant in (34)); (v) H from the $\sum_{h=1}^H \sum_{t=h+1}^H \rightarrow \sum_{t=1}^H \sum_{h=1}^{t-1}$ swap in (46); and (vi) H^2 from $L_{t+1}^2 \leq H^2 \lambda^2$ via Lemma C.2. The η^3 scaling comes from $\eta \cdot \eta^2$, where the η^2 originates from the squared stability (64).

Under the condition (7) we have $\eta^2 H^4 \lambda^2 \leq 1/16$ and hence $\eta^2 H^4 \leq 1/16$, so $\eta^3 H^{10} \leq \eta H^6/16$ and the ηH^6 term dominates. This concludes the proof of Theorem 6.3.

D. Proof of Optimization Convergence (Theorem 6.8)

D.1. Sequential Duality Gap and Decomposition

We evaluate the convergence of the algorithm using the Duality Gap defined on the learned policy pair $\pi^{(T)} = (\pi_1^{(T)}, \pi_2^{(T)})$ (generated by Algorithm 2). To handle general zero-sum games without assuming symmetric transitions, we define the duality gap as the sum of the exploitability of each player with respect to the initial distribution ρ :

$$\text{Gap}(\pi^{(T)}) := \mathbb{E}_{s_1 \sim \rho} \left[\underbrace{\left(V_1^{\pi_1^\dagger, \pi_2^{(T)}}(s_1) - V_1^{\pi_1^{(T)}, \pi_2^{(T)}}(s_1) \right)}_{\text{Player 1 Exploitability}} + \underbrace{\left(V_1^{\pi_1^{(T)}, \pi_2^\dagger}(s_1) - V_1^{\pi_1^{(T)}, \pi_2^\dagger}(s_1) \right)}_{\text{Player 2 Exploitability}} \right],$$

where $\pi^\dagger$ is the best response to $\pi_2^{(T)}$ (maximizing V_1) and $\pi^\ddagger$ is the best response to $\pi_1^{(T)}$ (minimizing V_1). As everywhere else in the paper, both best-response optima are taken over the reference-supported class Π_i^{ref} of (2), which is without loss (Appendix C.6).

Since the Nash equilibrium value $V_1^{\pi_1^*, \pi_2^*}$ is unique, we can decompose this gap into two separate regret terms relative to the Nash equilibrium:

$$\text{Gap}(\pi^{(T)}) = \underbrace{\mathbb{E}_{s_1 \sim \rho} \left[V_1^{\pi_1^\dagger, \pi_2^{(T)}}(s_1) - V_1^{\pi_1^*, \pi_2^*}(s_1) \right]}_{\text{Term A (Player 1 Regret)}} + \underbrace{\mathbb{E}_{s_1 \sim \rho} \left[V_1^{\pi_1^*, \pi_2^\dagger}(s_1) - V_1^{\pi_1^{(T)}, \pi_2^\dagger}(s_1) \right]}_{\text{Term B (Player 2 Regret)}}. \quad (95)$$

Our proof strategy is to bound **Term A** in full detail. Because the setting is genuinely asymmetric, **Term B** is *not* obtained by an informal symmetry appeal: in Appendix D.9 we construct an explicit *reflected game* obtained from the action-coordinate swap $\sigma(s, a_1, a_2) = (s, a_2, a_1)$, verify that every object entering the Term-A analysis is equivariant under this swap, and show that Term B of the original game *equals* Term A of the reflected game. The Term-A bound then transfers to Term B with the same exponents and the same α -dependence.

D.2. The Fitted Sequence of SOS-MD and Its Exact Equilibrium

Algorithm 1 and Algorithm 2 both write $\hat{Q}_h$ for their stage- h regression output, but the two objects are *different*: ROSE forms its stage- h target from the *saddle value* of its own $\hat{Q}_{h+1}$, whereas SOS-MD forms it from the value of its *last mirror-descent iterate* $\pi_{h+1}^{(T)}$, which is not a saddle value. Theorem 6.3 and Lemma C.6 are statements about the former sequence, so we make the SOS-MD sequence explicit and attach an exact stagewise regularized equilibrium to it.

Definition D.1 (SOS-MD fitted sequence and exact equilibrium). Write $J_s(\pi_1, \pi_2; Q) := \mathbb{E}_{a \sim \pi_1 \times \pi_2} [Q(s, a)] - \eta^{-1} \text{KL}(\pi_1 \| \pi_{h,1}^{\text{ref}}(\cdot|s)) + \eta^{-1} \text{KL}(\pi_2 \| \pi_{h,2}^{\text{ref}}(\cdot|s))$ for the stage- h regularized objective at state s . Set $\tilde{V}_{H+1} \equiv 0$ and, for $h = H, H-1, \dots, 1$, define:

- $\tilde{Q}_h \in \arg \min_{f \in \mathcal{Q}_h} \sum_{i \in [n]} (f(s_{i,h}, a_{i,h,1}, a_{i,h,2}) - r_{i,h} - \tilde{V}_{h+1}(s'_{i,h}))^2$, the stage- h regression output of Algorithm 2;
- $\{\pi_h^{(t)}\}_{t=0}^T$, the mirror-descent iterates of Algorithm 2 on the stage game with payoff $\tilde{Q}_h$, with last iterate $\pi_h^{(T)}$;
- $\tilde{V}_h(s) := J_s(\pi_{h,1}^{(T)}, \pi_{h,2}^{(T)}; \tilde{Q}_h)$, the value actually backed up by Algorithm 2 (no clipping is applied at any point);
- $\bar{\pi}_h(\cdot|s)$, the unique regularized Nash equilibrium of the stage game $J_s(\cdot, \cdot; \tilde{Q}_h)$ — the exact stagewise regularized equilibrium — with saddle value $\bar{V}_h(s) := J_s(\bar{\pi}_{h,1}, \bar{\pi}_{h,2}; \tilde{Q}_h)$, and *last-iterate defect*

$$\omega_h(s) := \tilde{V}_h(s) - \bar{V}_h(s), \quad \omega_{H+1} \equiv 0.$$

Notational convention for this appendix. Lemmas D.4, D.5, 6.7 and Theorem 6.8 describe the stage games that Algorithm 2 actually solves. Accordingly, *throughout Appendix D the symbols $\hat{Q}_h$ and $\hat{\pi}_h$ are to be read as $\tilde{Q}_h$ and $\bar{\pi}_h$* . Their proofs use only two properties of the pair $(\tilde{Q}_h, \bar{\pi}_h)$, both true by construction: that $\tilde{Q}_h \in \mathcal{Q}_h$, whence $\|\tilde{Q}_h\|_\infty \leq C_{\text{val}} = H\lambda$ by Assumption C.1; and that $\bar{\pi}_h(\cdot|s)$ is the saddle point of $J_s(\cdot, \cdot; \tilde{Q}_h)$. Only the final assembly (Appendix D.10), which writes $\bar{\pi}$ explicitly, must distinguish $\bar{\pi}$ from the ROSE output.

Lemma D.2 (Unclipped value envelope for SOS-MD). *For every $h \in [H]$ and $s \in \mathcal{S}$,*

$$|\tilde{V}_h(s)| \leq \|\tilde{Q}_h\|_\infty + \eta^{-1} \log(1/\alpha), \quad |\bar{V}_h(s)| \leq \|\tilde{Q}_h\|_\infty + \eta^{-1} \log(1/\alpha). \quad (96)$$

In particular $\|\tilde{V}_h\|_\infty, \|\bar{V}_h\|_\infty \leq H\lambda + (\lambda - 1) \leq (H + 1)\lambda \leq 2C_{\text{val}}$ under Assumption C.1; this unconditional bound is the only one used in the sequel. If moreover the stage- h class carries the envelope $\|f\|_\infty \leq 1 + (H - h)\lambda$ — which is exactly the range of $\mathcal{T}_h$ on values bounded by $(H - h)\lambda$, and hence the envelope that Bellman completeness produces — then the recursion closes at the sharper stagewise scale $\|\tilde{V}_h\|_\infty \leq (H - h + 1)\lambda$. (The coarser envelope $\|f\|_\infty \leq (H - h + 1)\lambda$ would not close it: it leaves a slack of $\lambda - 1$ at every stage.)

Proof. Fix h, s and a player i , and write $\mathcal{A}_{h,i}^{\text{ref}}(s) := \text{supp}(\pi_{h,i}^{\text{ref}}(\cdot|s))$. Every stage policy appearing in $\tilde{V}_h$ or $\bar{V}_h$ is supported inside $\mathcal{A}_{h,i}^{\text{ref}}(s)$: the update (5)–(6) carries the factor $\pi_{h,i}^{\text{ref}}(a|s)^{\gamma_t \eta^{-1}}$ with $\gamma_t \eta^{-1} = 2/(t + 2) \in (0, 1]$, which vanishes off the reference support, and $\bar{\pi}_{h,i}$ is Gibbs with base measure $\pi_{h,i}^{\text{ref}}$. For any such admissible $p \in \Delta(\mathcal{A}_{h,i}^{\text{ref}}(s))$, the definition $\alpha = \inf\{\pi_{h,i}^{\text{ref}}(a|s) : \pi_{h,i}^{\text{ref}}(a|s) > 0\}$ gives

$$0 \leq \text{KL}(p \| \pi_{h,i}^{\text{ref}}(\cdot|s)) = \sum_a p(a) \log \frac{p(a)}{\pi_{h,i}^{\text{ref}}(a|s)} \leq \sum_a p(a) \log \frac{1}{\alpha} = \log(1/\alpha), \quad (97)$$

using $p(a) \leq 1$ and $\pi_{h,i}^{\text{ref}}(a|s) \geq \alpha$ on the support. In $J_s(\pi_1, \pi_2; \tilde{Q}_h)$ the two KL terms enter with opposite signs and both lie in $[0, \log(1/\alpha)]$, so their combined contribution has absolute value at most $\eta^{-1} \log(1/\alpha)$, while the bilinear term is bounded by $\|\tilde{Q}_h\|_\infty$. This gives (96) for both $\tilde{V}_h$ and $\bar{V}_h$. Since $\tilde{Q}_h \in \mathcal{Q}_h$, Assumption C.1 gives $\|\tilde{Q}_h\|_\infty \leq H\lambda$ and $\eta^{-1} \log(1/\alpha) = \lambda - 1$, whence the unconditional consequence. For the last claim, argue by backward induction from $\tilde{V}_{H+1} \equiv 0$: if $\|\tilde{V}_{h+1}\|_\infty \leq (H - h)\lambda$ then, since $|r_h^*| \leq 1$ gives $\|\mathcal{T}_h V\|_\infty \leq 1 + \|V\|_\infty$, we get $\|\mathcal{T}_h \tilde{V}_{h+1}\|_\infty \leq 1 + (H - h)\lambda$, so the stage- h envelope gives $\|\tilde{Q}_h\|_\infty \leq 1 + (H - h)\lambda$, and (96) with $\eta^{-1} \log(1/\alpha) = \lambda - 1$ yields $\|\tilde{V}_h\|_\infty \leq 1 + (H - h)\lambda + (\lambda - 1) = (H - h + 1)\lambda$, closing the induction. Note also $1 + (H - h)\lambda \leq \lambda + (H - 1)\lambda = H\lambda = C_{\text{val}}$ for $h \geq 1$, since $\lambda \geq 1$. $\square$

No clipping is therefore required: $\|\tilde{V}_h\|_\infty \leq 2C_{\text{val}}$ keeps the unclipped backup on the same scale as the stagewise function classes $\{\mathcal{Q}_h\}_{h=1}^H$, so the least-squares concentration of Lemma C.9 is applicable to the SOS-MD sequence with its sub-Gaussian constant $C_{\text{sub}} = 1 + 2C_{\text{val}}$, i.e. with its constants changed by at most an absolute factor.

Lemma D.3 (Size of the last-iterate defect). *For every $h \in [H]$ and $s \in \mathcal{S}$,*

$$|\omega_h(s)| \leq 3H\lambda \left(\|\pi_{h,1}^{(T)}(\cdot|s) - \bar{\pi}_{h,1}(\cdot|s)\|_1 + \|\pi_{h,2}^{(T)}(\cdot|s) - \bar{\pi}_{h,2}(\cdot|s)\|_1 \right),$$

and hence, for every $T \geq 1$,

$$\|\omega_h\|_\infty \leq \frac{36\sqrt{2}\eta H^2 \lambda^2}{\sqrt{T+1}} = \tilde{\mathcal{O}}\left(\frac{\eta H^2 \lambda^2}{\sqrt{T}}\right). \quad (98)$$

Proof. The proof invokes Lemma D.4 and Theorem 6.8, proved below; neither depends on ω , so there is no circularity. Abbreviate $J_s := J_s(\cdot, \cdot; \tilde{Q}_h)$ and split

$$\omega_h(s) = [J_s(\pi_{h,1}^{(T)}, \pi_{h,2}^{(T)}) - J_s(\bar{\pi}_{h,1}, \pi_{h,2}^{(T)})] + [J_s(\bar{\pi}_{h,1}, \pi_{h,2}^{(T)}) - J_s(\bar{\pi}_{h,1}, \bar{\pi}_{h,2})],$$

so that each bracket perturbs one player only. In the first bracket, the bilinear part is at most $\|\tilde{Q}_h\|_\infty \|\pi_{h,1}^{(T)} - \bar{\pi}_{h,1}\|_1 \leq H\lambda \|\pi_{h,1}^{(T)} - \bar{\pi}_{h,1}\|_1$. For its KL part, observe that the log-linear class

$$\mathcal{P}_{h,i}(s) := \left\{ p \in \Delta(\mathcal{A}_{h,i}^{\text{ref}}(s)) : \left| \log(p(a)/\pi_{h,i}^{\text{ref}}(a|s)) \right| \leq C_{\log} \quad \forall a \right\}, \quad C_{\log} = 2\eta H\lambda,$$

is the intersection of the simplex with the finitely many half-spaces $\pi_{h,i}^{\text{ref}}(a|s)e^{-C_{\log}} \leq p(a) \leq \pi_{h,i}^{\text{ref}}(a|s)e^{C_{\log}}$, hence *convex*, and that it lies in the relative interior of $\Delta(\mathcal{A}_{h,i}^{\text{ref}}(s))$, where $\text{KL}(\cdot \| \pi_{h,i}^{\text{ref}})$ is smooth. It contains both $\pi_{h,i}^{(T)}(\cdot|s)$ and $\bar{\pi}_{h,i}(\cdot|s)$

by Lemma D.4. Since $\nabla_p \text{KL}(p||r) = \log(p/r) + 1$ and $\langle \mathbf{1}, p - q \rangle = 0$ for $p, q \in \Delta$, the mean-value theorem along the segment $[q, p] \subset \mathcal{P}_{h,i}(s)$ yields, for some ζ on that segment,

$$\eta^{-1} |\text{KL}(p||r) - \text{KL}(q||r)| = \eta^{-1} |\langle \log(\zeta/r), p - q \rangle| \leq \eta^{-1} C_{\log} \|p - q\|_1 = 2H\lambda \|p - q\|_1. \quad (99)$$

Adding the two contributions bounds the first bracket by $3H\lambda \|\pi_{h,1}^{(T)} - \bar{\pi}_{h,1}\|_1$; the second bracket is bounded by $3H\lambda \|\pi_{h,2}^{(T)} - \bar{\pi}_{h,2}\|_1$ identically. Finally Theorem 6.8 gives $\sup_s \|\bar{\pi}_{h,i}(\cdot|s) - \pi_{h,i}^{(T)}(\cdot|s)\|_1 \leq 6\sqrt{2} \eta H \lambda / \sqrt{T+1}$ for each i , and $3H\lambda \cdot 2 \cdot 6\sqrt{2} \eta H \lambda = 36\sqrt{2} \eta H^2 \lambda^2$. $\square$

The exact-equilibrium sequence obeys an inexact Bellman recursion. The triple $(\tilde{Q}_h, \bar{\pi}_h, \bar{V}_h)_{h=1}^H$ is exactly a ROSE-type sequence — $\bar{\pi}_h$ is the exact stagewise regularized equilibrium of $\tilde{Q}_h$ and $\bar{V}_h$ is its saddle value — except that the stage- h regression target was formed from $\tilde{V}_{h+1}$ rather than from $\bar{V}_{h+1}$. Since $\mathcal{T}_h$ is affine, writing $(P_h g)(s, a) := \mathbb{E}_{s' \sim P_h(\cdot|s, a)}[g(s')]$,

$$\tilde{Q}_h - \mathcal{T}_h \bar{V}_{h+1} = \underbrace{(\tilde{Q}_h - \mathcal{T}_h \tilde{V}_{h+1})}_{\text{least-squares regression residual}} + \underbrace{P_h \omega_{h+1}}_{\|\cdot\|_\infty \leq \|\omega_{h+1}\|_\infty}. \quad (100)$$

Accordingly, let $\bar{Z}_{t,s}^2$ denote the single-player Nash-averaged squared error (85) formed from $\tilde{Q}_t - \mathcal{T}_t \bar{V}_{t+1}$, and $Z_{t,s}^2$ the one formed from the regression residual $\tilde{Q}_t - \mathcal{T}_t \tilde{V}_{t+1}$. Then $(x+y)^2 \leq 2x^2 + 2y^2$ gives the *pointwise, measure-free* bound

$$\bar{Z}_{t,s}^2 \leq 2 Z_{t,s}^2 + 2 \|\omega_{t+1}\|_\infty^2. \quad (101)$$

The second summand is constant in s and in the action arguments, so — unlike the first — it requires neither the concentrability transfer of Assumption 6.2 nor the density-ratio transfer of Lemma C.5: it is invariant under every change of measure. Finally, the least-squares concentration of Lemma C.9 is to be applied to the residual $\tilde{Q}_t - \mathcal{T}_t \tilde{V}_{t+1}$ with the SOS-MD-induced next-step value class $\mathcal{V}_{t+1}^{\text{SOS}}$ in place of $\mathcal{V}_{t+1}^{\text{alg}}$, i.e. with the completeness hypothesis $\mathcal{T}_t V \in \mathcal{Q}_t$ for all $V \in \mathcal{V}_{t+1}^{\text{SOS}}$; the cardinality bound $|\mathcal{V}_{t+1}^{\text{SOS}}| \leq \prod_{j=t+1}^H |\mathcal{Q}_j| \leq |\mathcal{Q}|^{H-t}$ holds by the same argument, because the mirror-descent recursion is deterministic given π^{ref} and the stepsize schedule, so $\tilde{V}_{t+1}$ is a deterministic function of $(\tilde{Q}_{t+1}, \dots, \tilde{Q}_H)$, and $\|\tilde{V}_{t+1}\|_\infty \leq 2C_{\text{val}}$ by Lemma D.2.

D.3. Proof Strategy

Our strategy decouples the optimization analysis from the statistical analysis via a Lipschitz transfer argument, carried out against the exact equilibrium $\bar{\pi}$ of Definition D.1. Concretely,

$$\text{Gap}(\pi^{(T)}) \leq \text{Gap}(\bar{\pi}) + |\text{Gap}(\pi^{(T)}) - \text{Gap}(\bar{\pi})|. \quad (102)$$

The first term is a statistical error: the whole Term-A chain of Appendix C.6 — including the saddle-point non-expansion of Lemma C.6 — applies to $\bar{\pi}$, with the single modification (101). The second term is the additional optimization error incurred by running T mirror-descent steps instead of solving each stage game exactly; we control it via Lipschitzness of the duality gap (Lemma D.7), reducing it to a bound on $\sup_s \|\pi_h^{(T)}(\cdot|s) - \bar{\pi}_h(\cdot|s)\|_1$, which is in turn controlled by mirror-descent KL convergence (Lemma 6.7) and Pinsker's inequality.

The Lipschitzness argument requires both $\pi^{(T)}$ and $\bar{\pi}$ to lie in a uniformly bounded log-density-ratio class relative to π^{ref} , which is Lemma D.4, proved first.

D.4. Proof of Lemma D.4 (Log-Linear Boundedness)

Lemma D.4 (Log-Linear Boundedness of Self-Play Iterates). *Let $\{\pi^{(t)}\}_{t=0}^T$ be the iterates generated by Algorithm 2 with stepsize $\gamma_t = \frac{2\eta}{t+2}$. Then for every $t \geq 0$, every player $i \in \{1, 2\}$, every step $h \in [H]$, every state $s \in \mathcal{S}$, and every $a \in \text{supp}(\pi_{h,i}^{\text{ref}}(\cdot|s))$,*

$$\left| \log \frac{\pi_{h,i}^{(t)}(a|s)}{\pi_{h,i}^{\text{ref}}(a|s)} \right| \leq 2\eta H (1 + \eta^{-1} \log(1/\alpha)).$$

The empirical Nash equilibrium $\hat{\pi}_h$ also satisfies this bound.

Fix an arbitrary context (h, s) and, for each player $i \in \{1, 2\}$, let $\mathcal{A}_{h,i}^{\text{ref}}(s) := \text{supp}(\pi_{h,i}^{\text{ref}}(\cdot|s))$; when the player index is clear from the context we abbreviate $\mathcal{A}_x := \mathcal{A}_{h,i}^{\text{ref}}(s)$, and all maxima, minima and sums over actions below are restricted to the corresponding player's set. For $v \in \mathbb{R}^{\mathcal{A}_x}$, write $\text{osc}(v) := \max_{a \in \mathcal{A}_x} v(a) - \min_{a \in \mathcal{A}_x} v(a)$ for its oscillation.

Proof of Lemma D.4. For each player $i \in \{1, 2\}$, define the log-density ratio

$$u_i^{(t)}(a) := \log \frac{\pi_{h,i}^{(t)}(a|s)}{\pi_{h,i}^{\text{ref}}(a|s)}, \quad a \in \mathcal{A}_x.$$

We show by induction on t that $\text{osc}(u_i^{(t)}) \leq C_{\log} := 2\eta H(1 + \eta^{-1} \log(1/\alpha))$ for both players. At initialization, $\pi_{h,i}^{(0)}(\cdot|s) = \pi_{h,i}^{\text{ref}}(\cdot|s)$, so $u_i^{(0)} \equiv 0$ and the claim holds.

Consider Player 1; the argument for Player 2 is identical. The update (5) can be written equivalently as $u_1^{(t+1)}(a) = w_1^{(t)}(a) - \log Z_1^{(t)}$, where

$$w_1^{(t)}(a) := (1 - \gamma_t \eta^{-1}) u_1^{(t)}(a) + \gamma_t \hat{q}_{h,1}^{(t)}(s, a), \quad Z_1^{(t)} := \sum_{b \in \mathcal{A}_x} \pi_{h,1}^{\text{ref}}(b|s) \exp(w_1^{(t)}(b)).$$

Since $\hat{q}_{h,1}^{(t)}(s, a_1) = \mathbb{E}_{a_2 \sim \pi_{h,2}^{(t)}} [\hat{Q}_h(s, a_1, a_2)]$ and $\|\hat{Q}_h\|_{\infty} \leq C_{\text{val}} = H(1 + \eta^{-1} \log(1/\alpha))$ by Assumption C.1, we have $\text{osc}(\hat{q}_{h,1}^{(t)}(s, \cdot)) \leq 2H(1 + \eta^{-1} \log(1/\alpha))$. With the stepsize $\gamma_t = 2\eta/(t+2) \leq \eta$ for all $t \geq 0$, so $\gamma_t \eta^{-1} \in [0, 1]$, the inductive hypothesis $\text{osc}(u_1^{(t)}) \leq C_{\log}$ yields

$$\text{osc}(w_1^{(t)}) \leq (1 - \gamma_t \eta^{-1}) C_{\log} + \gamma_t \cdot 2H(1 + \eta^{-1} \log(1/\alpha)) = (1 - \gamma_t \eta^{-1}) C_{\log} + (\gamma_t \eta^{-1}) C_{\log} = C_{\log},$$

where the second equality substitutes $C_{\log} = 2\eta H(1 + \eta^{-1} \log(1/\alpha))$. Since subtracting a constant does not change oscillation, $\text{osc}(u_1^{(t+1)}) = \text{osc}(w_1^{(t)}) \leq C_{\log}$. Moreover, because $\pi_{h,1}^{\text{ref}}(\cdot|s)$ is a probability distribution on $\mathcal{A}_x$, $\min_{a \in \mathcal{A}_x} w_1^{(t)}(a) \leq \log Z_1^{(t)} \leq \max_{a \in \mathcal{A}_x} w_1^{(t)}(a)$, so $|u_1^{(t+1)}(a)| \leq \text{osc}(w_1^{(t)}) \leq C_{\log}$ for all $a \in \mathcal{A}_x$. The argument for Player 2 with $-\hat{q}_{h,2}^{(t)}(s, \cdot)$ replacing $\hat{q}_{h,1}^{(t)}(s, \cdot)$ is identical.

It remains to show that the empirical Nash equilibrium $\hat{\pi}_h$ also satisfies the bound. By the saddle-point first-order conditions for the regularized stage game with payoff $\hat{Q}_h$, $\hat{\pi}_{h,1}$ and $\hat{\pi}_{h,2}$ take the Gibbs (softmax) form

$$\hat{\pi}_{h,1}(a|s) \propto \pi_{h,1}^{\text{ref}}(a|s) \exp(\eta \mathbb{E}_{a_2 \sim \hat{\pi}_{h,2}} [\hat{Q}_h(s, a, a_2)]), \quad \hat{\pi}_{h,2}(a|s) \propto \pi_{h,2}^{\text{ref}}(a|s) \exp(-\eta \mathbb{E}_{a_1 \sim \hat{\pi}_{h,1}} [\hat{Q}_h(s, a_1, a)]).$$

The unnormalized log-density ratio in both cases has oscillation at most $2\eta H(1 + \eta^{-1} \log(1/\alpha)) = C_{\log}$, so $|\log(\hat{\pi}_{h,i}(a|s)/\pi_{h,i}^{\text{ref}}(a|s))| \leq C_{\log}$ by the same normalization argument. $\square$

D.5. State-wise Online Mirror Descent Lemma

We adapt the proof from Theorem 4 of Zhang et al. (2025b) into the following lemma, sharpened with the Hoeffding-type oscillation bound of Munos et al. (2024).

Lemma D.5 (State-wise OMD Lemma). *Fix a state s . Let $\phi(\pi) = \sum_a \pi(a) \log \pi(a)$ be the negative entropy regularizer. Note that the Bregman divergence generated by ϕ corresponds to the KL-divergence: $D_{\phi}(\pi, \pi') = \text{KL}(\pi \| \pi')$.*

Given a payoff vector $\delta(s, \cdot) \in \mathbb{R}^{|\mathcal{A}|}$ and a current policy $\pi^{-}(\cdot|s)$, let the updated policy $\pi^{+}(\cdot|s)$ be defined by the Mirror Descent step:

$$\pi^{+}(\cdot|s) = \arg \max_{\pi \in \Delta(\mathcal{A})} \left[\sum_a \pi(a) \delta(s, a) - D_{\phi}(\pi, \pi^{-}(\cdot|s)) \right].$$

Then for any $\pi \in \Delta(\mathcal{A})$, the following Hoeffding-type descent inequality holds:

$$\text{KL}(\pi \| \pi^{+}(\cdot|s)) \leq \text{KL}(\pi \| \pi^{-}(\cdot|s)) + \sum_a (\pi^{-}(a|s) - \pi(a)) \delta(s, a) + \frac{1}{8} \text{osc}(\delta(s, \cdot))^2,$$

where $\text{osc}(v) := \max_a v(a) - \min_a v(a)$ is the oscillation of the payoff vector.

Proof. Write $\delta(a) := \delta(s, a)$ and $\pi^-(a) := \pi^-(a|s)$ for brevity. The closed-form solution of the Mirror Descent step is the exponential-weights update

$$\pi^+(a) = \frac{\pi^-(a) \exp(\delta(a))}{\sum_b \pi^-(b) \exp(\delta(b))}.$$

Therefore, for any $\pi \in \Delta(\mathcal{A})$,

$$\text{KL}(\pi \| \pi^+) - \text{KL}(\pi \| \pi^-) = \sum_a \pi(a) \log \frac{\pi^-(a)}{\pi^+(a)} = - \sum_a \pi(a) \delta(a) + \log \sum_a \pi^-(a) \exp(\delta(a)).$$

By Hoeffding's lemma applied to the bounded random variable $\delta(a)$ with $a \sim \pi^-$,

$$\log \sum_a \pi^-(a) \exp(\delta(a)) \leq \sum_a \pi^-(a) \delta(a) + \frac{1}{8} \text{osc}(\delta)^2.$$

Substituting yields

$$\text{KL}(\pi \| \pi^+) - \text{KL}(\pi \| \pi^-) \leq \sum_a (\pi^-(a) - \pi(a)) \delta(a) + \frac{1}{8} \text{osc}(\delta)^2,$$

which proves the claim. $\square$

Remark D.6. The Hoeffding bound $\frac{1}{8} \text{osc}(\delta)^2$ improves on the looser bound $2\|\delta\|_\infty^2$ that follows from 1-strong convexity of ϕ with respect to the ℓ_1 -norm. The improvement factor is $16\|\delta\|_\infty^2 / \text{osc}(\delta)^2$, which is at least 4 because $\text{osc}(\delta) \leq 2\|\delta\|_\infty$.

D.6. Proof of Lemma 6.7 (Last-Iterate KL Convergence)

Proof. We argue pointwise in the state and then take the supremum.

Pointwise Analysis. Fix an arbitrary context (h, s) and let $\mathcal{A}_x := \text{supp}(\pi_{h,i}^{\text{ref}}(\cdot|s))$. We verify that the SOS-MD update (5)–(6) corresponds to the OMD step in Lemma D.5 with the direction

$$\delta_t(s, a) := \gamma_t \hat{q}_{h,i}^{(t)}(s, a) - \gamma_t \eta^{-1} \log \frac{\pi_{h,i}^{(t)}(a|s)}{\pi_{h,i}^{\text{ref}}(a|s)}, \quad (103)$$

where the player-1 update uses $+\hat{q}_{h,1}^{(t)}$ and the player-2 update uses $-\hat{q}_{h,2}^{(t)}$. Apply Lemma D.5 with $\pi^- = \pi_h^{(t)}(\cdot|s)$, $\pi^+ = \pi_h^{(t+1)}(\cdot|s)$, target policy $\pi = \hat{\pi}_h(\cdot|s)$, and direction δ_t as above; sum the resulting inequalities for both players to obtain

$$V_{t+1}(s) \leq V_t(s) + \underbrace{\sum_{i=1}^2 \sum_{a \in \mathcal{A}_x} (\pi_{h,i}^{(t)}(a|s) - \hat{\pi}_{h,i}(a|s)) \delta_{t,i}(s, a)}_{(I)} + \frac{1}{8} (\text{osc}(\delta_{t,1}(s, \cdot))^2 + \text{osc}(\delta_{t,2}(s, \cdot))^2), \quad (104)$$

where $V_t(s) := \text{KL}(\hat{\pi}_{h,1}(\cdot|s) \| \pi_{h,1}^{(t)}(\cdot|s)) + \text{KL}(\hat{\pi}_{h,2}(\cdot|s) \| \pi_{h,2}^{(t)}(\cdot|s))$.

Bounding the oscillation term. We have $\text{osc}(\hat{q}_{h,i}^{(t)}(s, \cdot)) \leq 2\|\hat{Q}_h\|_\infty \leq 2H(1 + \eta^{-1} \log(1/\alpha))$ by Assumption C.1, and by Lemma D.4, $\text{osc}(\log(\pi_{h,i}^{(t)}(\cdot|s)/\pi_{h,i}^{\text{ref}}(\cdot|s))) \leq 2C_{\log} = 4\eta H(1 + \eta^{-1} \log(1/\alpha))$. Therefore,

$$\text{osc}(\delta_{t,i}(s, \cdot)) \leq \gamma_t \cdot 2H(1 + \eta^{-1} \log(1/\alpha)) + \gamma_t \eta^{-1} \cdot 4\eta H(1 + \eta^{-1} \log(1/\alpha)) = 6\gamma_t H(1 + \eta^{-1} \log(1/\alpha)).$$

Hence the oscillation term in (104) is bounded by

$$\frac{1}{8} \sum_{i=1}^2 \text{osc}(\delta_{t,i}(s, \cdot))^2 \leq 9\gamma_t^2 H^2(1 + \eta^{-1} \log(1/\alpha))^2.$$

Bounding term (I) via the joint saddle-point property of $\hat{\pi}_h$. Substituting (103) into term (I) and using $\sum_a (\pi_{h,i}^{(t)}(a|s) - \hat{\pi}_{h,i}(a|s)) = 0$,

$$\begin{aligned} \text{(I)} &= \gamma_t [\langle \pi_{h,1}^{(t)} - \hat{\pi}_{h,1}, \hat{q}_{h,1}^{(t)} \rangle - \langle \pi_{h,2}^{(t)} - \hat{\pi}_{h,2}, \hat{q}_{h,2}^{(t)} \rangle] \\ &\quad - \gamma_t \eta^{-1} \sum_{i=1}^2 \left\langle \pi_{h,i}^{(t)} - \hat{\pi}_{h,i}, \log \frac{\pi_{h,i}^{(t)}}{\pi_{h,i}^{\text{ref}}} \right\rangle. \end{aligned}$$

Using the identity $-\langle \pi^{(t)} - \hat{\pi}, \log(\pi^{(t)}/\pi^{\text{ref}}) \rangle = -\text{KL}(\pi^{(t)} \parallel \pi^{\text{ref}}) + \text{KL}(\hat{\pi} \parallel \pi^{\text{ref}}) - \text{KL}(\hat{\pi} \parallel \pi^{(t)})$ on both players, term (I) becomes

$$\begin{aligned} \text{(I)} &= \gamma_t [\langle \pi_{h,1}^{(t)} - \hat{\pi}_{h,1}, \hat{q}_{h,1}^{(t)} \rangle - \langle \pi_{h,2}^{(t)} - \hat{\pi}_{h,2}, \hat{q}_{h,2}^{(t)} \rangle] \\ &\quad + \gamma_t \eta^{-1} [-\text{KL}(\pi_{h,1}^{(t)} \parallel \pi_{h,1}^{\text{ref}}) + \text{KL}(\hat{\pi}_{h,1} \parallel \pi_{h,1}^{\text{ref}}) - \text{KL}(\pi_{h,2}^{(t)} \parallel \pi_{h,2}^{\text{ref}}) + \text{KL}(\hat{\pi}_{h,2} \parallel \pi_{h,2}^{\text{ref}})] \\ &\quad - \gamma_t \eta^{-1} V_t(s). \end{aligned}$$

Now substitute $\hat{q}_{h,1}^{(t)}(s, \cdot) = \mathbb{E}_{a_2 \sim \pi_{h,2}^{(t)}} [\hat{Q}_h(s, \cdot, a_2)]$ and $\hat{q}_{h,2}^{(t)}(s, \cdot) = \mathbb{E}_{a_1 \sim \pi_{h,1}^{(t)}} [\hat{Q}_h(s, a_1, \cdot)]$ into the first line: the $\mathbb{E}_{a \sim \pi_{h,i}^{(t)} \times \pi_{h,j}^{(t)}} [\hat{Q}_h(s, a)]$ cross-terms cancel, leaving $\mathbb{E}_{\pi_{h,1}^{(t)} \times \pi_{h,2}^{(t)}} [\hat{Q}_h] - \mathbb{E}_{\hat{\pi}_{h,1} \times \pi_{h,2}^{(t)}} [\hat{Q}_h]$. Combining with the KL terms on the second line yields

$$\begin{aligned} \text{(I)} &= \gamma_t \cdot \underbrace{[J_s(\pi_{h,1}^{(t)}, \hat{\pi}_{h,2}; \hat{Q}_h) - J_s(\hat{\pi}_{h,1}, \pi_{h,2}^{(t)}; \hat{Q}_h)]}_{\leq 0 \text{ by joint saddle-point of } \hat{\pi}_h} - \gamma_t \eta^{-1} V_t(s), \end{aligned} \quad (105)$$

where $J_s(\pi_1, \pi_2; Q) := \mathbb{E}_{a \sim \pi_1 \times \pi_2} [Q(s, a)] - \eta^{-1} \text{KL}(\pi_1 \parallel \pi_{h,1}^{\text{ref}}(\cdot|s)) + \eta^{-1} \text{KL}(\pi_2 \parallel \pi_{h,2}^{\text{ref}}(\cdot|s))$ is the stage-game regularized objective at state s with payoff Q . Since $\hat{\pi}_h$ is the saddle point of $J_s(\cdot, \cdot; \hat{Q}_h)$, for all π_1, π_2 ,

$$J_s(\pi_1, \hat{\pi}_{h,2}; \hat{Q}_h) \leq J_s(\hat{\pi}_{h,1}, \hat{\pi}_{h,2}; \hat{Q}_h) \leq J_s(\hat{\pi}_{h,1}, \pi_2; \hat{Q}_h),$$

which (setting $\pi_1 = \pi_{h,1}^{(t)}, \pi_2 = \pi_{h,2}^{(t)}$) makes the bracketed term in (105) ≤ 0 . Hence

$$\text{(I)} \leq -\gamma_t \eta^{-1} V_t(s).$$

Combining and solving the recursion. Substituting both bounds into (104):

$$V_{t+1}(s) \leq (1 - \gamma_t \eta^{-1}) V_t(s) + 9 \gamma_t^2 H^2 (1 + \eta^{-1} \log(1/\alpha))^2. \quad (106)$$

With $\gamma_t = 2\eta/(t+2)$, we have $1 - \gamma_t \eta^{-1} = t/(t+2)$ and $\gamma_t^2 = 4\eta^2/(t+2)^2$. Let $D := 36 \eta^2 H^2 (1 + \eta^{-1} \log(1/\alpha))^2$. Then (106) becomes

$$V_{t+1}(s) \leq \frac{t}{t+2} V_t(s) + \frac{D}{(t+2)^2}.$$

We prove $V_t(s) \leq D/(t+1)$ for all $t \geq 1$ by induction. Base case ($t = 1$): $V_1(s) \leq 0 \cdot V_0(s) + D/4 \leq D/2$. Inductive step: assume $V_t(s) \leq D/(t+1)$. Then

$$V_{t+1}(s) \leq \frac{t}{t+2} \cdot \frac{D}{t+1} + \frac{D}{(t+2)^2} = D \cdot \frac{t(t+2) + (t+1)}{(t+1)(t+2)^2} = D \cdot \frac{t^2 + 3t + 1}{(t+1)(t+2)^2} \leq \frac{D}{t+2},$$

where the final inequality uses $t^2 + 3t + 1 \leq (t+1)(t+2) = t^2 + 3t + 2$. This proves $V_T(s) \leq D/(T+1)$ pointwise for every $T \geq 1$; the bound is not claimed at $T = 0$, where $\pi^{(0)} = \pi^{\text{ref}}$ and $V_0(s) = \text{KL}(\hat{\pi}_{h,1} \parallel \pi_{h,1}^{\text{ref}}) + \text{KL}(\hat{\pi}_{h,2} \parallel \pi_{h,2}^{\text{ref}})$ need not be smaller than D . Every use of Lemma 6.7 and Theorem 6.8 in this paper has $T \geq 1$.

Aggregation. The induction is pointwise in s , so taking the supremum over $s \in \mathcal{S}$,

$$\sup_{s \in \mathcal{S}} V_T(s) \leq \frac{36 \eta^2 H^2 (1 + \eta^{-1} \log(1/\alpha))^2}{T+1}.$$

Since $V_T(s) \geq \text{KL}(\hat{\pi}_{h,i}(\cdot|s) \parallel \pi_{h,i}^{(T)}(\cdot|s))$ for each player i ,

$$\sup_{s \in \mathcal{S}} \text{KL}(\hat{\pi}_{h,i}(\cdot|s) \parallel \pi_{h,i}^{(T)}(\cdot|s)) \leq \frac{36 \eta^2 H^2 (1 + \eta^{-1} \log(1/\alpha))^2}{T+1}.$$

□

D.7. Proof of Lemma D.7 (Lipschitzness of the Duality Gap)

Lemma D.7 (Lipschitzness of the Duality Gap). *Let $\pi = (\pi_1, \pi_2)$ and $\pi' = (\pi'_1, \pi'_2)$ be two policy pairs that both lie in the log-linear-bounded class of Lemma D.4. Then*

$$|\text{Gap}(\pi) - \text{Gap}(\pi')| \leq \mathcal{O}(H^2(1 + \eta^{-1} \log(1/\alpha))) \cdot \sum_{h=1}^H \sup_{s \in \mathcal{S}} (\|\pi_{h,1}(\cdot|s) - \pi'_{h,1}(\cdot|s)\|_1 + \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1).$$

Proof. Define the best-response value functionals

$$\mathcal{V}_1(\pi_2) := \mathbb{E}_{s_1 \sim \rho} [V_1^{\dagger, \pi_2}(s_1)], \quad \mathcal{V}_2(\pi_1) := \mathbb{E}_{s_1 \sim \rho} [V_1^{\pi_1, \dagger}(s_1)],$$

so that $\text{Gap}(\pi) = \mathcal{V}_1(\pi_2) - \mathcal{V}_2(\pi_1)$. By the triangle inequality, it suffices to show that $\mathcal{V}_1$ and $\mathcal{V}_2$ are each Lipschitz in their argument.

We focus on $\mathcal{V}_1$; $\mathcal{V}_2$ is treated below via the reflected game of Appendix D.9. Let π_2, π'_2 be two Player-2 policies in the log-linear-bounded class and let $\pi_1^\dagger(\pi_2)$ and $\pi_1^\dagger(\pi'_2)$ be the regularized best responses of Player 1 to π_2 and to π'_2 . By optimality of $\pi_1^\dagger(\pi_2)$ against π_2 and suboptimality of that policy against π'_2 , and then the reverse pairing,

$$\mathcal{V}_1(\pi_2) - \mathcal{V}_1(\pi'_2) \leq \mathbb{E}_{s_1 \sim \rho} [V_1^{\pi_1^\dagger(\pi_2), \pi_2}(s_1) - V_1^{\pi_1^\dagger(\pi_2), \pi'_2}(s_1)],$$

$$\mathcal{V}_1(\pi'_2) - \mathcal{V}_1(\pi_2) \leq \mathbb{E}_{s_1 \sim \rho} [V_1^{\pi_1^\dagger(\pi'_2), \pi'_2}(s_1) - V_1^{\pi_1^\dagger(\pi'_2), \pi_2}(s_1)],$$

so that

$$|\mathcal{V}_1(\pi_2) - \mathcal{V}_1(\pi'_2)| \leq \max_{\tilde{\pi}_1 \in \{\pi_1^\dagger(\pi_2), \pi_1^\dagger(\pi'_2)\}} \mathbb{E}_{s_1 \sim \rho} |V_1^{\tilde{\pi}_1, \pi_2}(s_1) - V_1^{\tilde{\pi}_1, \pi'_2}(s_1)|.$$

A supremum over all Player-1 policies would be ill posed here: it would include policies charging actions outside $\text{supp}(\pi_{h,1}^{\text{ref}})$, whose forward-KL penalty is infinite. Both $\pi_1^\dagger(\pi_2)$ and $\pi_1^\dagger(\pi'_2)$ are stagewise Gibbs distributions with base measure $\pi_{h,1}^{\text{ref}}$ and logits bounded in ℓ_∞ by $\eta H \lambda$, hence lie in the log-linear class with constant $C_{\log}$ by the second part of Lemma D.4. Fix such a $\tilde{\pi}_1$. For each $h \in [H]$ and state s , define $\Delta_h(s) := V_h^{\tilde{\pi}_1, \pi_2}(s) - V_h^{\tilde{\pi}_1, \pi'_2}(s)$. By the regularized Bellman recursion,

$$\begin{aligned} \Delta_h(s) &= \mathbb{E}_{a \sim \tilde{\pi}_{h,1} \times \pi_{h,2}} [Q_h^{\tilde{\pi}_1, \pi_2}(s, a)] - \mathbb{E}_{a \sim \tilde{\pi}_{h,1} \times \pi'_{h,2}} [Q_h^{\tilde{\pi}_1, \pi'_2}(s, a)] \\ &\quad + \eta^{-1} [\text{KL}(\pi_{h,2} \| \pi_{h,2}^{\text{ref}}) - \text{KL}(\pi'_{h,2} \| \pi_{h,2}^{\text{ref}})](s). \end{aligned}$$

The first line decomposes via add-and-subtract into a stage- h Player-2 policy difference (bounded by $\|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1 \cdot \|Q_h^{\tilde{\pi}_1, \pi_2}\|_\infty$) plus a continuation difference $\mathbb{E}_{a \sim \tilde{\pi}_{h,1} \times \pi_{h,2}} \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [\Delta_{h+1}(s')]$. The required payoff bound is $\|Q_h^{\tilde{\pi}_1, \nu}\|_\infty \leq H\lambda$ for any reference-supported pair $(\tilde{\pi}_1, \nu)$, which Lemma C.2 does not cover: it bounds best-response values, not the value of a fixed pair. We use the α -envelope of Lemma D.2: by (97) the two KL terms of (1) contribute at most $\eta^{-1} \log(1/\alpha) = \lambda - 1$ in absolute value at each stage, and $|\gamma_h^*| \leq 1$, so backward induction gives $\|V_h^{\tilde{\pi}_1, \nu}\|_\infty \leq (H - h + 1)\lambda$ and therefore $\|Q_h^{\tilde{\pi}_1, \nu}\|_\infty \leq 1 + (H - h)\lambda \leq H\lambda$. The stage- h payoff contribution is thus at most $H\lambda \cdot \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1$.

For the KL-regularization term, since $\nabla_p \text{KL}(p \| r) = \log(p/r) + \mathbf{1}$ and $\langle \mathbf{1}, p - q \rangle = 0$ for probability vectors, the leading $\mathbf{1}$ cancels and no bare η^{-1} appears. Because $\pi_{h,2}$ and $\pi'_{h,2}$ both lie in the log-linear class with $|\log(\pi_{h,2}/\pi_{h,2}^{\text{ref}})| \leq C_{\log} = 2\eta H \lambda$, and that class is convex, the whole segment joining them stays inside it, and (99) gives

$$\eta^{-1} |\text{KL}(\pi_{h,2}(\cdot|s) \| \pi_{h,2}^{\text{ref}}(\cdot|s)) - \text{KL}(\pi'_{h,2}(\cdot|s) \| \pi_{h,2}^{\text{ref}}(\cdot|s))| \leq \eta^{-1} C_{\log} \cdot \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1 = 2H\lambda \cdot \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1.$$

Combining the two contributions,

$$|\Delta_h(s)| \leq 3H\lambda \cdot \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1 + \mathbb{E}_{a \sim \tilde{\pi}_{h,1}(\cdot|s) \times \pi_{h,2}(\cdot|s)} \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [|\Delta_{h+1}(s')|].$$

Unrolling from $h = 1$ to H along trajectories,

$$|\Delta_1(s_1)| \leq \mathcal{O}(H(1 + \eta^{-1} \log(1/\alpha))) \sum_{h=1}^H \mathbb{E}_{s_h \sim \tilde{\pi}_1, \pi_2} [\|\pi_{h,2}(\cdot|s_h) - \pi'_{h,2}(\cdot|s_h)\|_1].$$

Bounding each visitation expectation by $\sup_s \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1$ and aggregating over ρ ,

$$|\mathcal{V}_1(\pi_2) - \mathcal{V}_1(\pi'_2)| \leq \mathcal{O}(H \cdot H(1 + \eta^{-1} \log(1/\alpha))) \cdot \sum_{h=1}^H \sup_s \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1.$$

The corresponding Lipschitz bound for $\mathcal{V}_2$ in its argument π_1 is *not* asserted by symmetry. Let $\tilde{\mathcal{M}}$ be the reflected game of Appendix D.9 and let $\tilde{\mathcal{V}}_1$ denote its Player-1 best-response functional. By (107) with $c_1 = H$ one has $\tilde{\mathcal{V}}_1(\pi_1) = H - \mathcal{V}_2(\pi_1)$ for every reference-supported π_1 , so $|\mathcal{V}_2(\pi_1) - \mathcal{V}_2(\pi'_1)| = |\tilde{\mathcal{V}}_1(\pi_1) - \tilde{\mathcal{V}}_1(\pi'_1)|$; the display just proved, applied inside $\tilde{\mathcal{M}}$ (whose reference-policy constant is again α , so $\tilde{\lambda} = \lambda$, and whose value scale is at most $2C_{\text{val}}$), gives the same $\mathcal{O}(H^2\lambda)$ coefficient up to an absolute factor. Combining,

$$|\text{Gap}(\pi) - \text{Gap}(\pi')| \leq \mathcal{O}(H^2(1 + \eta^{-1} \log(1/\alpha))) \sum_{h=1}^H \left(\sup_s \|\pi_{h,1}(\cdot|s) - \pi'_{h,1}(\cdot|s)\|_1 + \sup_s \|\pi_{h,2}(\cdot|s) - \pi'_{h,2}(\cdot|s)\|_1 \right).$$

□

D.8. Proof of Theorem 6.8

Proof. Let $T \geq 1$. By Lemma 6.7 — which, by the convention of Appendix D.2, is a statement about this exact equilibrium $\hat{\pi}_h$ of the payoff $\tilde{Q}_h$ actually fitted by Algorithm 2 — we have $\sup_s \text{KL}(\hat{\pi}_h(\cdot|s) \|\pi_h^{(T)}(\cdot|s)) \leq 36\eta^2 H^2(1 + \eta^{-1} \log(1/\alpha))^2 / (T + 1)$, where the stage policies are the product distributions of the two players, so that this KL is the sum of the two players' KL divergences and the same bound holds for each player's marginal. Pinsker's inequality gives, pointwise in s ,

$$\|\hat{\pi}_h(\cdot|s) - \pi_h^{(T)}(\cdot|s)\|_1 \leq \sqrt{2 \text{KL}(\hat{\pi}_h(\cdot|s) \|\pi_h^{(T)}(\cdot|s))}.$$

Taking the supremum over s on both sides and monotonicity of $\sqrt{\cdot}$,

$$\sup_{s \in \mathcal{S}} \|\hat{\pi}_h(\cdot|s) - \pi_h^{(T)}(\cdot|s)\|_1 \leq \sqrt{2 \cdot \sup_s \text{KL}(\hat{\pi}_h(\cdot|s) \|\pi_h^{(T)}(\cdot|s))} \leq 6\sqrt{2} \eta H (1 + \eta^{-1} \log(1/\alpha)) / \sqrt{T + 1} = \tilde{\mathcal{O}}(\eta H \lambda / \sqrt{T}),$$

where $\lambda = 1 + \eta^{-1} \log(1/\alpha)$. □

D.9. Term B via the Reflected Game

Term B of (95) is Player 2's regret. Rather than assert it “by symmetry”, we exhibit an isomorphism of the learning problem under which Term B of $\mathcal{M}$ becomes Term A of a second game, to which every step of the Term-A analysis applies unchanged.

The reflection. Let $\sigma(s, a_1, a_2) := (s, a_2, a_1)$ be the action-coordinate swap and put $c_h := H - h + 1$, so $c_{H+1} = 0$ and $c_h = 1 + c_{h+1}$. Define the *reflected game* $\tilde{\mathcal{M}} := (\mathcal{S}, \mathcal{A}_2, \mathcal{A}_1, \tilde{P}, \tilde{r}^*, H)$, with the same initial distribution ρ and the same regularization parameter η , by

$$\tilde{P}_h(\cdot | s, b_1, b_2) := P_h(\cdot | s, b_2, b_1), \quad \tilde{r}_h^*(s, b_1, b_2) := 1 - r_h^*(s, b_2, b_1) \in [0, 1], \quad \tilde{\pi}_{h,1}^{\text{ref}} := \pi_{h,2}^{\text{ref}}, \quad \tilde{\pi}_{h,2}^{\text{ref}} := \pi_{h,1}^{\text{ref}}.$$

For a joint policy $\pi = (\pi_1, \pi_2)$ of $\mathcal{M}$ write $\tilde{\pi} := (\pi_2, \pi_1)$ for the corresponding joint policy of $\tilde{\mathcal{M}}$. Reflection is an involution, and $\tilde{\alpha} = \alpha$, hence $\tilde{\lambda} = \lambda$ and $\tilde{C}_{\text{trans}} = C_{\text{trans}}$; the side condition (7) is unchanged.

(a) Values. For every h , every s and every reference-supported π ,

$$\tilde{V}_h^{\tilde{\pi}}(s) = c_h - V_h^\pi(s), \quad \tilde{Q}_h^{\tilde{\pi}}(s, b_1, b_2) = c_h - Q_h^\pi(s, b_2, b_1). \quad (107)$$

Indeed, both sides vanish at $h = H + 1$; assuming the claim at $h + 1$ and using $\tilde{P}_h(\cdot | s, b_1, b_2) = P_h(\cdot | s, b_2, b_1)$,

$$\tilde{Q}_h^{\tilde{\pi}}(s, b_1, b_2) = 1 - r_h^*(s, b_2, b_1) + \mathbb{E}_{s' \sim P_h(\cdot | s, b_2, b_1)} [c_{h+1} - V_{h+1}^\pi(s')] = c_h - Q_h^\pi(s, b_2, b_1),$$

and substituting into (1) for $\tilde{\mathcal{M}}$ — where Player 1 carries $-\eta^{-1}\text{KL}(\pi_{h,2}\|\pi_{h,2}^{\text{ref}})$ and Player 2 carries $+\eta^{-1}\text{KL}(\pi_{h,1}\|\pi_{h,1}^{\text{ref}})$ — gives $\tilde{V}_h^{\tilde{\pi}}(s) = c_h - V_h^{\pi}(s)$. Because $x \mapsto c_h - x$ is order-reversing, maximizing over Player 1 of $\tilde{\mathcal{M}}$ is minimizing over Player 2 of $\mathcal{M}$:

$$\tilde{V}_h^{\dagger, \pi_1}(s) = c_h - V_h^{\pi_1, \dagger}(s), \quad \tilde{V}_h^{\pi_2, \dagger}(s) = c_h - V_h^{\dagger, \pi_2}(s).$$

In particular $\tilde{\pi}^* = (\pi_2^*, \pi_1^*)$ is the regularized Nash equilibrium of $\tilde{\mathcal{M}}$, and with $c_1 = H$,

$$\text{Term B of } \mathcal{M} = \mathbb{E}_{s_1 \sim \rho} [V_1^{\pi^*}(s_1) - V_1^{\pi_1^{(T)}, \dagger}(s_1)] = \mathbb{E}_{s_1 \sim \rho} [\tilde{V}_1^{\dagger, \pi_1^{(T)}}(s_1) - \tilde{V}_1^{\tilde{\pi}^*}(s_1)] = \text{Term A of } \tilde{\mathcal{M}}, \quad (108)$$

the right-hand side being Term A of $\tilde{\mathcal{M}}$ formed with the reflected learned policy $\tilde{\pi}^{(T)} = (\pi_2^{(T)}, \pi_1^{(T)})$.

(b) Occupancies transform as pushforwards. If π induces $d_h^\pi \in \Delta(\mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2)$, then $\tilde{\pi}$ induces $\tilde{d}_h^{\tilde{\pi}} = \sigma_{\#} d_h^\pi$; in the countable case, $\tilde{d}_h^{\tilde{\pi}}(s, b_1, b_2) = d_h^\pi(s, b_2, b_1)$. The state marginals coincide. This follows by induction on h from $\tilde{P}_h = P_h \circ \sigma$ and $\tilde{\pi}_{h,1} \otimes \tilde{\pi}_{h,2} = (\pi_{h,1} \otimes \pi_{h,2}) \circ \sigma$.

(c) Function class, dataset and Bellman completeness. Set $\tilde{\mathcal{Q}}_h := \{(s, b_1, b_2) \mapsto c_h - q(s, b_2, b_1) : q \in \mathcal{Q}_h\}$, write $\tilde{\mathcal{Q}}$ for $\tilde{\mathcal{Q}}_h$ when the stage is clear, and set $|\tilde{\mathcal{Q}}| := \max_{h \in [H]} |\tilde{\mathcal{Q}}_h|$. Then $|\tilde{\mathcal{Q}}| \leq |\mathcal{Q}|$ and $\|\tilde{q}\|_\infty \leq c_h + C_{\text{val}} \leq 2C_{\text{val}}$. Thus Assumption C.1 holds in $\tilde{\mathcal{M}}$ with $\tilde{C}_{\text{val}} \leq 2C_{\text{val}}$; every regression constant therefore changes by at most an absolute factor, absorbed into $\mathcal{O}(\cdot)$. This raw-envelope inflation does not alter $\tilde{C}_{\text{trans}}$: the action-independent c_h cancels from Gibbs normalizations and logit differences, leaving centered payoff vectors bounded by the original $C_{\text{val}} = H\lambda$. Bellman backups correspond, $\tilde{T}_h(c_{h+1} - g) = c_h - (\mathcal{T}_h g) \circ \sigma$, so $\mathcal{T}_h g \in \mathcal{Q}_h$ implies $\tilde{T}_h \tilde{g} \in \tilde{\mathcal{Q}}_h$: completeness transfers. The reflected dataset is $\tilde{\mathcal{D}} := \{(s_{i,h}, a_{i,h,2}, a_{i,h,1}, 1 - r_{i,h}, s'_{i,h})\}$, whose reward noise $-\xi_{i,h}$ is again zero-mean and 1-sub-Gaussian and whose stage- h state-action law is $\tilde{\mu}_h = \sigma_{\#} \mu_h$. Since $\tilde{y}_{i,h} = c_h - y_{i,h}$ and $\tilde{f}(\sigma(s_{i,h}, a_{i,h,1}, a_{i,h,2})) = c_h - f(s_{i,h}, a_{i,h,1}, a_{i,h,2})$ for $\tilde{f} = c_h - f \circ \sigma$, the two least-squares residuals differ only by a sign and the squared objectives coincide: the regression is equivariant, $\tilde{\tilde{Q}}_h = c_h - \tilde{Q}_h \circ \sigma$.

(d) Coverage. Differences of class members are anti-equivariant, $(\tilde{q}_1 - \tilde{q}_2)(\sigma z) = -(q_1 - q_2)(z)$, so the constant c_h cancels and $D_{\tilde{\mathcal{Q}}}^2(\sigma z; \tilde{\mu}_h) = D_{\mathcal{Q}}^2(z; \mu_h)$ pointwise. Combining this with (b) and with $\tilde{\pi}^* = (\pi_2^*, \pi_1^*)$, the reflected unilateral class $\tilde{\Pi}_{\text{uni}} = (\{\pi_2^*\} \times \Pi_1) \cup (\Pi_2 \times \{\pi_1^*\})$ is precisely the image of Π_{uni} , and $\mathbb{E}_{\tilde{d}_h^{\tilde{\pi}^*}}[D_{\tilde{\mathcal{Q}}}^2] = \mathbb{E}_{d_h^{\pi^*}}[D_{\mathcal{Q}}^2]$. Hence Assumption 6.2 holds in $\tilde{\mathcal{M}}$ with $\tilde{C}_{\text{uni}} = C_{\text{uni}}$ — in particular the reflected assumption is again *ordinary* unilateral concentrability.

(e) The algorithms are equivariant. Running Algorithm 2 on $(\tilde{\mathcal{M}}, \tilde{\mathcal{D}}, \{\tilde{\mathcal{Q}}_h\}_{h=1}^H, \tilde{\pi}^{\text{ref}})$ reproduces the original run with the players interchanged. Indeed $\tilde{\pi}_h^{(0)} = (\pi_{h,2}^{\text{ref}}, \pi_{h,1}^{\text{ref}})$, and if $\tilde{\pi}_h^{(t)} = (\pi_{h,2}^{(t)}, \pi_{h,1}^{(t)})$ then

$$\tilde{q}_{h,1}^{(t)}(s, b_1) = \mathbb{E}_{b_2 \sim \tilde{\pi}_{h,2}^{(t)}} [\tilde{Q}_h(s, b_1, b_2)] = c_h - \mathbb{E}_{a_1 \sim \pi_{h,1}^{(t)}} [\tilde{Q}_h(s, a_1, b_1)] = c_h - \hat{q}_{h,2}^{(t)}(s, b_1),$$

and the additive constant c_h contributes the factor $\exp(\gamma_t c_h)$, which cancels in the normalization of (5), so the reflected Player-1 *ascent* step is literally the original Player-2 *descent* step (6); the same computation with $\tilde{q}_{h,2}^{(t)} = c_h - \hat{q}_{h,1}^{(t)}$ turns the reflected Player-2 descent step into the original Player-1 ascent step. By induction $\tilde{\pi}_h^{(t)} = (\pi_{h,2}^{(t)}, \pi_{h,1}^{(t)})$ for every t , and the value update of Algorithm 2 then gives $\tilde{\tilde{V}}_h = c_h - \tilde{V}_h$ by the computation in (a), closing the backward induction over h . The same computation with the exact equilibrium solver shows that Algorithm 1 is equivariant as well, and that this exact equilibrium of the reflected run is $(\tilde{\pi}_2, \tilde{\pi}_1)$.

Conclusion. Every hypothesis used in the Term-A analysis holds in $\tilde{\mathcal{M}}$ by (c)–(d), with α , λ , C_{trans} and the side condition (7) unchanged. By (e) the policy produced in $\tilde{\mathcal{M}}$ is the reflection of the policy produced in $\mathcal{M}$, so by (108) the Term-A bound applied inside $\tilde{\mathcal{M}}$ yields the same bound for Term B of $\mathcal{M}$, up to an absolute factor. The same argument licenses the appeals elsewhere in the paper to the “symmetric” unilateral marginal $d_h^{\tilde{\pi}_1, \tilde{\pi}_2^*}$.

D.10. Proof of Total Error Bound (Corollary 6.9)

Proof. Fix $T \geq 1$ and recall this exact equilibrium $\bar{\pi}$ of Definition D.1. By the transfer inequality (102),

$$\text{Gap}(\pi^{(T)}) \leq \text{Gap}(\bar{\pi}) + |\text{Gap}(\pi^{(T)}) - \text{Gap}(\bar{\pi})|.$$

Write $\text{Gap}(\bar{\pi}) = A(\bar{\pi}) + B(\bar{\pi})$ for the decomposition (95) applied to $\bar{\pi}$, with $A(\bar{\pi}) := \mathbb{E}_{s_1 \sim \rho}[V_1^{\dagger, \bar{\pi}_2}(s_1) - V_1^{\pi^*}(s_1)]$ and $B(\bar{\pi}) := \mathbb{E}_{s_1 \sim \rho}[V_1^{\pi^*}(s_1) - V_1^{\bar{\pi}_1, \dagger}(s_1)]$.

Step 1 (optimization term; whole gap). By Lemma D.4, both $\pi^{(T)}$ and $\bar{\pi}$ lie in the log-linear class, so Lemma D.7 applies to the full duality gap:

$$|\text{Gap}(\pi^{(T)}) - \text{Gap}(\bar{\pi})| \leq \mathcal{O}(H^2(1 + \eta^{-1} \log(1/\alpha))) \sum_{h=1}^H \sup_s (\|\pi_{h,1}^{(T)}(\cdot|s) - \bar{\pi}_{h,1}(\cdot|s)\|_1 + \|\pi_{h,2}^{(T)}(\cdot|s) - \bar{\pi}_{h,2}(\cdot|s)\|_1).$$

By Theorem 6.8 each summand is $\tilde{\mathcal{O}}(\eta H \lambda / \sqrt{T})$, so the sum over $h \in [H]$ is $\tilde{\mathcal{O}}(\eta H^2 \lambda / \sqrt{T})$, and multiplying by the Lipschitz constant $\mathcal{O}(H^2 \lambda)$,

$$|\text{Gap}(\pi^{(T)}) - \text{Gap}(\bar{\pi})| \leq \tilde{\mathcal{O}}\left(\frac{\eta H^4 \lambda^2}{\sqrt{T}}\right). \quad (109)$$

Step 2 (statistical term: Term A of this exact equilibrium). At every stage $\bar{\pi}_h$ is the exact regularized Nash equilibrium of Q_h and $\bar{V}_h$ is its saddle value, so Lemma C.6 and the Term-A chain of Appendix C.6 apply to the triple $(\bar{Q}_h, \bar{\pi}_h, \bar{V}_h)$, the only change being that the pointwise Bellman error is $\bar{Q}_h - \mathcal{T}_h \bar{V}_{h+1}$ rather than a pure regression residual. By (101) this error splits as $\bar{\mathcal{Z}}_{t,s}^2 \leq 2\mathcal{Z}_{t,s}^2 + 2\|\omega_{t+1}\|_\infty^2$, where the first part is the least-squares residual controlled by Lemma C.9 (applied with the SOS-MD-induced value class, see the remark after (100)) and the second is a constant. Substituting into the Term-A instance of (92) for the concrete unilateral marginal $d_t^{\pi_1^*, \bar{\pi}_2}$, which is covered by the supremum notation $\mathbb{E}_{s \sim d_t^{\text{uni}}}$ defined above, and bounding the first part by (94),

$$\begin{aligned} A(\bar{\pi}) &\leq \mathcal{O}(\eta H^2 \lambda^2 + \eta^3 H^6 \lambda^2) \sum_{t=1}^H \left(2 \mathbb{E}_{s \sim d_t^{\text{uni}}} [\mathcal{Z}_{t,s}^2] + 2\|\omega_{t+1}\|_\infty^2 \right) \\ &\leq \tilde{\mathcal{O}}\left(\frac{\eta H^6 \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n}\right) + \mathcal{R}_T, \end{aligned}$$

with probability at least $1 - \delta$, where

$$\mathcal{R}_T := \mathcal{O}(\eta H^2 \lambda^2 + \eta^3 H^6 \lambda^2) \cdot 2H \max_{h \in [H]} \|\omega_h\|_\infty^2. \quad (110)$$

The ω -part of (101) is measure-free, so it is *not* multiplied by C_{uni} and does not pass through Lemma C.5. No clipping is introduced anywhere: Lemma D.2 keeps the unclipped backups $\bar{V}_h$, $\bar{V}_h$ on the $\mathcal{O}(H\lambda)$ scale of $\mathcal{Q}$, which is what makes the least-squares concentration applicable to the SOS-MD sequence.

Step 3 (the residual is of lower order in T). By (98), $\max_h \|\omega_h\|_\infty^2 \leq \mathcal{O}(\eta^2 H^4 \lambda^4 / T)$. Under the side condition (7) we have $\eta^2 H^4 \leq 1/16$, so $\eta^3 H^6 \lambda^2 \leq \eta H^2 \lambda^2 / 16$ and the prefactor in (110) is $\mathcal{O}(\eta H^3 \lambda^2)$. The C_{trans} multipliers of (73) are powers of the absolute constant e and are absorbed into the universal constant, giving

$$\mathcal{R}_T \leq \tilde{\mathcal{O}}\left(\frac{\eta^3 H^7 \lambda^6}{T}\right) \leq \tilde{\mathcal{O}}\left(\frac{\eta H^3 \lambda^6}{T}\right),$$

using $\eta^2 H^4 \leq 1/16$ once more. This is $\tilde{\mathcal{O}}(1/T)$, hence of strictly lower order in T than the optimization term (109). Absorbing it into that term requires $\sqrt{T} \geq \eta^2 H^3 \lambda^4$, i.e. $T \geq \eta^4 H^6 \lambda^8$; under (7), $T \geq \lambda^4 / (256 H^2)$ is sufficient. We therefore carry the residual explicitly:

$$A(\bar{\pi}) \leq \tilde{\mathcal{O}}\left(\frac{\eta H^6 \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n}\right) + \tilde{\mathcal{O}}\left(\frac{\eta H^4 \lambda^2}{\sqrt{T}}\right) + \tilde{\mathcal{O}}\left(\frac{\eta^3 H^7 \lambda^6}{T}\right). \quad (111)$$

For $T \geq c \lambda^4 / H^2$ the third term is dominated by the second and may be dropped; for all $T \geq 1$ the second and third may instead be merged into $\tilde{\mathcal{O}}((\eta H^4 \lambda^2 + \eta^3 H^7 \lambda^6) / \sqrt{T})$.

Step 4 (Term B of this exact equilibrium, via the reflected game). For $B(\bar{\pi})$: by (108) applied to $\bar{\pi}$, $B(\bar{\pi})$ equals Term A of the reflected game $\check{\mathcal{M}}$ of Appendix D.9 evaluated at the reflected policy $(\bar{\pi}_2, \bar{\pi}_1)$. By parts (c)–(d) all hypotheses and constants carry over, with $\check{C}_{\text{uni}} = C_{\text{uni}}$, $\check{C}_{\text{trans}} = C_{\text{trans}}$, $\check{\lambda} = \lambda$ and $\check{C}_{\text{val}} \leq 2C_{\text{val}}$, and by part (e) the SOS-MD run in $\check{\mathcal{M}}$

is the reflection of the run in $\mathcal{M}$, so its exact equilibrium is $(\bar{\pi}_2, \bar{\pi}_1)$ and its last-iterate defect is $-\omega$. Running Steps 2–3 inside $\check{\mathcal{M}}$ therefore gives the bound (111) for $B(\bar{\pi})$ as well, with only absolute constants changed.

Step 5 (combining). Adding Steps 3 and 4 gives $\text{Gap}(\bar{\pi}) = A(\bar{\pi}) + B(\bar{\pi})$ bounded by twice the right-hand side of (111), and combining with the optimization transfer (109),

$$\text{Gap}(\pi^{(T)}) \leq \tilde{\mathcal{O}}\left(\frac{\eta H^4 \lambda^2}{\sqrt{T}}\right) + \tilde{\mathcal{O}}\left(\frac{\eta H^6 \lambda^4 C_{\text{uni}} \log(|\mathcal{Q}|/\delta)}{n}\right) + \tilde{\mathcal{O}}\left(\frac{\eta^3 H^7 \lambda^6}{T}\right),$$

which is the displayed form of Corollary 6.9; the third term may be dropped once $T \geq c \lambda^4 / H^2$. □